

INTERNATIONAL STANDARD

Calibration of optical time-domain reflectometers (OTDR) – Part 1: OTDR for single mode fibres



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IEC Central Office
3, rue de Varembe
CH-1211 Geneva 20
Switzerland
Email: inmail@iec.ch
Web: www.iec.ch

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IEC 61746-1

Edition 1.0 2009-12

INTERNATIONAL STANDARD

**Calibration of optical time-domain reflectometers (OTDR) –
Part 1: OTDR for single mode fibres**

INTERNATIONAL
ELECTROTECHNICAL
COMMISSION

INTERNATIONAL
ELECTROTECHNICAL
COMMISSION

PRICE CODE **XC**

ICS 33.180.01

ISBN 2-8318-1072-9

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INTERNATIONAL ELECTROTECHNICAL COMMISSION

**CALIBRATION OF OPTICAL TIME-DOMAIN
REFLECTOMETERS (OTDR) –****Part 1: OTDR for single mode fibres**

FOREWORD

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International Standard IEC 61746-1 has been prepared by IEC technical committee 86: Fibre optics.

This first edition of IEC 61746-1 cancels and replaces the second edition of IEC 61746, published in 2005. It constitutes a technical revision.

The main technical changes are the adaptation of Clause 4, the suppression of Clause 10, the improvement and the addition of some definitions, the change of some calculations and the change of graphical symbology to IEC/TR 61930.

The text of this standard is based on the following documents:

FDIS	Report on voting
86/347/FDIS	86/362/RVD

Full information on the voting for the approval of this standard can be found in the report on voting indicated in the above table.

This publication has been drafted in accordance with the ISO/IEC Directives, Part 2.

The committee has decided that the contents of this publication will remain unchanged until the maintenance result date indicated on the IEC web site under "<http://webstore.iec.ch>" in the data related to the specific publication. At this date, the publication will be

- reconfirmed,
- withdrawn,
- replaced by a revised edition, or
- amended.

A bilingual version of this standard may be issued at a later date.

INTRODUCTION

In order for an Optical time-domain reflectometer (OTDR) to qualify as a candidate for complete calibration using this standard, it must be equipped with the following minimum feature set:

- a) a programmable index of refraction, or equivalent parameter;
- b) the ability to present a display of a trace representation, with a logarithmic power scale and a linear distance scale;
- c) two markers/cursors, which display the loss and distance between any two points on a trace display;
- d) the ability to measure absolute distance (location) from the OTDR's zero-distance reference;
- e) the ability to measure the displayed power level relative to a reference level (for example, the clipping level);
- f) the ability to evaluate the reflectance of a reflective event.

CALIBRATION OF OPTICAL TIME-DOMAIN REFLECTOMETERS (OTDR) –

Part 1: OTDR for single mode fibres

1 Scope

This part of IEC 61746 provides procedures for calibrating single-mode optical time domain reflectometers (OTDR). It only covers OTDR measurement errors and uncertainties.

This standard does not cover correction of the OTDR response.

2 Normative references

The following referenced documents are indispensable for the application of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

IEC 60793-1-40, *Optical fibres – Part 1-40: Measurement methods and test procedures – Attenuation*

IEC 60793-2-50, *Optical fibres – Part 2-50: Product specifications – Sectional specification for class B single-mode fibres*

ISO/IEC 17025, *General requirements for the competence of testing and calibration laboratories*

ITU-T Recommendation G.650.1:2002, *Definitions and test methods for linear, deterministic attributes of single-mode fibre and cable*

ITU-T Recommendation G.650.2:2002, *Definitions and test methods for statistical and non-linear attributes of single-mode fibre and cable*

3 Terms, definitions and symbols

For the purposes of this document, the following terms, definitions and symbols apply.

NOTE For more precise definitions, the references to IEC 60050-731 should be consulted.

3.1 attenuation loss

A

optical power decrease in decibels (dB)

NOTE If P_{in} (watts) is the power entering one end of a segment of fibre and P_{out} (watts) is the power leaving the other end, then the attenuation of the segment is

$$A = 10 \log_{10} \left(\frac{P_{\text{in}}}{P_{\text{out}}} \right) \quad \text{dB} \quad (1)$$

[IEV 731-01-48, modified]

3.2 attenuation coefficient

α

attenuation (3.1) of a fibre per unit length

[IEV 731-03-42, modified]

3.3 attenuation dead zone

for a reflective or attenuating event, the region after the event where the displayed trace deviates from the undisturbed backscatter trace by more than a given vertical distance ΔF

NOTE The attenuation dead zone (see Figure 1 below) will depend on the following event parameters: reflectance, loss, displayed power level and location. It may also depend on any fibre optic component in front of the event.

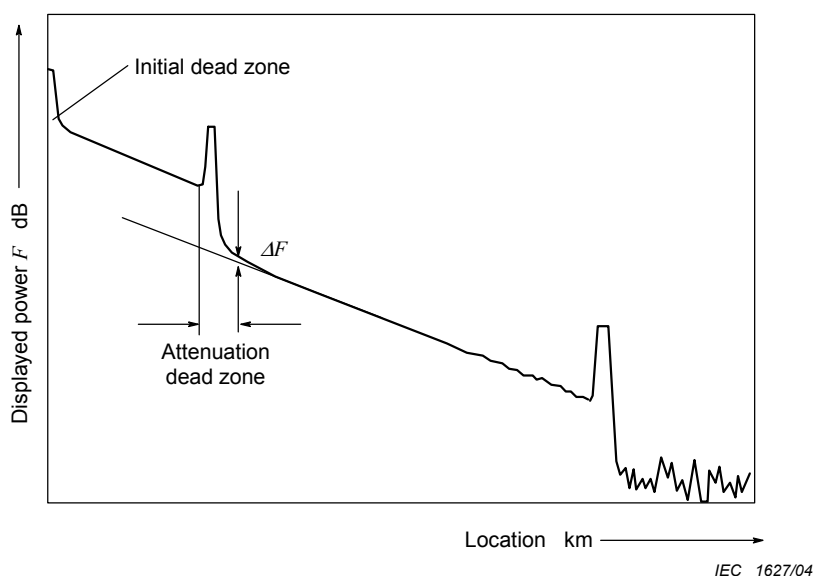


Figure 1 – Definition of attenuation dead zone

3.4 backscatter parameter

K

at a given point along the fibre, the backscattered propagating power per unit incident energy

NOTE 1 K is given by the following formula:

$$K = S\alpha_s \frac{V}{2} \text{ s}^{-1} \quad (2)$$

where

α_s is the scattering coefficient, e.g.; in m^{-1} ;

S is the backscatter capture fraction. It depends on other standard fibre parameters such as the mode field diameter in single mode fibre;

V is the group velocity (in m/s);

$= c / N$ where c is the speed of the light in vacuum, N the group index of the fibre.

NOTE 2 See also Annex G.

3.5**backscatter coefficient** **C**

for a given pulse, the ratio of backscattered power at the input side of the fibre to the pulse input power

NOTE 1 It represents the backscatter parameter for a given pulse width. The backscatter coefficient is defined from the backscatter parameter (3.4) using the following formula:

$$C(\Delta T) = K\Delta T \quad (3)$$

where ΔT is the pulse width, e.g. in seconds.

Usually the backscatter coefficient is expressed in dB for a given pulse width, ΔT .

$$C_{\text{dB}}(\Delta T) = 10 \log_{10}(K\Delta T) \quad (4)$$

NOTE 2 The pulse width, ΔT in the previous formula is used to normalise $C(\Delta T)$. Usual values for ΔT are 1 ns and 1 μ s. See also Annex G.

3.6**calibration**

set of operations which establish, under specified conditions, the relationship between the values indicated by the measuring instrument and the corresponding known values of that quantity

NOTE See ISO/IEC Guide 99 [11] in the bibliography.

3.7**centroidal wavelength** **λ_{avg}**

power-weighted mean wavelength of a light source in vacuum

[IEC 61280-1-3, definition 2.1.4]

3.8**displayed power level** **F**

level displayed on the OTDR's power scale

NOTE 1 Unless otherwise specified, F is defined in relation to the clipping level (see Figure 8).

NOTE 2 Usually, the OTDR scale displays five times the logarithm of the received power, plus a constant offset.

3.9**distance** **D**

spacing between two features

NOTE Usually expressed in metres.

3.10**distance sampling error** **ΔL_{sample}**

maximum distance (3.9) error attributable to the distance between successive sample points

NOTE 1 Usually expressed in metres.

NOTE 2 The distance sampling error is repetitive in nature; therefore, one way of quantifying this error is by its amplitude.

3.11 distance scale deviation

ΔS_L

difference between the average displayed distance (3.9) $\langle D_{\text{otdr}} \rangle$ and the correspondent reference distance (3.27) D_{ref} divided by the reference distance (3.27)

NOTE 1 Usually expressed in m/m

NOTE 2 ΔS_L is given by the following formula:

$$\Delta S_L = \frac{\langle D_{\text{otdr}} \rangle - D_{\text{ref}}}{D_{\text{ref}}} = \frac{\langle D_{\text{otdr}} \rangle}{D_{\text{ref}}} - 1 \quad (5)$$

where $\langle D_{\text{otdr}} \rangle$ is the displayed distance on a fibre averaged over at least one sample spacing.

3.12 distance scale factor

S_L

average displayed distance (3.9) distance divided by the correspondent reference distance (3.27)

NOTE S_L is given by the following formula:

$$S_L = \frac{\langle D_{\text{otdr}} \rangle}{D_{\text{ref}}} \quad (6)$$

where $\langle D_{\text{otdr}} \rangle$ is the displayed distance between two features on a fibre averaged over at least one sample spacing.

3.13 distance scale uncertainty

$u_{\Delta S_L}$

uncertainty of the distance scale deviation (3.11)

NOTE 1 Usually expressed in m/m.

NOTE 2 $u_{\Delta S_L}$ is given by the following formula:

$$u_{\Delta S_L} = u\left(\frac{\langle D_{\text{otdr}} \rangle}{D_{\text{ref}}} - 1\right) = u\left(\frac{\langle D_{\text{otdr}} \rangle}{D_{\text{ref}}}\right) \quad (7)$$

NOTE 3 In the above formula, $u()$ is understood as the standard uncertainty of ().

3.14 dynamic range at 98 %(one-way)

amount of fibre attenuation (3.1) that causes the backscatter signal to equal the noise level at 98 % (3.24)

NOTE It can be represented by the difference between the extrapolated point of the backscattered trace (taken at the intercept with the power axis) and the noise level expressed in decibels, using a standard category B fibre (see IEC 60793-2-50).

3.15 group index

N

factor by which the speed of light in vacuum has to be divided to yield the propagation velocity of light pulses in the fibre

3.16 location

L

spacing between the front panel of the OTDR and a feature in a fibre

NOTE Usually expressed in metres.

3.17 location deviation

ΔL

displayed location (3.16) of a feature L_{otdr} minus the reference location (3.28) L_{ref}

NOTE 1 Usually expressed in metres.

NOTE 2 This deviation is a function of the location.

3.18 location offset

ΔL_0

constant term of the location deviation (3.17) model

NOTE 1 Usually expressed in metres.

NOTE 2 This is approximately equivalent to the location of the OTDR front panel connector on the instrument's distance scale.

NOTE 3 For a perfect OTDR, the location offset is zero.

3.19 location offset uncertainty

$u_{\Delta L_0}$

uncertainty of the location offset (3.18)

3.20 location readout uncertainty

$u_{L_{\text{readout}}}$

uncertainty of the location (3.16) measurement samples caused by both the distance sampling error (3.10) and the uncertainty type A of the measurement samples

3.21 loss deviation

ΔA

difference between the displayed loss of a fibre component A_{otdr} and the reference loss (3.29), in dB

NOTE 1 ΔA is given by the following formula:

$$\Delta A = A_{\text{otdr}} - A_{\text{ref}} \quad (8)$$

NOTE 2 The loss deviation usually depends on the displayed power level, F .

3.22 loss uncertainty

$u_{\Delta A}$

uncertainty of the loss deviation (3.21), in dB

3.23 loss scale deviation

ΔS_A

difference between the displayed loss of a fibre component A_{otdr} and the reference loss (3.29) A_{ref} , divided by the reference loss (3.29), in dB/dB

NOTE 1 ΔS_A is given by the following formula:

$$\Delta S_A = \frac{A_{\text{otdr}} - A_{\text{ref}}}{A_{\text{ref}}} \quad (9)$$

NOTE 2 Refer to 7.1 for more details.

3.24 noise level at 98 %

upper limit of a range which contains at least 98 % of all noise data points

3.25 non-linearity

NL_{loss}

difference between the maximum and minimum values of the loss deviation (3.21) ΔA for a given range of power levels, in dB

NOTE 1 This is the non linearity of a logarithmic power scale.

NOTE 2 Non-linearity is one contribution to loss deviation; it usually depends on the displayed power level and the location.

3.26 received power level

P

power received by the OTDR's optical port

3.27 reference distance

D_{ref}

distance (3.9) precisely determined by measuring equipment with calibration traceable to international or national standards

NOTE Usually expressed in metres.

3.28 reference location

L_{ref}

location (3.16) precisely determined by measuring equipment with calibration traceable to international or national standards

NOTE Usually expressed in metres.

3.29 reference loss

A_{ref}

loss of a fibre optic component precisely determined by measuring equipment with calibration traceable to international or national standards

3.30 reflectance

R

the ratio, expressed in dB, of the reflected power (in watts), to the incident power (in watts), at a discrete location in a fibre optic component

NOTE 1 R is given by the following formula:

$$R=10\log_{10}\left(\frac{P_{\text{refl}}}{P_{\text{inc}}}\right) \quad (10)$$

where P_{refl} is the reflected power, e.g. in watts;

P_{inc} is the incident power, e.g. in watts.

NOTE 2 In this standard, reflectance is expressed in decibels.

NOTE 3 Reflectance values are negative.

NOTE 4 For the purpose of this standard, the reflectivity, ρ , is defined as the linear value of the reflectance:

$$\rho=\frac{P_{\text{refl}}}{P_{\text{inc}}} \quad (11)$$

3.31

reflectance deviation

ΔR

difference between the reflectance (3.30) of a fibre component R_{otdr} and the reflectance R_{ref} , in dB

$$\Delta R=R_{\text{otdr}}-R_{\text{ref}} \quad \text{dB} \quad (12)$$

3.32

rms dynamic range (one-way)

amount of fibre attenuation (3.1) that causes the backscatter signal to equal the rms noise level (3.33)

NOTE Assuming a Gaussian distribution of noise, the rms dynamic range can be calculated adding 1,56 dB to the one way dynamic range.

3.33

rms noise level

quadratic mean of the noise

NOTE 1 On a general basis, the rms noise level can not be read or extracted from the logarithm data of the OTDR. This is because of the linear to logarithm conversion used to display the power level on a dB scale removes the negative part of the noise.

NOTE 2 Assuming a Gaussian distribution of noise, a relation between the noise level and the rms noise level can be found using the following formula:

$$\text{noise}_{98}-\text{noise}_{\text{rms}}=5\times\log_{10}(2,053\,75)=1,56 \text{ dB} \quad (13)$$

where noise_{98} is the noise level at 98 %, e.g. in dB;

$\text{noise}_{\text{rms}}$ is the rms noise level, e.g. in dB;

2,053 75 is the value of the reverse standard normal distribution for 98 %.

3.34

sample spacing

distance of consecutive data points digitized by the OTDR

NOTE 1 Usually expressed in metres.

NOTE 2 Sample spacing may be obtainable from instrument set-up information. Sample spacing may depend on the measurement span and other OTDR instrument settings.

3.35

spectral width

$\Delta\lambda_{FWHM}$

full-width half-maximum (FWHM) spectral width of the source

[IEC 61280-1-3, modified]

4 Preparation for calibration

4.1 Organization

The calibration laboratory should satisfy requirements of ISO/IEC 17025.

There should be a documented measurement procedure for each type of calibration performed, giving step-by-step operating instructions and equipment to be used.

4.2 Traceability

The requirements of ISO/IEC 17025 should be met.

All standards used in the calibration process shall be calibrated according to a documented program with traceability to national standards laboratories or to accredited calibration laboratories. It is advisable to maintain more than one standard on each hierarchical level, so that the performance of the standard can be verified by comparisons on the same level. Make sure that any other test equipment which has a significant influence on the calibration results is calibrated. Upon request, specify this test equipment and its traceability chain(s). The re-calibration period(s) shall be defined and documented.

4.3 Preparation

Perform all tests at an ambient room temperature of $23\text{ °C} \pm 3\text{ °C}$, unless otherwise specified. Give the test equipment a minimum of 2 h prior to testing to reach equilibrium with its environment. Allow the OTDR a warm-up period according to the manufacturer's instructions.

4.4 Test conditions

The test conditions usually include the following OTDR external conditions: date, temperature, connector-adaptor combination and use of a lead-in fibre.

Perform the calibration in accordance with the manufacturer's specifications and operating procedures. Where practical, select a range of test conditions and parameters so as to emulate the actual field operating conditions of the OTDR under test. Choose these parameters so as to optimize the OTDR's accuracy and resolution capabilities (for example, view windows, zoom features, etc.), as specified by the manufacturer's operating procedures.

The test conditions usually include the following OTDR parameters: averaging time, pulse width, sample spacing, centroidal wavelength. Unless otherwise specified, set the OTDR group index to exactly 1,46.

NOTE 1 The calibration results only apply to the set of test conditions used in the calibration process.

NOTE 2 Because of the potential for hazardous radiation, be sure to establish and maintain conditions of laser safety. Refer to IEC 60825-1 and IEC 60825-2.

4.5 Documentation

Calibration certificates shall include the following data and their uncertainties:

- a) the location offset ΔL_0 and its uncertainty $\pm 2 u_{\Delta L_0}$ as well as the distance scale deviation ΔS_L and its uncertainty $\pm 2 u_{\Delta S_L}$, or the location deviations ΔL_i and their uncertainties $\pm 2 u_{\Delta L_i}$;
- b) the loss deviations ΔA and their uncertainties $\pm 2 u_{\Delta A}$ or the loss scale deviation ΔS_A and their uncertainties $\pm 2 u_{\Delta S_A}$;
- c) the instrument configuration (pulse width, measurement span, wavelength, averaging time,...) used during calibration;
- d) other calibration certificate requirement as per ISO/IEC 17025.

5 Distance calibration – General

5.1 General

The objective of distance calibration is to determine deviations (errors) between the measured and actual distances between points on a fibre, and to characterize the uncertainties of these deviations.

An OTDR measures the location L of a feature from the point where a fibre is connected to the instrument, by measuring the round-trip transit time T for a light pulse to reach the feature and return. L is calculated from T using the speed of light in vacuum c ($2,997\,924\,58 \times 10^8$ m/s) and the group index N of the fibre:

$$L = \frac{cT}{2N} \quad (14)$$

Errors in measuring L will result from scale errors, from offsets in the timebase of the OTDR and from errors in locating a feature relative to the timebase. Placing a marker in order to measure the location may be done manually or automatically by the instrument. The error will, generally, depend on both the marker placement method and the type of feature (for example, a point loss, a large reflection that saturates the receiver or a small reflection that does not).

Even larger errors in measuring L may result from the uncertainty in determining the fibre's group index N . The determination of N is beyond the scope of this standard. Consequently, the calibration procedures below only discuss the OTDR's ability to measure T correctly. For the purpose of this standard, a default value $N = 1,46$ is used and the uncertainty of N is considered to be 0.

5.2 Location deviation model

In order to characterize location deviations, a specific model will be assumed that describes the behaviour of most OTDRs. Let L_{ref} be the reference location of a feature from the front panel connector of the OTDR and let L_{otdr} be the displayed location. It is assumed that the displayed location L_{otdr} , using OTDR averaging to eliminate noise, depends functionally on the reference location L_{ref} in the following way:

$$L_{\text{otdr}} = S_L \times L_{\text{ref}} + \Delta L_0 + f(L_{\text{ref}}) \quad (15)$$

where

S_L is the scale factor, which ideally should be 1;

ΔL_0 is the location offset, which ideally should be 0;

$f(L_{\text{ref}})$ represents the distance sampling error, which is also ideally 0. The distance sampling error is a periodic function with a mean of zero and a period equal to the distance interval between sampled points on the OTDR. As an example, if the location of a large reflection is measured by placing a marker on the first digitized point that shows an increase in signal and the position of the reflection is incremented in fine steps, then $f(L_{\text{ref}})$ may be shaped like a periodic ramp waveform.

Equation (15) is meant to characterize known errors in location measurements, but there may still be an additive uncertainty type A. This will affect both the distance measurements and the accuracy with which parameters describing the errors can be determined by the procedures below.

S_L and ΔL_0 may be determined by measuring L_{otdr} for different values of L_{ref} , then fitting a straight line to the data by the least squares method. S_L and ΔL_0 are the slope and intercept, respectively.

Equivalently, a line may be fitted to the location deviation function, that is the difference between L_{otdr} and L_{ref} :

$$\Delta L = L_{otdr} - L_{ref} = \Delta S_L \times L_{ref} + \Delta L_0 + f(L_{ref}) \quad (16)$$

where

ΔS_L is the slope; and

ΔL_0 is still the intercept, as illustrated in Figure 2.

After finding the linear approximation, the distance sampling error $f(L_{ref})$ respectively its half-amplitude $\Delta L_{readout}$ may be determined by measuring departures from the line for different values of L_{ref} . The distance sampling error amplitude ΔL_{sample} is taken as half the amplitude of $f(L_{ref})$.

In this standard, the distance sampling error amplitude ΔL_{sample} is treated as part of the location readout uncertainty type A. The stated uncertainty result thus ignores the repetitive nature of the sampling error, that is it does not distinguish between the relative contributions of the sampling error and the uncertainty type A.

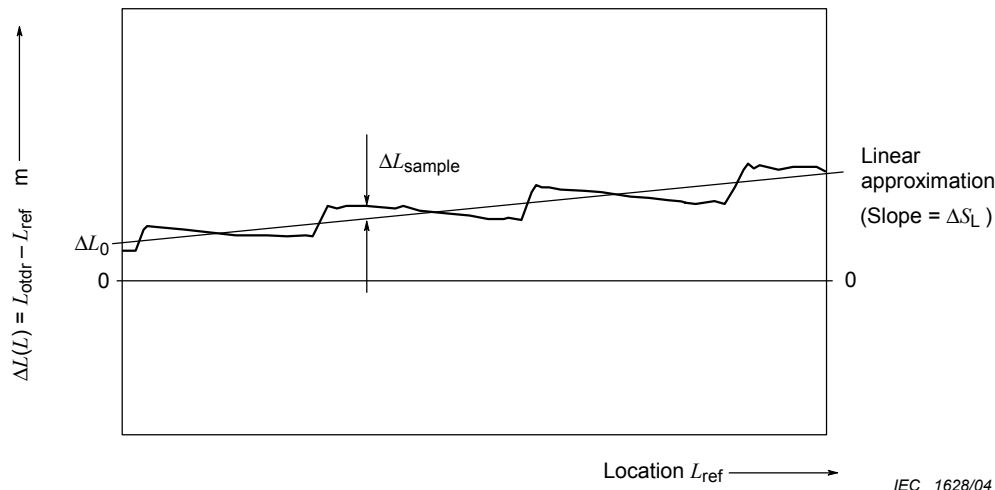


Figure 2 – Representation of the location deviation $\Delta L(L)$

Therefore, the result of the distance calibration shall be stated by the following parameters:

- $\Delta S_L, u_{\Delta S_L}$ the distance scale deviation and its uncertainty;
- $\Delta L_0, u_{\Delta L_0}$ the location offset and its uncertainty;
- $u_{L_{\text{readout}}}$ the location readout uncertainty, that is the combined uncertainty due to the distance sampling error and the uncertainty type A of the measurement samples, in the form of a standard deviation.

In compliance with the "mathematical basis", divide the largest excursions from the least-squares approximation by the square root of 3 for stating $u_{L_{\text{readout}}}$. Note that the uncertainty will depend on the distance, the displayed power level and the instrument settings.

NOTE ΔL_{sample} represents the physical sampling error of the instrument. This error is accessible for the user as $u_{L_{\text{readout}}}$ that includes distance calculation and displaying errors.

5.3 Using the calibration results

The error in the location of a feature $\Delta L = L_{\text{otdr}} - L_{\text{ref}}$ can be calculated from the calibration results:

$$\Delta L = \Delta L_0 + L_{\text{ref}} \Delta S_L \quad (17)$$

with the uncertainty in ΔL given by the following formula, in which the recommended confidence level of 95 % is used:

$$\pm 2u_{\Delta L} = \pm 2 \left(u_{\Delta L_0}^2 + L_{\text{ref}}^2 u_{\Delta S_L}^2 + u_{L_{\text{readout}}}^2 \right)^{1/2} \quad (17a)$$

where the displayed location L_{otdr} can be used instead of the reference location L_{ref} without serious consequences.

Similarly, the error in the distance between two features ΔD and its uncertainty can be calculated from the following formula:

$$\Delta D = D_{\text{ref}} \Delta S_L \quad (18)$$

with uncertainty in ΔD given by the following formula:

$$\pm 2u_{\Delta D} = \pm 2 \left(D_{\text{ref}}^2 u_{\Delta S_L}^2 + u_{L_{\text{readout}}}^2 \right)^{1/2} \quad (18a)$$

where the displayed distance D_{otdr} can be used instead of the reference distance D_{ref} .

NOTE The 2 in front of $u_{L_{\text{readout}}}^2$ is due to combining two uncorrelated uncertainties.

Additional uncertainties may have to be taken into account if the type of feature is different from the feature used in the calibration. Specify the type of feature as part of the calibration result.

5.4 Measuring fibre length

As indicated above, one of the methods of OTDR distance calibration is to measure fibres of known length with the OTDR. In several instances in this standard, it is required that fibre length be determined using the fibre's transit time, in contrast to a mechanical length measurement. This method is directly compatible with the measurement principle of the OTDR itself. In addition, the transit time can usually be measured with better accuracy than its mechanical length, particularly when the fibre is long. Therefore, in this standard, it is suggested that fibre transit time instead of fibre length be used whenever accuracy is important.

Measure the transit time of the fibre T_{transit} with the help, for example, of a pulse generator, a triggerable laser source, an optical-to-electrical converter (O/E converter) and a time interval counter. It is important that the laser source has approximately the same centroidal wavelength λ_{avg} as the test OTDR, because a difference in wavelength may result in a difference of transit time due to the chromatic dispersion of the fibre. An alternative to the laser source is using the OTDR itself to produce optical pulses; in this case, the centroidal wavelengths automatically coincide. Record the transit time as the difference between the arrival times with and without the fibre inserted between the laser source and the O/E converter.

When this fibre is used for OTDR distance calibrations, then the reference distance D_{ref} can be calculated by

$$D_{\text{ref}} = \frac{cT_{\text{transit}}}{N} \text{ m} \quad (19)$$

In this equation, use a group index N which is identical with the OTDR's group index setting. The time measurement principle makes it possible to use D_{ref} as the reference distance.

6 Distance calibration methods

6.1 General

Each of the three alternative calibration methods described below is capable of determining all of the necessary calibration results: location offset, distance scale deviation, and their uncertainties.

6.2 External source method

6.2.1 Short description and advantage

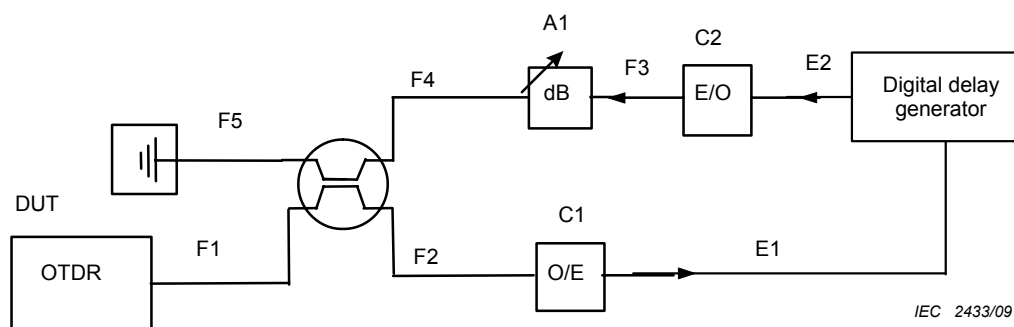
The external source method uses a calibrated time-delay generator to simulate the time delay in a fibre and an optical source to simulate the reflected or scattered signal from a fibre.

The method is well suited to automated laboratory testing under computer control. For simplicity, only reflective features are discussed in this standard. To calibrate the OTDR for features other than reflection, the pulsed E/O converter described below should be replaced by an optical source that simulates the appropriate feature.

6.2.2 Equipment

In addition to the OTDR, the measurement equipment includes, as shown in Figure 3:

- a) an optical coupler;
- b) an optical-to-electrical converter;
- c) a digital delay generator with pulse capability;
- d) an electrical to optical converter;
- e) a variable optical attenuator, for reduction of the pulse amplitude to just below the clipping level.

**Key**

F1, F2, F3, F4 and F5	fibres
E1 and E2	electric cables
E/O	electrical-to-optical converter
O/E	optical-to-electrical converter
A1	variable attenuator
C1	optical-to-electrical converter
C2	electrical-to-optical converter

Figure 3 – Equipment for calibration of the distance scale – External source method

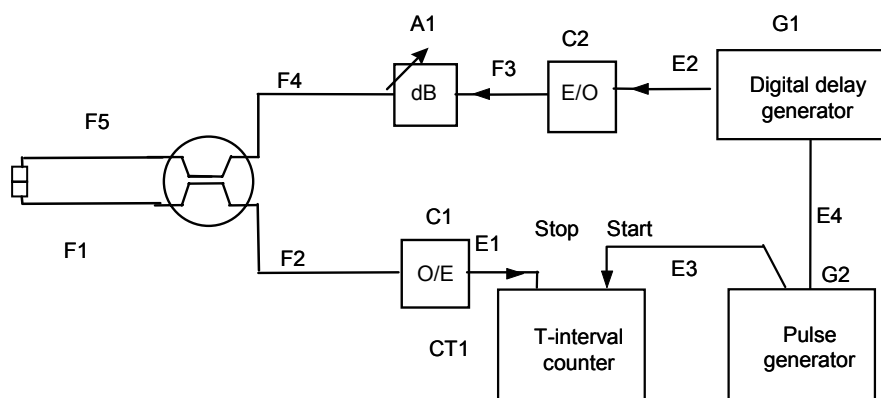
The coupler routes the OTDR signal to the O/E converter (detector). The detector triggers the delay generator, which, after a known time delay, causes an optical pulse to be generated. This pulse is then coupled back to the OTDR.

The E/O converter can be a simple pulsed laser that simulates a reflection. Constant pulse amplitude and pulse width are considered adequate to calibrate the distance scale for reflective features. However, the attenuator makes it possible to adjust the pulse amplitude based on the distance of the reflection from the front panel of the OTDR, in order to simulate the change of reflection amplitude caused by the attenuation of the fibre.

To allow accurate calibration of the set-up, fibres F1 and F5 should have the same length (see below). F5 is terminated to absorb reflections.

6.2.3 Calibration of the equipment

Before using the "external source" equipment, it shall be properly calibrated. It is assumed that the digital delay generator is regularly calibrated. For computing the location offset ΔL_0 from the measured data, it is also necessary to determine the insertion delay T_{delay} of the apparatus. This can be accomplished by adding a pulse generator and a calibrated time interval counter to the equipment, as shown in Figure 4.



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Key

F1, F2, F3, F4 and F5	fibres
E1, E2, E3 and E4	electric cables
C2	electrical-to-optical converter
C1	optical-to-electrical converter
A1	variable attenuator

Figure 4 – Set-up for calibrating the system insertion delay

To calibrate the insertion delay T_{delay} , proceed as follows.

Set the pulse generator to square wave, with a repetition period more than twice as long as the delay time to be measured. Use the output pulse of the pulse generator as the start pulse on the time interval counter, and to externally trigger the delay generator. Set the digital delay generator for external triggering and zero delay for the leading edge of the pulse generator signal. Set the trigger levels of the delay generator and the counter.

The external source will then generate an optical square wave which, after re-conversion to an electrical pulse, will stop the time interval counter. To ensure lowest uncertainty, the electrical cables E3 and E4 should have equal length. Also, fibres F1 and F5 should have equal lengths. Note that identical cable numbers in Figures 3 and 4 mean the same physical cables. Adjust the optical attenuator for best triggering of the time interval counter. Record the displayed time interval (between start and stop) as the insertion delay T_{delay} .

6.2.4 Measurement procedure

6.2.4.1 Preparation

Select the technique (automatic or manual) for locating the feature on the OTDR. Program the attenuator to generate the desired pulse amplitude(s). Select the pulse width on the digital delay generator, for example 1 μs .

Choose the time settings of the delay generator T_i so that the samples are distributed over a wide distance range with some randomness, to accomplish averaging over the OTDR's distance sampling interval. The first time setting should be chosen so that the pulse appears close to the front panel of the OTDR, but sufficiently out of the initial dead zone for good measurements. If the testing laboratory does not determine and analytically justify a different distance sampling scheme, one of the two schemes below shall be chosen.

- a) In the first scheme, evaluate the sample spacing D_{sample} (for the appropriate OTDR instrument setting), for example by zooming into the OTDR trace. Then calculate the corresponding delay difference of the delay generator T_{sample} using:

$$T_{\text{sample}} = \frac{2ND_{\text{sample}}}{c} \quad (20)$$

where N is the OTDR's group index setting and c is the speed of light in vacuum.

Then calculate a total number of i delay generator settings, grouped in k clusters of n settings each ($i = kn$), where each cluster uniformly covers one sample spacing. Each cluster shall have the form:

$$T_K, T_K + \frac{T_{\text{sample}}}{n}, T_K + 2\frac{T_{\text{sample}}}{n}, \dots, T_K + (n-1)\frac{T_{\text{sample}}}{n} \quad (21)$$

where the number of settings in each cluster n is at least four and is the same for every cluster. The centres of the clusters are uniformly spaced, from just beyond the initial dead zone to a large distance over which the instrument is to be calibrated. The number of clusters k may be as small as two.

- b) In the second scheme, there are no clusters, and the sample spacing D_{sample} does not need to be known except very approximately. Calculate T_{sample} from Equation (20). Choose the time settings so that they are uniformly spaced between the initial dead zone and a large distance and each has a random time interval added. The random intervals should have a uniform probability density in the interval $-T_1$ to T_1 , where T_1 is at least $20 T_{\text{sample}}$ but less than 10 % of the longest time delay for the tests. The number of measurements i (that is different settings) should be at least 20.

Alternatively, prior knowledge of the magnitude of the uncertainty type A and the tolerable uncertainty in the measurements may lead the testing laboratory to select a different systematic or random distance sampling scheme.

6.2.4.2 Taking the measurement results

Select the first time setting of the time T_i of the series T_1 as defined above. Record the time T_1 of the delay generator and the measured location $L_{\text{otdr},1}$ of the event on the OTDR. Proceed with the time settings as selected in 6.2.4.1 Always record the time T_i and the measured location $L_{\text{otdr},i}$. Continue until all time settings are completed.

6.2.5 Calculations and results

Following the concept of Clause 5, use the time settings to calculate i reference locations $L_{\text{ref},i}$:

$$L_{\text{ref},i} = \frac{c(T_i + T_{\text{delay}})}{2N} \quad (22)$$

where

N is the group index setting of the OTDR;

T_i are the time settings defined in 6.2.3;

T_{delay} is the calibrated insertion delay of the test equipment (see 6.2.2).

Then, use the reference locations and the displayed locations $L_{\text{otdr},i}$ to calculate the set of i location deviations ΔL_i :

$$\Delta L_i = L_{\text{otdr},i} - L_{\text{ref},i} \quad (23)$$

To determine the location offset ΔL_0 and the distance scale deviation ΔS_L , fit the location deviation data to the simplified location deviation model (in which the distance sampling error is momentarily neglected):

$$\Delta L_{i, \text{model}} = \Delta S_L L_{\text{ref},i} + \Delta L_0 \quad (24)$$

Specifically, minimize the difference between the model and the data using the least-squares criterion that is, choose ΔS_L and ΔL_0 so that the summation

$$\sum_i (\Delta L_i - \Delta S_L L_{\text{ref}, i} - \Delta L_0)^2 \quad (25)$$

is minimized. Record ΔL_0 and ΔS_L obtained from the approximation.

As in Figure 2, the slope of the linear approximation represents the distance scale deviation ΔS_L . The intercept with the vertical axis represents the location offset ΔL_0 . Record ΔS_L and ΔL_0 obtained from the calculation.

6.2.6 Uncertainties

6.2.6.1 General

A general discussion of the distance uncertainties can be found in Clause 5.

Note that the following list of uncertainties may not be complete. Additional contributions may have to be taken into account, depending on the measurement set-up and procedure. The mathematical basis given in Annex D should be used to calculate and state the uncertainties.

6.2.6.2 Distance scale uncertainty

The least-squares approximation outlined in 6.2.5 effectively uses the displayed distances between the measurement samples to calculate the distance scale deviation. It is assumed that the measurement samples near $L = 0$ and near the farthest location $L = L_{\text{max}}$ have the strongest influence on the distance scale deviation because the samples in the middle of the range have no influence on the slope of the distance error model.

Applying the standard formula for the propagation of errors to Equation (7) yields the distance scale uncertainty $u_{\Delta S_L}$ in which $\langle D_{\text{otdr}} \rangle \cong D_{\text{ref}}$ was used.

$$u_{\Delta S_L} \cong \left[\left(\frac{u_{\langle D_{\text{otdr}} \rangle}}{\langle D_{\text{otdr}} \rangle} \right)^2 + \left(\frac{u_{D_{\text{ref}}}}{D_{\text{ref}}} \right)^2 \right]^{1/2} \quad \text{m/km} \quad (26)$$

where

D_{otdr}	is $D_{\text{ref}} \approx L_{\text{ref}}$ (for the long distances discussed here);
$u_{\langle D_{\text{otdr}} \rangle}$	is the standard deviation expressing the uncertainty of the distance samples (on the basis of the location samples);
$u_{\langle D_{\text{otdr}} \rangle} / \langle D_{\text{otdr}} \rangle$	represents the slope uncertainty due to inaccurate distance readout; it is equivalent to the standard deviation of the slope, ΔS_L in the location model of Equation (16) which includes the marker placement uncertainty and the distance sampling error; the least-squares algorithm used for the determination of ΔS_L can be used to determine $u_{\langle D_{\text{otdr}} \rangle}$; if applicable, ΔL_i may be averaged over the corresponding sampling interval;
$u_{D_{\text{ref}}}$	is the uncertainty of the reference distances;
$u_{D_{\text{ref}}} / D_{\text{ref}}$	represents the slope uncertainty caused by the digital delay generator and is equal to the relative timing uncertainty of the delay generator.

6.2.6.3 Location offset uncertainty

The location offset ΔL_0 is equal to the intercept of the least-squares approximation with the vertical axis. This intercept mostly depends on the first few samples, that is those samples which are closest to the location $L = 0$, and on the accuracy of the insertion delay T_{delay} .

The location offset uncertainty $u_{\Delta L_0}$ can be calculated by using the standard formula for the propagation of errors:

$$u_{\Delta L_0} = \left[u\sigma_{\Delta L}^2 + \left(\frac{c}{2N} \right)^2 u_{T_{\text{delay}}}^2 \right]^{1/2} \quad (27)$$

where

$u_{\Delta L}$ is the uncertainty of the differences between ΔL_i and the least-squares approximation near $L = 0$, which includes the marker placement uncertainty and the distance sampling error; it is equivalent to the standard deviation of $(\Delta L_i - \Delta L_{i, \text{model}})$ near $L = 0$; if applicable, ΔL_i may be averaged over the correspondent sampling interval; the least-squares algorithm used for the determination of ΔL_0 can be used to determine $\sigma_{\Delta L}$.

$u_{T_{\text{delay}}}$ is the uncertainty of the system insertion delay; the assumption is that the first-time setting will be very short or even zero, reducing the delay generator uncertainty to one of the insertion delays only.

6.2.6.4 Location readout uncertainty

As outlined in Clause 5, determine the largest difference between the location deviation samples ΔL_i and the least-squares approximation near $L = 0$. Then calculate the location readout uncertainty $u_{L_{\text{readout}}}$ (which includes the distance sampling error) by dividing the largest difference by the square root of 3. Alternatively, $u_{L_{\text{readout}}}$ can be determined either with the least-squares algorithm used for the determination of ΔS_L and ΔL_0 or with the following formula:

$$u_{L_{\text{readout}}} = \left[\frac{1}{n-1} \sum_{i=1}^n (\Delta L_i - \Delta L_{i, \text{model}})^2 \right]^{1/2} \quad (28)$$

6.3 Concatenated fibre method

6.3.1 Short description and advantages

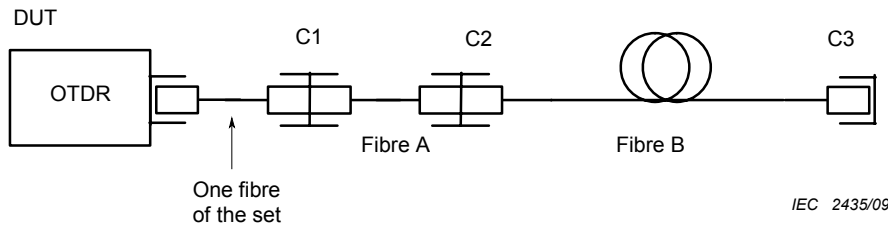
This method uses calibrated fibres with transit times precisely measured at the wavelength of the OTDR under test to calibrate the distance scale.

The method requires only connectorized lengths of fibre, and is thus both inexpensive and well suited to testing in locations where equipment such as that used in 6.2 cannot be carried. It may be viewed as a manual test method because it requires connecting and disconnecting short lengths of fibre a number of times to vary the locations of features. However, this process can be automated with optical switches, if desired.

6.3.2 Equipment

In addition to the test OTDR, the equipment includes, as shown in Figure 5:

- fibre A, to determine the location offset;
- fibre B, to determine the distance scale deviation;
- a set of incremental fibres, to determine the distance sampling error.



Key

C1, C2 and C3 optical connectors

Figure 5 – Concatenated fibres used for calibration of the distance scale

Normally these fibres will be cabled or packaged in some way for protection and connectorized for easy connection and disconnection. Alternatively, they may be used with a fibre optic switching mechanism that does not require connectors.

The requirements on these fibres are indicated below.

- a) Fibre A may be a simple fibre with an end reflection, or a fibre with an internal reflection or splice. Its length is not very important, as long as it puts the feature to be measured on a backscatter trace which is essentially undisturbed from the initial reflection(s) near the OTDR port.

Fibre A may be split into a short jumper fibre and a length as outlined before, with connectorized ends. In this way, the incremental fibres (see below) can be inserted after the first section of fibre A without having access to the front panel of the OTDR. The reflection from the near-end connectors shall be small enough to avoid placing the feature at the end of fibre A into the dead zone of the OTDR.

This alternative may be important in automated systems.

Fibre A can also be used as a lead-in fibre for measuring the distance scale deviation with the help of fibre B.

- b) Fibre B shall have reflective ends, for example by using the reflections from its connectors. Because the uncertainty is reduced by making this fibre longer, it is recommended that fibre B be at least 2 km long.

To calibrate the fibre, measure its optical transit time T_b as described in Clause 5.

Caution: for a correct distance calibration, the reflections from the two ends of fibre B (connectors C2 and C3) should be approximately equal. For example, if one end produces a reflection that saturates the OTDR and the other end does not, then the difference in waveforms can lead to inaccurate distance measurements. The effect of this difference on the measured distance scale deviation will be small if the fibre is long, however.

- c) When the sampling interval is not precisely known from the manufacturer and expected to be large, the set of incremental fibres will be used to vary the locations of the two reflections of the long fibre B by amounts less than the distance sampling interval of the OTDR. Their lengths should be selected in order to generate at least four distance increments which are evenly spaced over the distance sampling interval. As an example, if the distance sampling interval is 10 m, this can be accomplished with two fibres of lengths 2,5 m and 5 m. Separately and in combination, these can produce increments of 0 m (no fibre), 2,5 m, 5 m, and 7,5 m (both fibres). More generally, the fibres should generate length increments of

$$0, D_x, 2D_x, \dots, (n-1)D_x \quad (29)$$

where

$n \geq 4$, and

$n D_x$ equals the distance sampling interval of the OTDR under the conditions to be tested.

Except in unusual cases, calibrating the transit time of these fibres according to Clause 5 is unnecessary. Instead, their physical lengths should be measured. The difference between the true group index and the OTDR group index setting will be negligible on such short fibres.

6.3.3 Measurement procedures

6.3.3.1 General

The contribution of random noise to location deviation is usually small except when the displayed power level comes close to the noise limit of the instrument. Longer OTDR averaging is recommended in this case.

6.3.3.2 Preparation

Select the type of feature (in or at the end of fibre A) for which the location offset is to be determined and select the fibres accordingly. Also, establish the technique (automatic or manual) for placing the markers on the feature in fibre A and the reflective ends of fibre B.

Connect fibre A to the front of the OTDR, so that the feature at or near the far end of the fibre can be seen on the OTDR. Connect fibre B to the far end of fibre A, so that the reflections from both ends of fibre B may be seen on the OTDR.

6.3.3.3 Taking the measurements

Measure the location of the feature in fibre A with the OTDR. Record this first measured location as $L_{\text{otdr},1}$. Measure the length of fibre B with the OTDR, using the two reflections generated by this fibre. Record this first measured distance as $D_{\text{otdr},1}$.

Insert the shortest of the incremental fibres between the OTDR and the beginning of fibre A. Measure the location $L_{\text{otdr},2}$ and the distance $D_{\text{otdr},2}$. If fibre A is a split fibre, as described in 6.3.2, the incremental fibres may be inserted into the split instead of between the OTDR and the beginning.

Continue inserting successively increasing length combinations of the incremental fibres. Measure the location $L_{\text{otdr},i}$ and the distance $D_{\text{otdr},i}$ until $i = n$ and the total length of the incremental fibres is $(n - 1) D_X$.

6.3.4 Calculations and results

6.3.4.1 Distance scale deviation

Compute the distance $\langle D_{\text{otdr}} \rangle$ (the length of fibre B) as the average of the n values of $D_{\text{otdr},i}$. Then compute the distance scale deviation as

$$\Delta S_L = \frac{\langle D_{\text{otdr}} \rangle}{D_{\text{ref}}} - 1 = \frac{N \langle D_{\text{otdr}} \rangle}{c T_b} - 1 \quad (30)$$

where

D_{ref} is the reference distance;

N is the group index setting of the OTDR;

T_b is the one-way transit time for fibre B, as measured according to Clause 5.

6.3.4.2 Location offset

Let $\langle L_{\text{otdr}} \rangle$ be the average of all n values of $L_{\text{otdr},i}$. Compute the location offset on the basis of Equation (16):

$$\Delta L_0 = \langle L_{\text{otdr}} \rangle - (1 + \Delta S_L) \langle L_{\text{ref}} \rangle \quad (31)$$

$$\Delta L_0 = \langle L_{\text{otdr}} \rangle - (1 + \Delta S_L) \left(\frac{cT_a}{N} + \frac{(n-1)D_x}{2} \right)$$

where

$\langle L_{\text{ref}} \rangle$ is the average reference location corresponding to the first reflection, to be calculated with the help of the average length of the incremental fibres;

N is the group index setting of the OTDR;

T_a is the one-way transit time for fibre A, as measured according to Clause 5;

ΔS_L is the distance scale deviation as determined by the Equation (30); if fibre A is sufficiently short, then the ΔS_L term can be neglected.

6.3.5 Uncertainties

6.3.5.1 General

A general discussion of the distance uncertainties is given in Clause 5. Note that the following list of uncertainties may not be complete. Additional contributions may have to be taken into account, depending on the measurement set-up and procedure.

The mathematical basis given in Annex D should be used to calculate and state the uncertainties.

6.3.5.2 Distance scale uncertainty

The distance scale uncertainty $u_{\Delta S_L}$ should be calculated with the following formula which is derived from Equation (30):

$$u_{\Delta S_L} = \left[\left(\frac{u_{\text{Dotdr}}}{D_{\text{otdr}}} \right)^2 + \left(\frac{u_{T_b}}{T_b} \right)^2 \right]^{1/2} \text{ m/km} \quad (32)$$

where u_{Dotdr} is the uncertainty of the displayed length of fibre B, for example, as caused by the marker placement uncertainty and the distance sampling error.

The transit time uncertainty of fibre B σ_{T_b} should itself be calculated by root-sum-squaring.

$u_{T_b, \text{counter}}$ is the transit time uncertainty of fibre B, due to the time interval counter;

$u_{T_b, \lambda}$ is the transit time uncertainty of fibre B, due to a difference between the wavelength used in determining the transit time and the OTDR wavelength;

$u_{T_b, \Theta}$ is the transit time uncertainty of fibre B, due to its temperature coefficient; typical value: 1 cm/(km °C).

6.3.5.3 Location offset uncertainty

The location offset uncertainty $u_{\Delta L_0}$ should be calculated from the following formula which is derived from Equation (31), by neglecting the ΔS_L and the $(n-1) D_x/2$ terms:

$$u_{\Delta L_0} = \left[u_{\text{Lotdr}}^2 + \left(\frac{c}{N} \right)^2 u_{T_a}^2 \right]^{1/2} \quad (33)$$

where u_{Lotdr} is the uncertainty of measuring the location of the feature at the end (or inside) of fibre A, that is mainly the uncertainty of the marker placement; it is assumed that the distance sampling error is effectively removed by averaging over one sampling interval.

The transit time uncertainty of fibre A u_{Ta} should itself be calculated by root-sum-squaring:

- $u_{Ta,counter}$ is the transit time uncertainty of fibre A, due to the time interval counter;
 $u_{Ta,\lambda}$ is the transit time uncertainty of fibre A, due to a difference between the wavelength used in determining the transit time and the OTDR wavelength;
 $u_{Ta,\Theta}$ is the transit time uncertainty of fibre A, due to its temperature; typical value: 1 cm/(km °C).

6.3.5.4 Location readout uncertainty

Calculate the following two sets of data, one for the location deviation and one for the distance error, using the measurement samples given in 6.3.3.

$$L_{otdr,i} - L_{ref,i} = L_{otdr,i} - \left(\frac{cT_a}{N} + iD_x \right) \quad i = 0 \text{ to } n-1 \quad (34)$$

and

$$D_{otdr,i} - D_{ref,i} = D_{otdr,i} - \frac{cT_b}{N} \quad i = 0 \text{ to } n-1 \quad (35)$$

It is recommended that half the difference between the largest and the smallest value of the L set or the D set, whichever is larger, be divided by the square root of 3 to calculate the location readout uncertainty $u_{Lreadout}$.

6.4 Recirculating delay line method

6.4.1 Short description and advantage

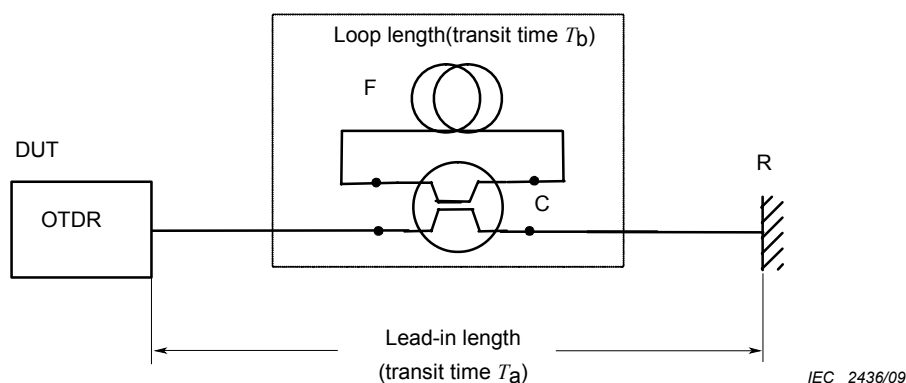
The recirculating delay line method uses a calibrated loop of fibre, made with a coupler and a reflector, to generate periodic reflections.

The method is similar to the concatenated fibre method: a fibre artifact is used, and there is no need for electronic equipment. The artifact generates many calibrated distance samples; this has the potential of reducing type A uncertainties affecting the distance scale deviation.

The measurements of location offset are limited to the reflective features generated by the recirculating delay line.

6.4.2 Equipment

In addition to the test OTDR, the measurement equipment only includes a recirculating delay line manufactured and calibrated according to Annex A, as shown in Figure 6.



Key

F	fibre in the loop
C	four-port coupler
R	reflection

Figure 6 – Distance calibration with a recirculating delay line

The recirculating delay line places a number of reflective features on the OTDR display, as shown in Figure 7. The first feature is the one obtained from the optical pulse travelling direct to the mirror, then back direct to the OTDR. The second feature is generated by the optical pulse travelling once through the loop, then to the mirror, then back direct to the OTDR (this pulse coincides with the pulse travelling direct to the mirror, then back through the loop, then back to the OTDR). The third pulse travels through the loop twice, etc.

Accordingly, the ideal displayed locations would be

$$\begin{aligned}
 L_{\text{otdr},0} &= L_a \\
 L_{\text{otdr},1} &= L_a + L_b/2 \\
 L_{\text{otdr},2} &= L_a + L_b \rightarrow \text{etc}
 \end{aligned}
 \tag{36}$$

where L_a is the length of the lead-in fibre and L_b is the length of the fibre loop.

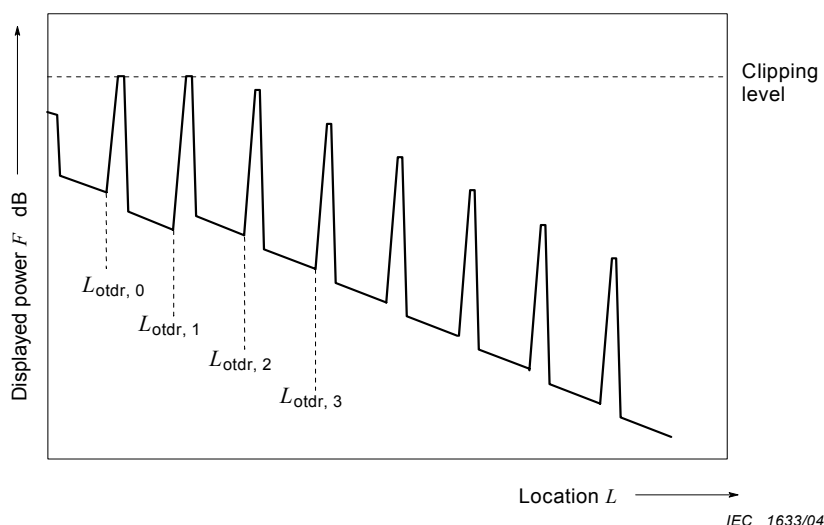


Figure 7 – OTDR trace produced by recirculating delay line

As an optional addition to the measurement set-up, it may be advantageous to use one or more incremental fibres as in 6.3.3. The need for this is reduced because the multiple reflections from the delay line are likely to generate averaging over the distance sampling interval. However, this averaging effect is not controlled and systematic control may be preferred. Using the notation of 6.3.3, it may be sufficient to let $n = 2$, so there is just one incremental fibre with length equal to half of the distance sampling interval.

6.4.3 Measurement procedure

6.4.3.1 General

The procedure assumes that no incremental fibres are used. If they are used, this simply increases the number of distance samples that are recorded, and it is straightforward to modify the notation and computations. The method then becomes very similar to that of 6.3, with the lead-in fibre being equivalent to fibre A and the loop length equivalent to fibre B.

6.4.3.2 Preparation

Establish the technique (automatic or manual) of placing the markers at the leading edges of the reflections from the recirculating delay line, following the manufacturer's recommendations.

Connect the recirculating delay line assembly directly to the OTDR so that the reflective features can be seen on the OTDR.

6.4.3.3 Taking the measurement results

Measure the locations of successive reflections from the recirculating delay line with the OTDR. Record these as $L_{\text{otdr},i}$, where the index i goes from 0 to k and represents the number of passes through the loop. A large number k will presumably increase the accuracy of the result, but it will be limited by loss and the noise floor of the OTDR.

6.4.4 Calculations and results

Using the calibration data of the recirculating delay line T_a and T_b the series of reference locations is:

$$\begin{aligned} i = 1: L_{\text{ref},1} &= \frac{c(T_a + T_b/2)}{N} \\ i = 2: L_{\text{ref},2} &= \frac{c(T_a + T_b)}{N} \\ i = i: L_{\text{ref},i} &= \frac{c(T_a + \frac{iT_b}{2})}{N} \end{aligned}$$

where N is the group index setting of the OTDR.

Then use the displayed locations $L_{\text{otdr},i}$ and the reference locations to calculate the series of location deviations ΔL_i :

$$\Delta L_i = L_{\text{otdr},i} - L_{\text{ref},i} = \Delta S_L L_{\text{ref},i} + \Delta L_0 + f(L_{\text{ref},i}) \quad (37)$$

To determine the location offset ΔL_0 and the distance scale deviation ΔS_L , fit the location deviation data to the simplified location deviation model (in which the location readout uncertainty is momentarily neglected):

$$\Delta L_{i,\text{model}} = \Delta S_L L_{\text{ref},0} + \Delta L_0 \quad (38)$$

Specifically, minimize the difference between the model and the data using the least-squares criterion, that is choose ΔS_L and ΔL_0 so that the summation:

$$\sum_i (\Delta L_i - \Delta S_L L_{\text{ref},i} - \Delta L_0)^2 \quad (39)$$

is minimized. Record ΔL_0 and ΔS_L obtained from the approximation.

6.4.5 Uncertainties

6.4.5.1 General

A general discussion of the distance uncertainties is given in Clause 5.

Note that the following list of uncertainties may not be complete. Additional contributions may have to be taken into account, depending on the measurement set-up and procedure. The mathematical basis given in Annex D should be used to calculate and state the uncertainties.

6.4.5.2 Distance scale uncertainty

The least-squares approximation outlined in 6.4.4 effectively uses the displayed distances between the measurement samples to calculate the distance scale deviation. It is assumed that the group of measurement samples near $L = 0$ and the one near the farthest location $L = L_{\text{max}}$ have the strongest influence on the distance scale deviation because the samples in the middle of the range have no influence on the slope of the distance error model.

Applying the standard formula for the propagation of errors to Equation (7) yields the distance scale uncertainty $u_{\Delta S_L}$ in which $\langle D_{\text{otdr}} \rangle \cong D_{\text{ref}}$ was used.

$$u_{\Delta S_L} \cong \left[\left(\frac{u_{\langle D_{\text{otdr}} \rangle}}{\langle D_{\text{otdr}} \rangle} \right)^2 + \left(\frac{u_{D_{\text{ref}}}}{D_{\text{ref}}} \right)^2 \right]^{1/2} \text{ m/km} \quad (40)$$

where

- D_{otdr} is $D_{\text{ref}} \approx L_{\text{ref}}$ (for the long distances discussed here);
- $u_{\langle D_{\text{otdr}} \rangle}$ is the standard deviation expressing the uncertainty of the distance samples (on the basis of the location samples); this is equivalent to the standard deviation of $(\Delta L_i - \Delta L_{i,\text{model}})$ which includes the marker placement uncertainty and the distance sampling error; the least squares algorithm used for the determination of ΔS_L can be used to determine $u_{\langle D_{\text{otdr}} \rangle}$; if incremental fibres are used, then ΔL_i may be averaged over the correspondent sampling interval;
- $u_{D_{\text{ref}}}$ is the uncertainty of the reference distances; it can be calculated from the formula $u_{D_{\text{ref}}}/D_{\text{ref}} = u_{T_b}/T_b$, where u_{T_b} is the uncertainty of the loop transit time, as documented in the calibration certificate of the recirculating delay line (see Annex A).

6.4.5.3 Location offset uncertainty

The location offset ΔL_0 is equal to the intercept of the least-squares approximation with the vertical axis. This intercept mostly depends on the first few samples, that is those which are closest to the location $L = 0$, and on the accuracy of the transit time T_a .

The location offset uncertainty $u_{\Delta L_0}$ can be calculated by applying the standard formula for the propagation of errors to Equation (38):

$$u_{\Delta L_0} = \left[u_{\Delta L}^2 + \left(\frac{c}{N} \right)^2 u_{T_a}^2 \right]^{1/2} \quad (41)$$

where

- $u_{\Delta L}$ is the uncertainty of the differences between ΔL_i and the least squares approximation near $L = 0$; this includes the marker placement uncertainty and the distance sampling error; this is equivalent to the standard deviation of $(\Delta L_i - \Delta L_{i, \text{model}})$ near $L = 0$; the least squares algorithm used for the determination of ΔL_0 can be used to determine $\sigma_{\Delta L}$; if incremental fibres are used, then ΔL_i may be averaged over the correspondent sampling interval;
- u_{Ta} is the documented uncertainty of the delay time of the lead-in fibre of the recirculating delay line (see Annex A);
- $u_{Ta, \Theta}$ is the uncertainty of the delay time of the lead-in fibre due to the temperature coefficient of the fibre; typical value: 1 cm/(km °C).

6.4.5.4 Location readout uncertainty

The principle of determining the location readout uncertainty is shown in Figure 2. With the recirculating delay line, it is expected that not enough data are produced to show the repetitive nature of the measurement samples. Nevertheless, it is recommended that the largest differences between the location deviations ΔL_i (L_{ref}) and the least-squares approximation be determined. Then divide this difference by the square root of 3 to obtain the location readout uncertainty $u_{L\text{readout}}$ (which includes the distance sampling error).

7 Loss calibration – General

7.1 General

The objective of the loss calibration is to determine the loss deviation ΔA for power levels F within the OTDR backscatter regime and to evaluate the measurement uncertainties. ΔA is a function of the displayed power level F ; it includes both the inaccuracy of the displayed loss and the non-linearity of the OTDR power scale.

The loss scale deviation ΔS_A can be used to present the calibration results as an alternative. This is especially useful, on previous generation of OTDR, when the error can be a function of the attenuation. The user should take care of the use of this parameter for attenuation lower than 1 dB: the uncertainty can be higher than the one provided by the formula.

The calibration can be carried out with any of the following four methods: fibre standard, external source, splice simulator or power reduction. The following subclauses outline the principles of the loss calibration.

7.2 Determination of the displayed power level F

For each measured loss, determine the displayed power level or an equivalent parameter that can be used to reproduce the vertical position of a measurement sample. This level is termed F .

Unless otherwise specified, use the OTDR's clipping level, determined at the front panel connector location, as the (default) reference point for determining F : $F_{\text{ref}} = 0$ dB. State all values of F in relation to this reference point (for example, if the displayed power level is x dB below the clipping level, then $F = -x$ dB). The clipping level can be found by introducing a sufficiently large reflection into a length of fibre, as indicated in Figure 8. Alternatively, a sufficiently strong optical pulse from an external source can be used (see 8.3). The clipping level was chosen because it represents the most reproducible level on most OTDR power scales.

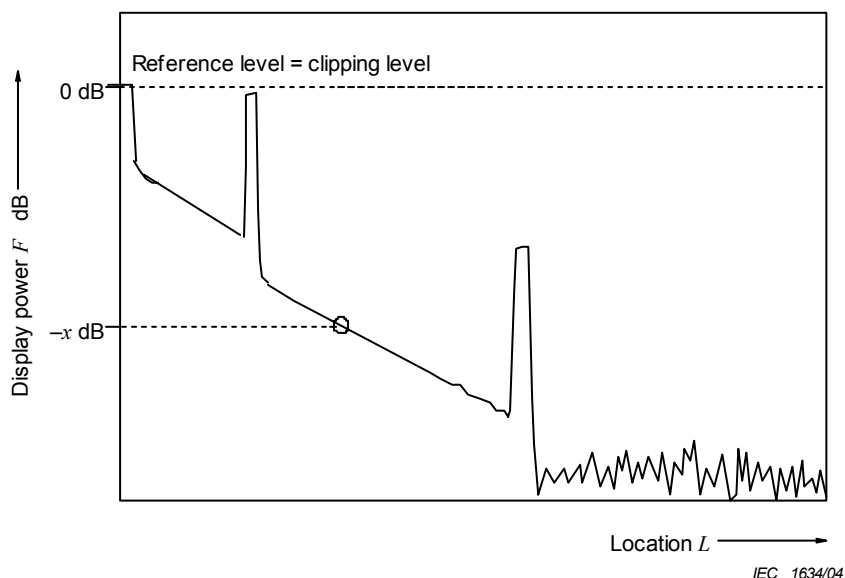


Figure 8 – Determining the reference level and the displayed power level

Alternative solutions may be to state the value of F in dB, relatively to a fixed level, if the OTDR provides the capability of reading out the displayed power in dB, or to use the start level of the backscatter trace from a specified type of fibre, at a specified pulse width, as the reference level. Note that, in the latter case, the reproducibility of the reference level is usually affected by the non-reproducibility of the connection to the OTDR port.

7.3 Selection of an appropriate reference loss A_{ref}

The principle of the loss calibration is to apply a device of known (reference) loss A_{ref} to the OTDR and to measure the displayed loss $A_{\text{otdr},i}$ as a function of the displayed power level F_i , as shown in Figure 9. Notice that the F_i values denote the high power ends of $A_{\text{otdr},i}$.

From a theoretical point of view, an infinitesimally small value of A_{ref} would be desirable. In practice, values of A_{ref} being too small may result in additional measurement uncertainty due to OTDR noise, whereas large values of A_{ref} tend to obscure fine details. Therefore, the specific value of A_{ref} used in the calibration has to be documented. The recommended range for A_{ref} is 0,5 dB to 2 dB.

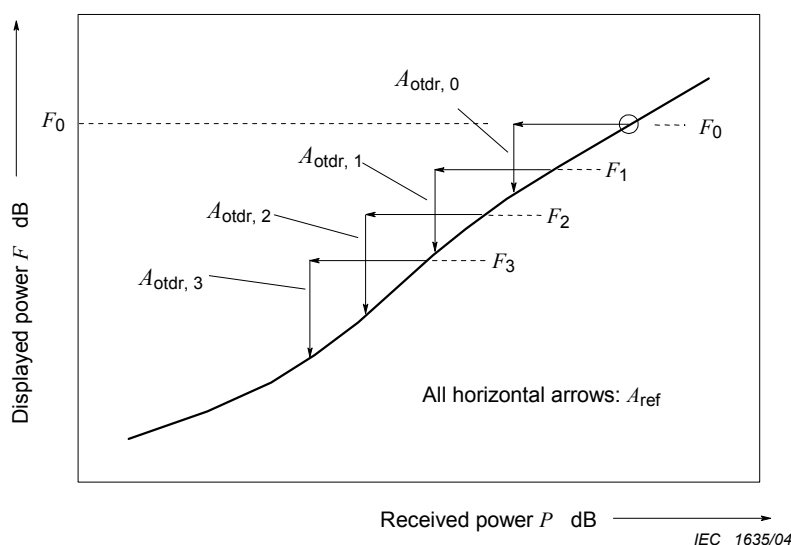


Figure 9 – Measurement of the OTDR loss samples

The reference loss can be an actual or simulated fibre optic component. Note that fibre optic components usually exhibit wavelength-dependent losses. Therefore, it is necessary to know the reference loss at the centroidal wavelength of the OTDR. It is also important that the reference loss has a small polarization dependence.

7.4 Development of a test plan

The loss samples may depend not only on the power level, but also on the distance and the history of the signal (that is, the shape of the fibre's OTDR signature) prior to the feature that is being measured. In particular, the detector and electronics may be affected by recovery from the initial firing of the laser and from signals due either to scattering or reflections in the fibre. The calibration will apply only to the distance and signal conditions for which it is performed.

This standard does not require specific conditions of signal history. For an aid in describing power levels and distance, this standard defines an OTDR display region A as an approximation to the region where the user normally takes measurements. For the purpose of this standard, region A is defined by four quantities, as illustrated in Figure 10: the extrapolated start of the backscatter trace for the specific pulse width used F_0 , a lowest and a highest attenuation as defined in the Table 1 below, and 3 dB margins on both sides.

Table 1 – Attenuation coefficients defining region A

Wavelength nm	Fibre attenuation coefficients	
	Lowest ($\alpha_{\min}$) dB/km	Highest ($\alpha_{\max}$) dB/km
1 310	0,33	0,43
1 550	0,18	0,28

On the same basis, attenuation coefficient values for other wavelengths may be chosen to represent typical single-mode fibres. An analytical description of region A is given by

$$\begin{aligned}
 F_{\max}(L) &= F_0 - \alpha_{\min} L + 3 \text{ dB} \\
 F_{\min}(L) &= F_0 - \alpha_{\max} L - 3 \text{ dB}
 \end{aligned}
 \tag{42}$$

$F_{\max}$ should not exceed an upper limit of 1 dB below the clipping level, unless otherwise specified by the OTDR manufacturer. The loss calibration points F should lie inside region A. Calibration data in regions B and C can be provided on a voluntary basis. Region B is applicable when the fibre path includes components with high loss. Region C is applicable when the fibre path includes components with strong reflection.

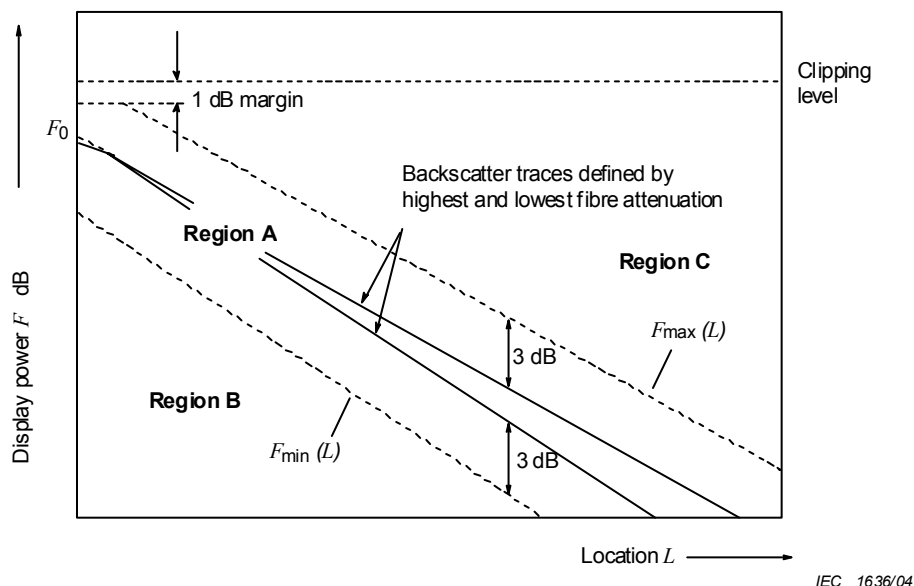


Figure 10 – Region A, the recommended region for loss measurement samples

For each of the methods outlined below, develop a test plan of sample placements, each of which is a combination of location and displayed power level. The goal is a vertical sample spacing of 0,5 dB to 1 dB, and not more than the reference loss A_{ref} . An attenuation range from F_0 down to the noise level and an even distribution of samples inside region A should be chosen. Overlapping measurement samples, that is samples at the same displayed power level but at different locations, are desirable, as indicated in Figure 11.

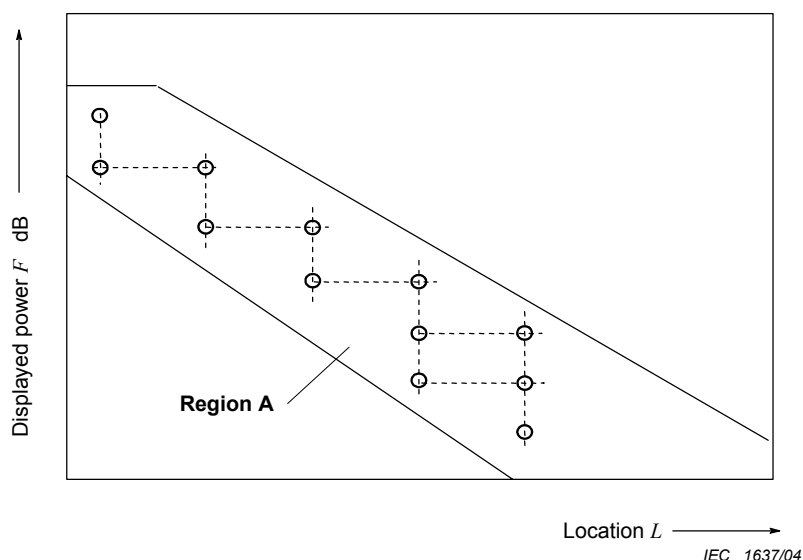
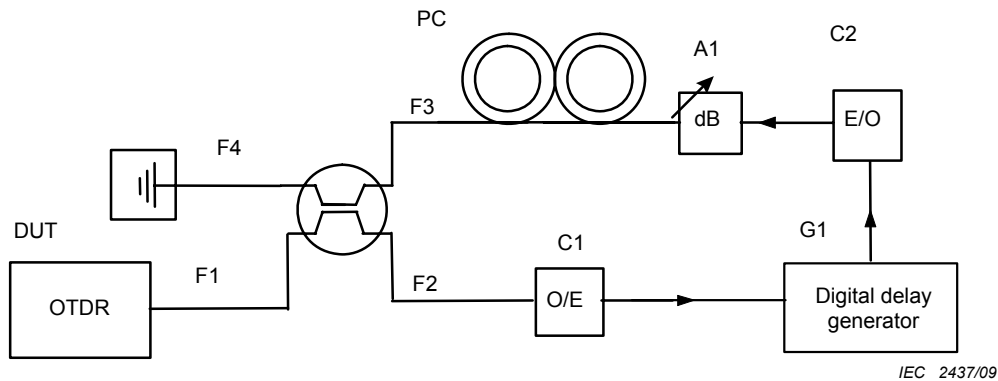


Figure 11 – Possible placement of sample points within region A

7.5 Polarization dependence

Polarization dependence (PDL) testing can be accomplished by feeding a polarized external signal with variable polarization state and constant power level to the OTDR port (see IEC 61300-3-2). One possibility is including a PDL test into the external source method outlined in 8.3, by replacing one attenuator by a polarization controller. Figure 12 shows an example. The polarization controller should be able to generate all states of polarization while leaving the output power and degree of polarization essentially unchanged.

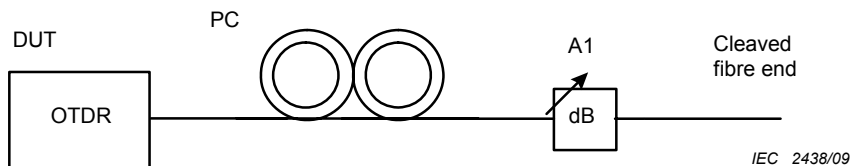


Key

- F1, F2, F3 and F4 fibres
- C1 optical-to-electrical converter
- C2 electrical-to-optical converter
- A1 variable attenuator
- PC polarisation controller
- G1 digital delay generator

Figure 12 – External source method for testing the polarization dependence of the OTDR

Another possibility for polarization dependence testing is using the reflected signal from, for example, a cleaved fibre end. See Figure 13.



Key

- A1 variable attenuator
- PC polarisation controller

Figure 13 – Reflection method for testing the polarization dependence of the OTDR

Be aware that the loss of the polarization controller may depend on its settings (rotation dependence). The coupler used in the external source method may also exhibit polarization-dependent loss. It is essential that these dependences be smaller than the OTDR polarization dependence. Both will introduce a measurement uncertainty. It is recommended to test the set-up for this uncertainty as follows.

- a) In the case of the external source method, evaluate the uncertainty by setting the E/O converter to continuous-wave operation and replacing the OTDR with a polarization-insensitive optical power meter.

In the case of the reflection method, evaluate the uncertainty by setting the OTDR to continuous-wave mode if possible; otherwise, replace the OTDR with a continuous-wave (polarized) laser source. Connect the fibre end to a polarization-insensitive optical power meter.

- b) Adjust the polarization controller so that a large number of polarization states is produced which cover essentially the entire Poincaré sphere. Observe the peak-to-peak change of the displayed power level caused by the change of polarization state, $\Delta A_{\text{pdl, set-up}}$; this quantity will also include the power meter's polarization dependence. Following the mathematical basis, record the result in the form of an uncertainty for the PDL measurement set-up, $u_{\text{pdl, set-up}}$:

$$u_{\text{pdl, set-up}} = \frac{\Delta A_{\text{pdl, set-up}}}{2\sqrt{3}} \quad (43)$$

To test the OTDR polarization dependence, place the optical pulse on the centre of the screen. Avoid displayed power levels near the OTDR's clipping level by proper setting of the attenuator. Adjust the polarization controller so that a large number of polarization states is produced. Observe the change of displayed power level caused by changing the polarization state. Record the variation of the pulse amplitude $\pm \Delta A_{\text{pdl, otdr}}$.

Include the PDL measurement result into the documentation. If PDL testing was not carried out, then this is to be specifically stated in the documentation.

7.6 Calculation of the calibration results

From the measured values $A_{\text{otdr, i}}$, calculate the loss deviation ΔA_i :

$$\Delta A_i = A_{\text{otdr, i}} - A_{\text{ref}} \quad \text{dB} \quad (44)$$

The loss uncertainties $u_{\Delta A}$ are discussed in conjunction with the calibration methods.

7.7 Using the calibration results

The error of a loss value, ΔY_i measured with the OTDR ΔA_i and its uncertainty can be calculated from the calibration results by the following formula, in which the recommended confidence level of 95 % is used:

$$\Delta Y_i = \Delta A_i \pm 2u_{\Delta A, i} \quad \text{dB} \quad (45)$$

Notice that this error applies, in a rigorous sense, only to measured losses at the displayed power level and displayed location for which ΔA_i was recorded.

8 Loss calibration methods

8.1 General

In all loss calibration methods, a reference loss is produced outside the OTDR, and the OTDR response to this reference loss is tested. In some cases, the reference loss is the difference between two signals of equal polarization state (see 8.3 and 8.5). In these cases, the loss calibration result is not influenced by the OTDR's polarization dependence. In other cases, the reference loss is the difference between two signals of potentially different polarization states (see 8.2 and 8.4). In these cases, it may be advisable to add a polarization controller between the OTDR and the reference loss device, in order to be able to average out the influence of the OTDR polarization dependence. Generally, it is desirable that the loss calibration results are not influenced by the OTDR polarization dependence.

8.2 Fibre standard method

8.2.1 Short description and advantage

The fibre standard method describes the calibration of the OTDR power scale with the help of an optical fibre standard of precisely determined loss (see Annex B). This reference loss is used to determine $A_{\text{otdr},i}$. This calibration method closely reflects the conditions in measuring fibre attenuation with an OTDR; it is particularly well suited for the calibration of region A of Figure 10.

8.2.2 Equipment

8.2.2.1 General

In addition to the test OTDR, the measurement equipment includes

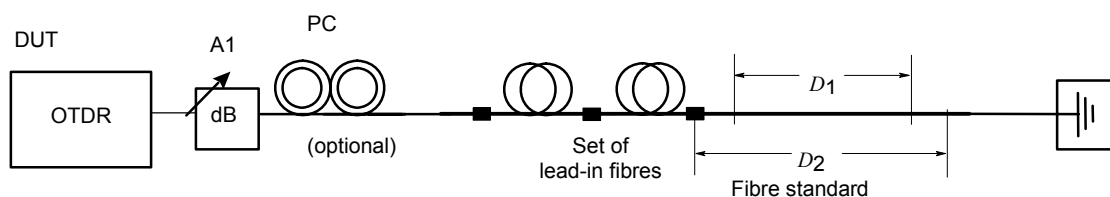
- a fibre standard as described in Annex B;
- a set of lead-in fibres, these shall be type B single-mode fibres (see IEC 60793-2-50);
- a variable attenuator;
- optionally, a polarization controller.

The purpose of the attenuator and lead-in fibres is to place the fibre standard at a number of different locations within region A (see Clause 7) of the OTDR display. Example: the displayed power level can be varied within a 30 dB range with 2 dB steps by a proper combination of four lead-in fibres with attenuation values of 2 dB, 4 dB, 8 dB and 16 dB.

These fibres should be equipped with low-reflectance connectors; the attenuation numbers include the typical connector losses. In order to obtain the recommended (finer) sample spacing of 0,5 dB, the attenuator will have to be varied between 0 dB and 1,5 dB in 0,5 dB steps.

It should be noted that the attenuator is not necessary if the recommended steps of 0,5 dB are generated with a larger number of lead-in fibres.

The purpose of the optional polarization controller is to reduce the possible influence of the OTDR polarization dependence (see Figure 14).



IEC 2439/09

Key

- | | |
|----|------------------------------------|
| A1 | variable attenuator |
| PC | polarisation controller (optional) |

Figure 14 – Loss calibration with a fibre standard

Both the attenuator and the polarization controller shall have low reflections, because reflections can influence the attenuation measurement results.

8.2.2.2 Calibration of the reference loss

Measure the total length D_2 of the fibre standard according to the OTDR manufacturer's instructions for length measurement. Using the OTDR markers, select a section of the fibre standard, of length D_1 , outside the attenuation dead zone caused primarily by any connectors in front of the fibre standard (see Figure 15). Choose the beginning of the section so that the difference between the actual backscatter trace and its linear extrapolation $\Delta F_{\max}$ is sufficiently small at that point (a lead-in fibre may be necessary to accomplish this). Measure the length of the section D_1 .

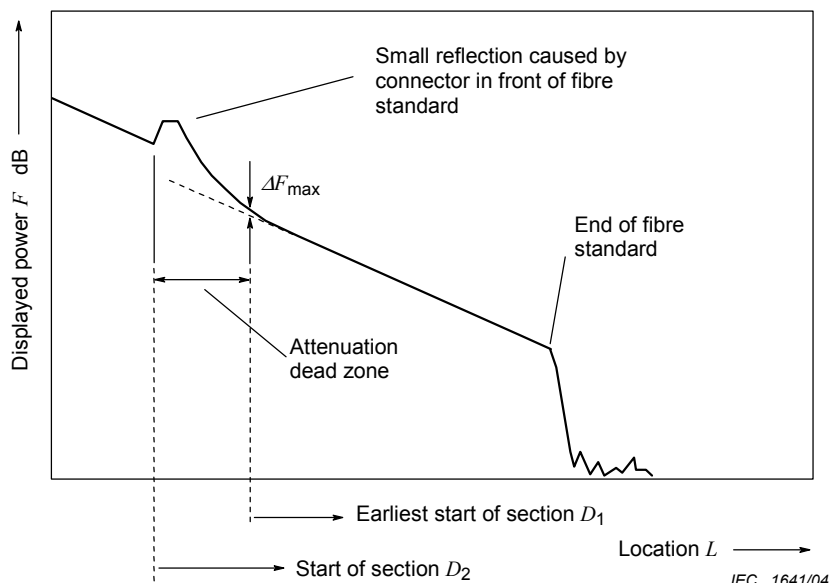


Figure 15 – Placing the beginning of section D_1 outside the attenuation dead zone

A length D_1 that corresponds to a reference loss of about 0,5 dB is recommended. Back reflections from the fibre standard's far end should be carefully avoided, because they can influence the preceding backscatter trace. Due to the longitudinal uniformity of the fibre standard and the directional independence of its backscatter loss, the reference loss A_{ref} at the marker locations is given with high accuracy by

$$A_{\text{ref}} = A_{\text{std}} (\lambda_{\text{avg}}) \frac{D_1}{D_2} \quad \text{dB} \quad (46)$$

where $A_{\text{std}} (\lambda_{\text{avg}})$ is the calibrated loss of the fibre standard at the centroidal wavelength of the OTDR λ_{avg} (see Annex B). A_{ref} shall remain constant during the subsequent measurements, or the distance between the marker locations shall be kept fixed. Note that the influence of the group index is eliminated in the calculation.

8.2.3 Measurement procedure

8.2.3.1 Preparation

First, develop a test plan of fibre/attenuator settings so that a vertical sample spacing of approximately 0,5 dB is achieved and all measurement samples fall within the region A of the OTDR display; an example is described in 8.2.2. Overlapping measurement samples, that is samples at the same displayed power level but at different locations, are desirable. That part of the dynamic range which might be inaccessible due to the insertion loss of the attenuator (including both connectors) may be covered by directly connecting the shortest lead-in fibre to the OTDR and varying the insertion loss by bending the lead-in fibre.

8.2.3.2 Taking the measurement results

For each displayed power level F_i , measure the loss value $A_{\text{otdr},i}$. In order to reduce the uncertainty, it is recommended to average around the markers or over the entire length D_1 , instead of using single power levels. Longer OTDR averaging may be advisable at low displayed power levels in order to reduce the uncertainty type A; all applied averaging times should be reported in this case. Record all $A_{\text{otdr},i}$ displayed power levels F_i and locations L_i .

The measurement results are expected to be influenced by the OTDR polarization dependence (PDL), because the signals from the beginning and the end of section D_1 may have different polarization states. A way of reducing the PDL influence is inserting a polarization controller, as shown in Figure 14, and measuring n samples of $A_{\text{otdr},i}$ at different positions of the polarization controller. Finally, all of the n samples can be averaged to a single $A_{\text{otdr},i}$, in which the PDL influence is reduced.

8.2.4 Calculations and results

Calculate the loss deviation samples ΔA_i using Equation (47):

$$\Delta A_i = A_{\text{otdr},i} - A_{\text{ref}} \quad \text{dB} \quad (47)$$

Alternatively report the loss scale deviation using Equation (48):

$$\Delta S_{A,i} = \frac{A_{\text{otdr},i} - A_{\text{ref}}}{A_{\text{ref}}} \quad \text{dB/dB} \quad (48)$$

8.2.5 Uncertainties

Note that the following list of uncertainties may not be complete. Additional contributions may have to be taken into account, depending on the measurement set-up and procedure.

The guidelines of the mathematical basis given in Annex D should be used to calculate and state the uncertainties.

The standard deviation characterizing the loss uncertainty $u_{\Delta A}$ of an individual ΔA can be calculated from Equation (47), using the standard formula for the propagation of errors.

$$u_{\Delta A} = \left(u_{A_{\text{ref}}}^2 + u_{A_{\text{otdr}}}^2 \right)^{1/2} \quad \text{dB} \quad (49)$$

The uncertainty $u_{A_{\text{ref}}}$ of the reference loss, that is of the fibre standard, and $u_{A_{\text{otdr}}}$ of the OTDR response shall be determined.

The uncertainty $u_{A_{\text{ref}}}$ should be accumulated, by root-sum-squaring, from the following contributions:

- $u_{A,\text{std}}$ the loss uncertainty, in decibels, of the fibre standard as documented according to Annex B;
- $u_{A,\text{distance}}$ the loss uncertainty, in decibels, due to not using the full length of the fibre standard; this uncertainty should itself be accumulated by root-sum-squaring the uncertainty in determining the length relation D_1/D_2 (see Equation (46)) and the uncertainty due to the longitudinal attenuation uniformity (see Annex B).

The uncertainty $u_{A_{\text{otdr}}}$ should be accumulated, by root-sum-squaring, from the following contributions:

- $u_{A,\lambda}$ the loss uncertainty, in decibels, introduced by the uncertainty of the OTDR centroidal wavelength u_{λ} .

NOTE This uncertainty may be negligible at wavelengths near 1 550 nm because of the small wavelength dependence of the fibre loss at these wavelengths; in the wavelength region where the loss is predominantly given by Rayleigh scattering, for example at 1 300 nm, the uncertainty is

$$u_{A,\lambda} = \frac{4 A_{\text{ref}}}{\lambda} u_{\lambda} \quad (50)$$

$u_{A,\text{deadzone}}$	the loss uncertainty, in decibels, caused by the attenuation dead zone following the connector(s);
$u_{A,\text{attenuator}}$	the loss uncertainty of the optical attenuator caused by the possibility that the backscatter signal is polarized and the attenuator may have polarization-dependent loss;
$u_{A,\text{pdl}}$	the loss uncertainty caused by the OTDR's polarization dependence, due to the fact that the backscatter signal exhibits different polarization states; this uncertainty can be reduced (see 8.2.3.2);
$u_{A,\text{type A}}$	the loss uncertainty type A, in decibels, to be experimentally determined. It will depend on the displayed power level and OTDR averaging time.

Additional contributions may have to be taken into account, depending on the measurement set-up and procedure.

8.3 External source method (see Figure 16)

8.3.1 Short description and advantage

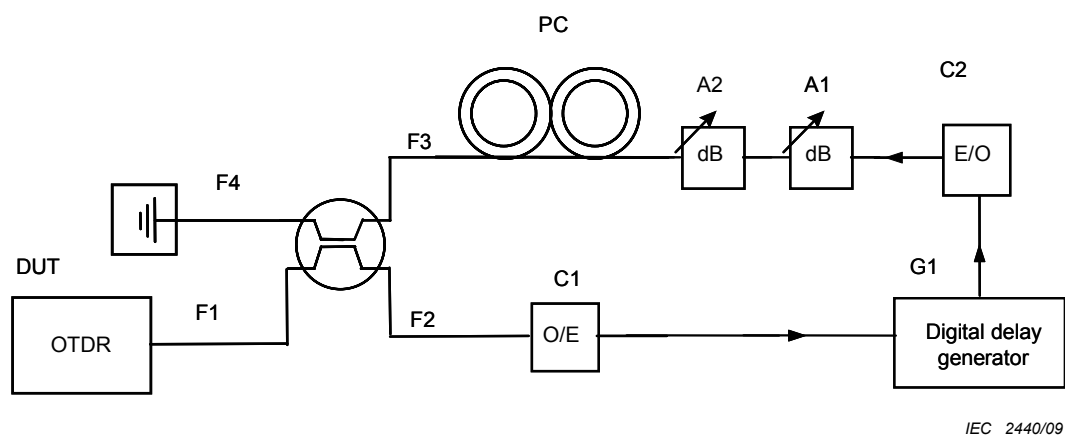
This method uses a calibrated delay generator to simulate the time delay in a fibre, and an optical source to generate a known change of optical pulse amplitude. The equipment is similar to the one used for the external source method for distance calibration (see 6.2), except that control of the optical amplitude is added. The combination of variable delay and variable pulse amplitude allows complete characterization of the OTDR power scale, including its dependence on distance.

The method is well suited to fully automated laboratory testing under computer control. It can calibrate all regions A, B and C of Figure 10. For simplicity, the text below describes only a pulsed signal source generating points in region A with no backscatter light in the signal history. To calibrate the OTDR for other conditions, the pulsed light source described below should be replaced by an optical source that simulates the appropriate signal.

8.3.2 Equipment

In addition to the test OTDR, the measurement equipment includes

- an optical fibre coupler to interconnect the system;
- an optical to electrical (O/E) converter;
- a delay generator with variable delay, variable pulse width and variable pulse amplitude;
- an electrical to optical (E/O) converter, with a centroidal wavelength which is approximately equal to the OTDR's centroidal wavelength, and preferably with the capability of generating continuous-wave optical power (for the calibration of the reference loss);
- a variable optical attenuator (attenuator 1);
- an optical attenuator (attenuator 2) with good repeatability to toggle in and out a specific quantity of attenuation (the recommended value is 1 dB to 4 dB);
- optical fibre jumper cables to connect the system;
- an optical power meter.



Key

F1, F2, F3 and F4	fibres
C2	electrical-to-optical converter
C1	optical-to-electrical converter
A1, A2	variable attenuators
PC	polarisation controller
G1	digital delay generator

Figure 16 – Loss calibration with the external source method

The combination of variable attenuator A1 and variable delay generator allows the peak of an optical pulse (or simulated backscatter trace) to be positioned anywhere in the two-dimensional space of the OTDR display. Attenuator A2 is used to toggle a fixed attenuation step in and out so that the optical pulse amplitude changes by a known amount. Note that the reference loss A_{ref} equals half the attenuation step, due to the OTDR's vertical scaling factor of 0,5.

The purpose of the optional polarization controller is to reduce the possible influence of the polarization dependence of the two attenuators and the power meter used in taking the reference loss. If applicable, it should be used in both calibrating the reference loss and taking the actual measurements, by taking the average of several polarization states as described in 8.2.3.2.

8.3.3 Calibration of the reference loss

Select a suitable value for the reference loss A_{ref} . A value of 0,5 dB to 2 dB is recommended – this equals an attenuation step of 1 dB to 4 dB.

Set the E/O converter to continuous wave (CW) operation. Connect the optical power meter to the OTDR end of fibre F1. Set the wavelength correction of the optical power meter to the wavelength of the E/O converter. Measure the attenuation step of attenuator 2 a number of times to determine A_{ref} and its uncertainty type A $u_{A,\text{step}}$. Note that optical interference, caused by reflections in the measurement set-up and by a narrow spectral width of the source, may cause undesired variability of the measurement results.

Calculate the reference loss as the average attenuation step divided by 2.

8.3.4 Measurement procedure

8.3.4.1 Preparation

Select appropriate combinations of pulse amplitudes and delay settings so that the measurement samples are spaced by 0,5 dB to 1 dB and distributed evenly over the entire length of region A (see Figure 10). Overlapping measurement samples are desirable, that is samples at the same displayed power level but at different locations. Note that measurement samples outside of region A can be provided on a voluntary basis.

Select the E/O pulse width, for example, 1 μ s, so that the displayed pulse has an essentially flat top. Select the OTDR pulse width so that it is shorter than the return path, to prevent it from overlapping the pulse from the E/O. Program attenuators 1 and 2 and the delay generator accordingly. Another possibility is to change the pulse width with each new location, so that the pulse always starts at $L = 0$, and determine the attenuation step at the end of the pulse.

Define the method of using the OTDR to measure the change of power level caused by toggling attenuator 2. For example, the method may involve averaging a number of points in the vicinity of the measurement location. Also, define the method of measuring the displayed power level (see Clause 7; the displayed power level is defined as the upper level of the two in question).

Switch on all equipment. Specifically, allow sufficient warm-up time for the E/O converter to achieve a stable pulse amplitude. Toggle attenuator 2 a few times and verify the stability of the displayed power level and of the attenuation step A_{otdr} with the OTDR.

8.3.4.2 Taking the measurement results

Determine the 0 dB reference level as described in Clause 7.

For each of the preselected combinations of attenuator 1 settings and delay generator settings proceed as follows.

Determine the measurement location on the pulse L_i , for example, as the centre of the pulse width. To reduce the uncertainty, do not change L_i during the set of measurements at this particular location.

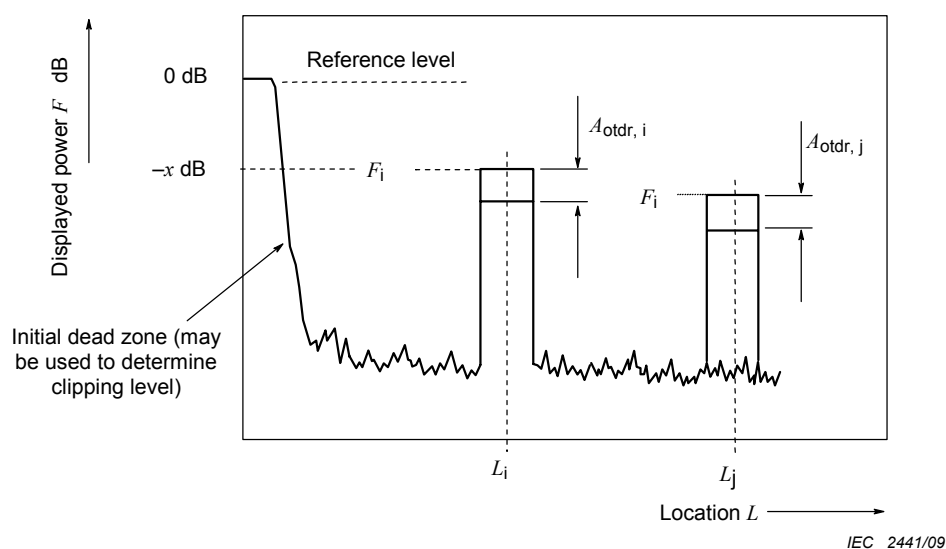


Figure 17 – Location and measurements for external source method

Toggle attenuator 2 between its predefined settings, record the displayed power level F_i (the upper level of the two in question) and the change of displayed power level $A_{\text{otdr},i}$ as shown in Figure 17.

It may be advisable to repeat the measurement of $A_{\text{otdr},i}$ several times and to record the average of these samples. It may also be desirable to generate more than one sample per location, at different displayed power levels, by adjusting the pulse power with attenuator 1.

Proceed by setting the delay generator to the next location.

8.3.5 Calculations and results

For each measurement sample, calculate the loss deviation using Equation (44):

$$\Delta A_i = A_{\text{otdr},i} - A_{\text{ref}} \quad \text{dB} \quad (51)$$

Record the displayed power level F_i and the location L_i with each loss deviation value. It may be advisable to plot the loss deviation values as a function of the displayed power level F_i .

Alternatively report the loss scale deviation using Equation (52):

$$\Delta S_{A,i} = \frac{A_{\text{otdr},i} - A_{\text{ref}}}{A_{\text{ref}}} \quad \text{dB/dB} \quad (52)$$

8.3.6 Uncertainties

Note that the following list of uncertainties may not be complete. Additional contributions may have to be taken into account, depending on the measurement set-up and procedure. The mathematical basis given in Annex D should be used to calculate and state the uncertainties.

The standard deviation characterizing the loss uncertainty can be calculated from Equation (51) using the standard formula for the propagation of errors:

$$u_{\Delta A} = \left(u_{A_{\text{ref}}}^2 + u_{A_{\text{otdr}}}^2 \right)^{1/2} \quad \text{dB} \quad (53)$$

The uncertainty of the reference loss $u_{A_{\text{ref}}}$ should be accumulated, by root-sum-squaring, from the following contributions:

- $u_{A,\text{pm}}$ the uncertainty, in decibels, due to calibrating the reference loss with the power meter, for example, due to the power meter's non-linearity, polarization dependence, inhomogeneity and noise. This uncertainty should itself be accumulated by root-sum-squaring;
- $u_{A,\text{refl}}$ the uncertainty due to differences in the reflection characteristics of the set-up. The return loss of the power meter will not be the same as the return loss of the coupler/OTDR combination; this may have an influence on the magnitude of the reference loss;
- $u_{A,\text{step}}$ the loss uncertainty, in decibels, introduced by the instability of the attenuation step, for example caused by the instability of the external source, optical interference, or the coupler's polarization dependence in conjunction with instability of the state of polarization.

The uncertainty of the OTDR measurement samples $u_{A_{\text{otdr}}}$ should be accumulated, by root-sum-squaring, from the following contributions:

- $u_{A,\lambda}$ the uncertainty, in decibels, caused by the difference between the centroidal wavelengths of the OTDR and the optical source used in the calibration;
- $u_{A,\text{pdl}}$ the uncertainty, in decibels, introduced by changing polarization state as a result of the attenuation step;

$u_{A,type A}$ the uncertainty, in decibels, introduced by the variability of the attenuation step measurements, for example, caused by limited readout resolution and displayed power levels when they approach the noise limit. This uncertainty can be determined together with $u_{A,step}$ above.

NOTE The uncertainties due to polarization dependence can be reduced or eliminated by the optional polarization controller.

8.4 Splice simulator method

8.4.1 Short description and advantage

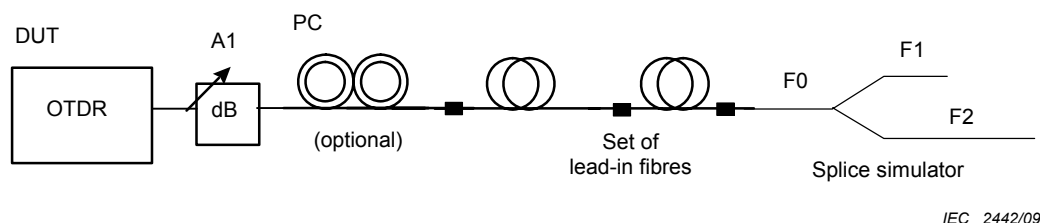
This method uses a splice simulator to calibrate the OTDR power scale. The splice simulator has a fixed reference loss of approximately 1,5 dB. This method closely reflects the condition in measuring splice attenuation with an OTDR. No electronic equipment is needed for the calibration. The method is well suited for the calibration of the regions A and B of Figure 10.

8.4.2 Equipment

In addition to the test OTDR, the measurement equipment includes

- a) a splice simulator as described in Annex C;
- b) a set of lead-in fibres. The lead-in fibres shall be type B of IEC 60793-2-50;
- c) a variable attenuator, if necessary;
- d) optionally, a polarization controller.

The test set-up is shown in Figure 18.



Key

- F0, F1 and F2 fibres
- A1 variable attenuator
- PC polarisation controller (optional)

Figure 18 – Set-up for loss calibration with splice simulator

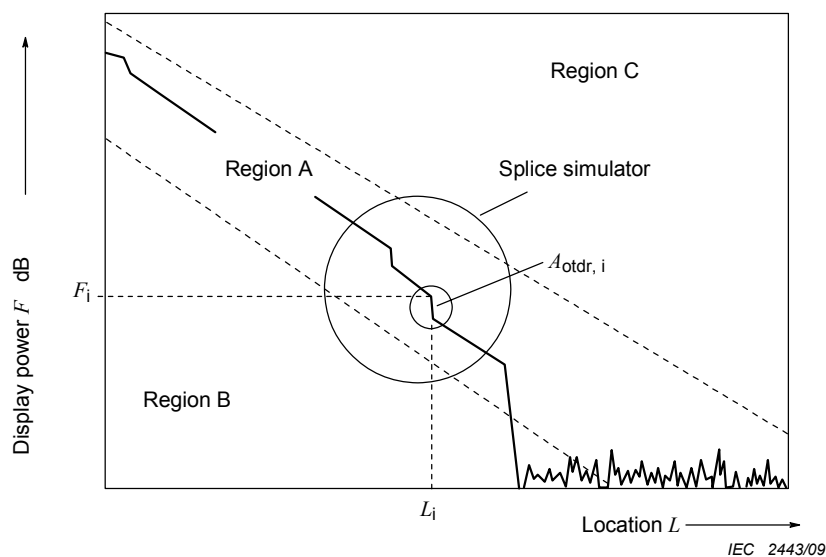
According to Annex C, the calibrated reference loss (approximately 1,5 dB) appears at the end of fibre F1. The purpose of the lead-in fibres is to generate measurement data within region A (see Clause 7 and Figure 19).

The loss produced by the splice simulator is the difference between

- a) the sum of the two backscatter signals from fibres F1 and F2, and
- b) the backscatter signal from fibre F2.

The two signals a) and b) will usually have different polarization states. Therefore, it may be advisable to use a polarization controller to produce a number of different polarization states of signals a) and b), which will then allow to average out the effect from the OTDR's polarization dependence.

It is important that the attenuator and the polarization controller have low reflections, because they can influence the calibration result.



NOTE The smaller circle represents the OTDR response to the reference loss.

Figure 19 – OTDR display with splice simulator

8.4.3 Procedure

8.4.3.1 Preparation

Select appropriate combinations of lead-in fibre and attenuator settings so that a vertical sample spacing of 0,5 dB to 1 dB is achieved and all measurement samples fall within the region A of the OTDR display. Overlapping measurement samples, that is samples which are taken at the same displayed power level but at different locations, are desirable. Note that the variable attenuator can be used in order to increase the number of different displayed power levels F .

Connect the splice simulator to the OTDR as shown in Figure 18.

8.4.3.2 Taking the splice loss measurements

As illustrated in Figure 20, the splice loss A_{otdr} is defined as the vertical distance between the two lines representing the attenuation coefficients.

Follow the instructions below for splice loss measurement with two cursors (two points method) or with the least squares approximation (LSA method). Note that other methods such as the five cursors method or automatic measurement can also be used. However, the method used shall be clearly specified.

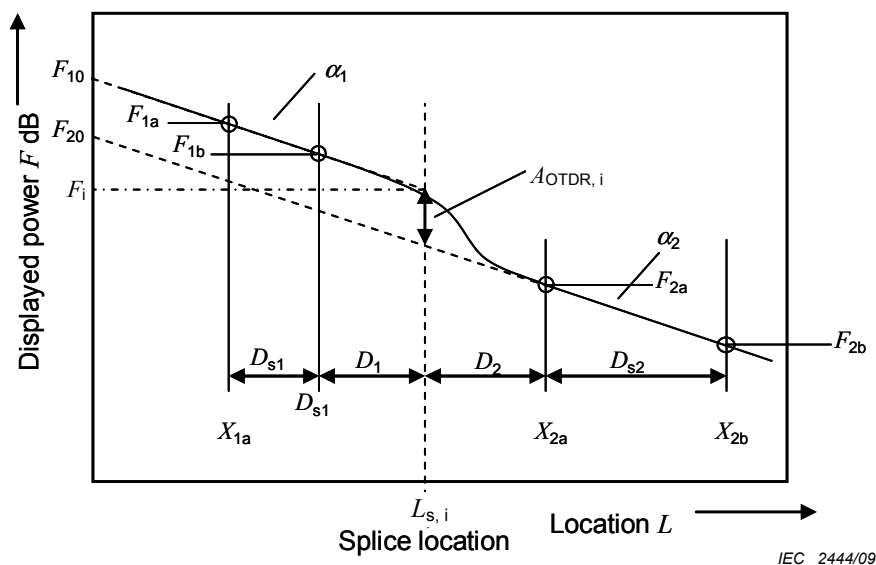


Figure 20 – Measurement of the splice loss

If the optional polarization controller is used, then repeat the following procedure n times for different positions of the polarization controller. In the calculations, average all of the n samples to a single $A_{\text{otdr},i}$, in which the PDL influence is reduced.

For each combination of lead-in fibres and attenuator settings, proceed as follows:

- define point X_{1b} near the splice, but to the left on a straight, clean backscatter trace;
- define point X_{1a} to the left of X_{1b} at least 500 m away. A straight, clean backscatter trace should connect the two points. Record the distance between the two points D_{s1} ;

NOTE A distance of 500 m was chosen because it is essential that the backscatter loss between the two points is at least in the order of 0,1 dB.

- according to the method used – the two-point method or the LSA method:
 - if using the two-point method, record the two power levels, F_{1a} and F_{1b} , and compute the displayed attenuation coefficient of the fibre as:

$$\alpha_1 = \frac{F_{1a} - F_{1b}}{D_{s1}} \quad (54)$$

Note that the attenuation coefficient given by the OTDR may be used;

- if using the LSA method, compute the best-fit line and determine the slope α_1 and the intercept with the vertical axis F_{10} . The fit should be performed on all points between X_{1a} and X_{1b} . Record α_1 and F_{10} ;
- define point X_{2a} near the splice, but to the right, after any effects caused by the drop of signal on a straight, clean backscatter trace;
- define point X_{2b} to the right of X_{2a} at least 500 m away. A straight, clean backscatter trace should connect the two points. Record the distance between the two points D_{s2} ;
- according to the method used – the two-point method or the LSA method:
 - if using the two-point method, record the two power levels, F_{2a} and F_{2b} , and compute the displayed attenuation coefficient of the fibre as

$$\alpha_2 = \frac{F_{2a} - F_{2b}}{D_{s2}} \quad (55)$$

Note that the attenuation coefficient given by the OTDR may be used;

- 2) if using the LSA method, compute the best-fit line and determine the slope α_2 and the intercept with the vertical axis F_{20} . The fit shall be performed on all points between X_{2a} and X_{2b} . Record α_2 and F_{20} ;
- g) define the splice location $L_{s,i}$ as the turning point of the backscatter trace;
- h) compute the distance D_1 between point X_{1b} and the splice location. Compute the distance D_2 between the splice location and point X_{2a} .

8.4.4 Calculations and results

a) According to the method used – the two-point method or the LSA method:

- 1) if using the two-point method, compute the splice loss $A_{otdr,i}$ and the correspondent power level F_i (the subscript i indicates the i^{th} splice loss) as:

$$A_{otdr,i} = (F_{1b} - \alpha_1 D_1) - (F_{2a} + \alpha_2 D_2) \quad (56)$$

$$F_i = F_{1b} - \alpha_1 D_1$$

- 2) if using the LSA method, compute the splice loss $A_{otdr,i}$ and the correspondent power level F_i , as:

$$A_{otdr,i} = F_{10} - F_{20} - L_{s,i}(\alpha_1 - \alpha_2) \quad (57)$$

$$F_i = F_{1b} - \alpha_1 D_1$$

b) from the loss values A_{otdr} , calculate the loss deviations ΔA :

$$\Delta A_i = A_{otdr,i} - A_{ref} \quad \text{dB} \quad (58)$$

Record the loss deviations, the displayed power levels and the splice locations. State the method used for splice loss analysis (for example two-point method or least-squares approximation).

Alternatively report the loss scale deviation using Equation (59):

$$\Delta S_{A,i} = \frac{A_{otdr,i} - A_{ref}}{A_{ref}} \quad \text{dB/dB} \quad (59)$$

8.4.5 Uncertainties

Note that the following list of uncertainties may not be complete. Additional contributions may have to be taken into account, depending on the measurement set-up and procedure. The mathematical basis given in Annex D should be used to calculate and state the uncertainties.

The standard deviation characterizing the loss uncertainty $u_{\Delta A}$ of an individual ΔA can be calculated from Equation (58), using the standard formula for propagation of uncertainties:

$$u_{\Delta A} = \left(u_{A_{ref}}^2 + u_{A_{otdr}}^2 \right)^{1/2} \quad \text{dB} \quad (60)$$

The uncertainty $u_{A_{ref}}$ can be extracted from the calibration certificate of the splice simulator (see Annex C). The uncertainty $\sigma_{A_{otdr}}$ should be accumulated, by root-sum-squaring, from the following contributions:

- $u_{A,\lambda}$ the splice loss uncertainty, in decibels, introduced by the difference between the centroidal wavelength used for the calibration of the splice simulator and the OTDR's centroidal wavelength;
- $u_{A,pdl}$ the splice loss uncertainty caused by the OTDR polarization dependence, due to the possibility that the lead-in fibres and/or the attenuation step introduce a change of the signal's polarization state (when the optional polarization controller is used, then this uncertainty is reduced);

$u_{A,pos}$ the splice loss uncertainty, in decibels, introduced by the uncertainty of the splice location σ_L . If not available elsewhere, use one half of the OTDR pulse width, expressed in metres; the relation between these uncertainties is given by:

$$u_{A,pos} = |\alpha_1 - \alpha_2| \sigma_L \quad (61)$$

$u_{A,slope}$ the splice loss uncertainty introduced by the uncertainty of the slope measurement σ_s . The relation between these uncertainties is given by:

$$u_{A,slope} = \sqrt{(D_1 u_{\alpha 1})^2 + (D_2 u_{\alpha 2})^2} \quad (62)$$

8.5 Power reduction method

8.5.1 Short description and advantage

This method uses a known change of optical signal to calibrate the OTDR power scale. The correspondent change of displayed power level is a measure of the loss deviation. Three variants of this method are in compliance with this standard:

- the "fibre-end" variant, in which the power steps are recorded as the change of reflection amplitude;
- the "multiple-reflections" variant, in which the power steps are recorded as the changes of all reflection amplitudes produced;
- the "long-fibre" variant, in which the power steps are recorded as the change of backscatter signal along the length of the fibre.

The equipment includes an attenuator which is toggled between two attenuation settings to produce the change of displayed power level. The equipment is lightweight and inexpensive.

The advantage of the "fibre-end reflection" variant is that it can measure all regions A, B and C. A well-cleaved fibre end or polished connector, a second attenuator and a number of lead-in fibres are needed, in addition to the equipment of the "long fibre" variant, to place the measurement samples into the two-dimensional space of power and distance.

The "multiple-reflections" variant can also measure all regions A, B and C. Its advantage over the "single" fibre-end reflection is that no lead-in fibres and no fibre reconnections are necessary during the calibration. The multiple reflections can be generated by an optical cavity with partially transparent mirrors. This variant is well suited for full automation.

The advantage of the "long-fibre" variant is that this method is close to measuring fibre loss with the OTDR and that the measurement samples automatically fall into region A. Placing measurement samples into region C and the upper part of region A is not possible. No fibre reconnections are necessary during the calibration. This variant is also well suited for full automation.

It is expected that toggling the attenuation step in and out will not change the state of polarization of the wave returning to the OTDR, thereby allowing OTDR loss calibration without an uncertainty contribution from the OTDR polarization dependence. Appropriate polarization characteristics of the "toggle" attenuator are necessary to accomplish this. Care should be taken not to alter the position of the fibres in the measurement set-up while toggling the attenuator, because otherwise the measurement results will be affected by the OTDR polarization dependence.

An optional polarization controller, such as in the "fibre-standard" and "splice-simulator" methods, is not needed here because each loss measurement result is the difference of two samples representing essentially the same polarization state.

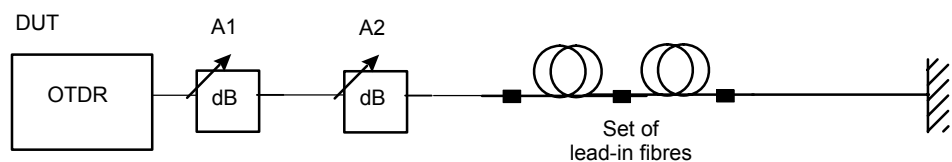
NOTE The set up must be mechanically and thermally stable over the period of the two measurements to avoid changes in the polarisation state caused by changes in the fibres.

8.5.2 Equipment

8.5.2.1 "Fibre-end" variant

The measurement equipment includes, as shown in Figure 21:

- a set of fibres, type B, IEC 60793-2-50, the end of which is equipped with a well-cleaved fibre end or polished connector; the shortest fibre of the set shall be long enough to ensure that the reflected pulse arrives after the recovery from the initial pulse;
- a variable optical attenuator (attenuator 1) with a resolution of $\leq 0,1$ dB, to set the displayed power level;
- an optical attenuator (attenuator 2) with good repeatability to toggle in and out a specific quantity of attenuation (recommended values are 0,5 dB to 2 dB); this attenuator should have high attenuation repeatability (preferably $< 0,05$ dB) when toggled between the two attenuation settings; it is also important that there is no change in the polarization state due to toggling, and that the reflectances are sufficiently low to avoid amplitude changes of the main reflection;
- optical jumper cables to connect the system;
- an optical power meter.



IEC 2445/09

Key

A1 and A2 attenuator

Figure 21 – Loss calibration with "fibre-end" variant of the power reduction method

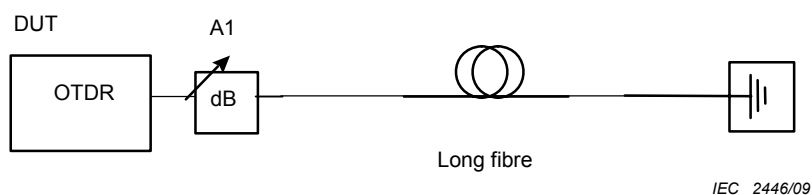
8.5.2.2 "Multiple-reflections" variant

The set of fibres and the cleaved fibre end are replaced by an optical cavity constructed, for example, from two partially transparent mirrors with a length of fibre in between. The intention is to produce a series of reflections which are separated by some distance.

8.5.2.3 "Long-fibre" variant

Attenuator A2 is not necessary. Instead, a long fibre and a device which is capable of reducing the fibre-end reflection to insignificance (absorber, for example, fibre knots) are required, as shown in Figure 22.

The long fibre should have a total attenuation corresponding to the OTDR measurement range.



Key

A1 attenuator

Figure 22 – Loss calibration with "long-fibre" variant of the power reduction method

8.5.2.4 Calibration of the reference loss

The attenuation step of the "toggle" attenuator represents the reference loss A_{ref} . Select a suitable value for A_{ref} . A value of 0,5 dB to 2 dB is recommended. Switch the OTDR to continuous wave mode, if possible. Otherwise, use a continuous wave laser source, with a centroidal wavelength as close as possible to the OTDR centroidal wavelength. A second attenuator can be considered as part of the source. Connect the source to the input of the "toggle" attenuator. Connect the output of the "toggle" attenuator to the optical power meter. Set the wavelength correction of the optical power meter.

Measure the attenuation step a number of times. Calculate the reference loss as the average attenuation step.

8.5.3 Measurement procedure

8.5.3.1 Preparation

In the "fibre-end" and "multiple-reflections" variants, determine the 0 dB reference level, for example the OTDR clipping level, as described in Clause 7. The clipping level can be determined by setting both attenuators to their respective minimum attenuations. Select a set of fibres so that the optical pulse is displayed in the centre of the screen. Select the pulse width, for example 10 μs , so that the displayed pulse has a flat top. Adjust attenuator A1 so that the peak of the generated optical pulse is just below (within 1 dB) the clipping level. Following the guidelines of Clause 7, develop a test plan of sample placements based on attenuator A1 settings and fibre combinations from the set.

In the "long-fibre" variant, determine the 0 dB reference level, for example the start of the backscatter signal (without the attenuator), as described in Clause 7. Select the pulse width, for example 10 μs . Set the attenuator to the smallest attenuation value possible, so that the displayed power is not significantly reduced in comparison with no attenuator inserted. Following the guidelines of Clause 7, develop a test plan of sample placements based on the selected attenuation step and the length of fibre connected.

8.5.3.2 Measurements

Find a method to measure the change in height of the reflection or backscatter signal. One technique is to place one cursor on a stable power level such as the initial OTDR pulse and a second cursor on the pulse top of the reflection or backscatter signal. To reduce the uncertainty of distance, do not change either marker's location during the set of measurements at a particular location.

The measurements are similar to those outlined in 8.3:

- a) select the first sample placement according to the test plan;

- b) toggle the attenuator between its predetermined settings;
- c) record the displayed power level F_i , the change of displayed power level $A_{\text{otdr},i}$ and the correspondent location L_i . It may be advisable to repeat the measurement of $A_{\text{otdr},i}$ several times and to record the average of these samples;
- d) select the next sample placement according to the test plan and go back to step b). Continue until the test plan is finished.

8.5.4 Calculations and results

For each measurement sample, calculate the loss deviation from the equation:

$$\Delta A_i = A_{\text{otdr},i} - A_{\text{ref}} \quad \text{dB} \quad (63)$$

Record the displayed power level F_i and the location L_i , with each loss deviation value. It may be advisable to plot the loss deviation values as a function of the displayed power level F_i .

Alternatively report the loss scale deviation using Equation (64):

$$\Delta S_{A,i} = \frac{A_{\text{otdr},i} - A_{\text{ref}}}{A_{\text{ref}}} \quad \text{dB/dB} \quad (64)$$

8.5.5 Uncertainties

Note that the following list of uncertainties may not be complete. Additional contributions may have to be taken into account, depending on the measurement set-up and procedure. The mathematical basis given in Annex D should be used to calculate and state the uncertainties.

The standard deviation characterizing the loss uncertainty can be calculated from Equation (63), using the standard formula for the propagation of errors:

$$u_{\Delta A} = \left(u_{A_{\text{ref}}}^2 + u_{A_{\text{otdr}}}^2 \right)^{1/2} \quad \text{dB} \quad (65)$$

The uncertainty of the reference loss $u_{A_{\text{ref}}}$ should be accumulated, by root-sum-squaring, from the following contributions:

$u_{A,\text{pm}}$	the uncertainty, in decibels, due to calibrating the reference loss with the power meter, for example due to the non-linearity of the power meter, polarization dependence, spatial inhomogeneity and noise. This uncertainty should itself be accumulated by root-sum-squaring;
$u_{A,\text{refl}}$	the uncertainty, in decibels, due to changes in the reflection characteristics of the set-up. For example, the power step measured with the power meter may differ from the power step with the test device connected, due to reflections from the power meter;
$u_{A,\lambda}$	the uncertainty, in decibels, caused by the difference between the centroidal wavelengths of the OTDR and the optical source used in the calibration of the reference loss;
$u_{A,\text{deadzone}}$	the uncertainty, in decibels, caused by the possibility that the attenuation step is influenced by the recovery from attenuator reflections (attenuation dead zone);
$u_{A,\text{step}}$	the uncertainty type A, in decibels, introduced by the instability of the attenuation step.

The uncertainty $u_{A_{\text{otdr}}}$ includes the following:

$u_{A,\text{pdl}}$	the uncertainty, in decibels, introduced by changing a polarization state as a result of the attenuation step;
$u_{A,\text{type A}}$	the uncertainty type A, in decibels, introduced by the variability of the attenuation step measurements, caused for example by limited readout resolution, tilted pulse

top in the case of the "fibre-end" and "multiple-reflections" variants, and displayed power levels when they approach the noise limit.

9 Reflectance calibration

9.1 Objective

The objective of the reflectance calibration is to determine the reflectance deviation, ΔS_R between the measured and actual reflectance and to evaluate its uncertainty.

9.2 Reflectance measurements (see Figure 23)

An OTDR measures the reflectance R_{otdr} of a fibre optic component, using the amplitude of the reflection over the backscatter trace, ΔF . The reflectance R_{otdr} can be expressed by a function such as:

$$R_{\text{otdr}} \approx f(\Delta F, K, \Delta Td) \quad (66)$$

where K is the backscatter parameter of the fibre, ΔF the amplitude of the reflection and ΔTd the displayed OTDR's pulsewidth.

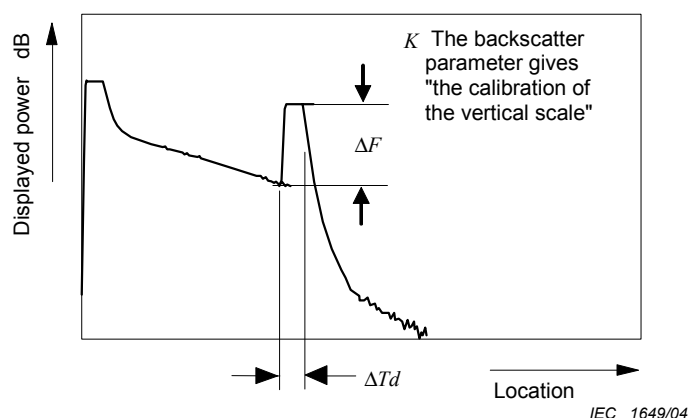


Figure 23 – Parameters involved in reflectance measurements

NOTE 1 The displayed OTDR's pulsewidth includes all parameters affecting the displayed pulsewidth; such as:

- a) the actual pulsewidth;
- b) the photodiode response;
- c) the amplifier bandwidth.

NOTE 2 It is assumed that there is no perturbation on the displayed pulse amplitude. For example, the top of the reflection is below the clipping level, and the noise of the backscatter trace before the pulse is very low with respect to the pulse amplitude.

NOTE 3 Other parameters may affect reflectance measurements e.g. displayed power level, event location, associated attenuation, and averaging time. These parameters should be included in the calibration process.

9.3 Use of the backscatter parameter, K

On a general basis the vertical scale of an OTDR is not calibrated for absolute optical power measurements. When an OTDR is used for reflectance measurements the backscatter parameter, K , effectively acts as reference level for the vertical scale. Therefore any change in K directly changes the displayed reflectance value, as shown in Figure 24. This is discussed in more detail in Annex G where it is shown that

$$\Delta F_{\text{dB}}(L) = 5 \log_{10} \left\{ \frac{R}{K \Delta T} + 1 \right\} \quad (67)$$

and

$$R_{\text{dB}} = 10 \log_{10}(K) + 10 \log_{10}(\Delta T) + 10 \log_{10} \left(10^{\frac{\Delta F_{\text{dB}}(L)}{5}} - 1 \right) \quad (68)$$

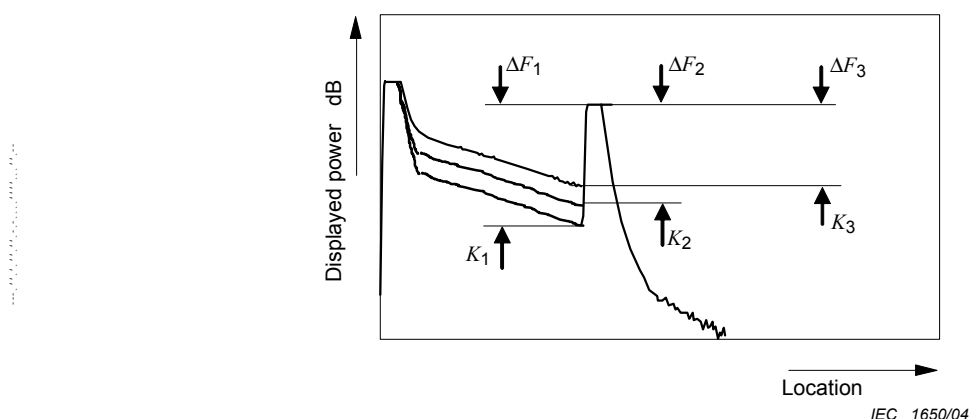


Figure 24 – The same reflectance at the end of three fibres with different values of the backscatter parameter shows different pulse amplitudes

The backscatter parameter is related to the fibre under test. This leads to the following consequences:

- a) The calibration is only valid if the OTDR uses the same backscatter parameter in measuring the test device as the backscatter parameter of the test device itself. Alternatively, the use of the calibration results shall take into account the different values of the backscatter parameter using for example Equation (68).
By using the same backscatter parameter as the one stated in the documentation of the reflection standard, the uncertainty of K is effectively eliminated in the calibration.
- b) Additional errors in using the OTDR for reflectance measurements may result from the uncertainty in determining the backscatter parameter of the fibre, K . The determination of K is beyond the scope of this standard. Consequently, the calibration procedure below only discusses the OTDR's ability to measure ΔF and compute R_{otdr} correctly.
As the fibre backscatter parameter is needed for reflectance measurements, mainly to reduce uncertainties, a method to determine K is provided in informative Annex G.

9.4 Range of reflectance measurement

For a given OTDR setting it will not be possible to measure all reflectance values. Analysis of the available range is useful to prepare the test plan. For a given OTDR setting, the pulse amplitude, ΔF is a function of the reflectance. The range (from the minimum to the maximum value) of the pulse amplitude is directly related to the available range of reflectance.

The minimum value of the pulse amplitude should be equal to zero. Generally a small minimum value, e.g. 3 dB, is applicable. This is because for small values of ΔF , small errors in this quantity cause large errors in R_{dB} , Equation G.10. Taking a minimum value of ΔF of 3 dB, the minimum reflectance that can be measured with a 1 μs pulse is ~ -45 dB and for a 10 ns pulse ~ -65 dB.

The maximum value of the pulse amplitude is limited at the top, by the clipping level, F_c , and at the bottom by the level of the backscatter trace, F , approaching the noise floor. As F depends on the attenuation of the fibre before the event, the maximum value of the pulse amplitude depends on the location of the pulse (see Figure 25). From Equation G.9 it can be seen that ΔF is minimised by using the longest pulse width.

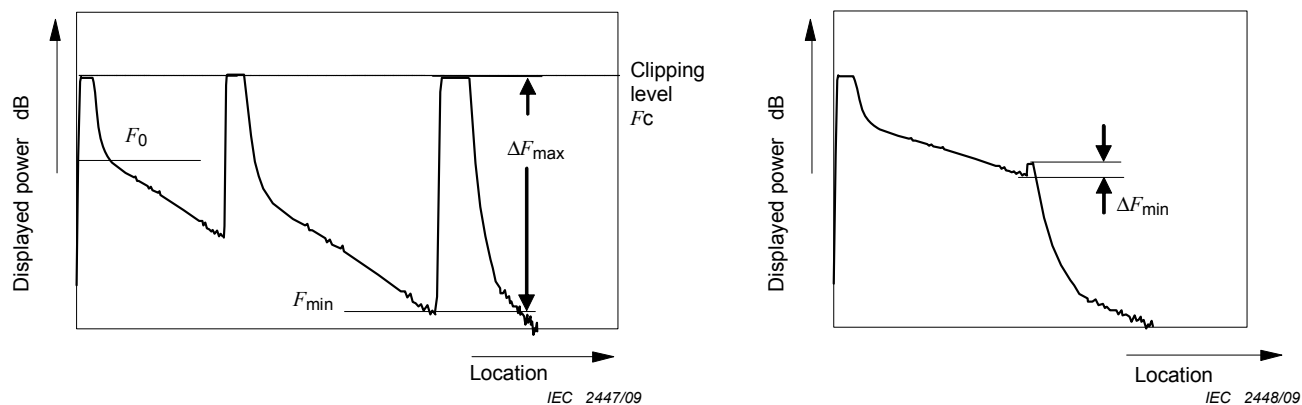
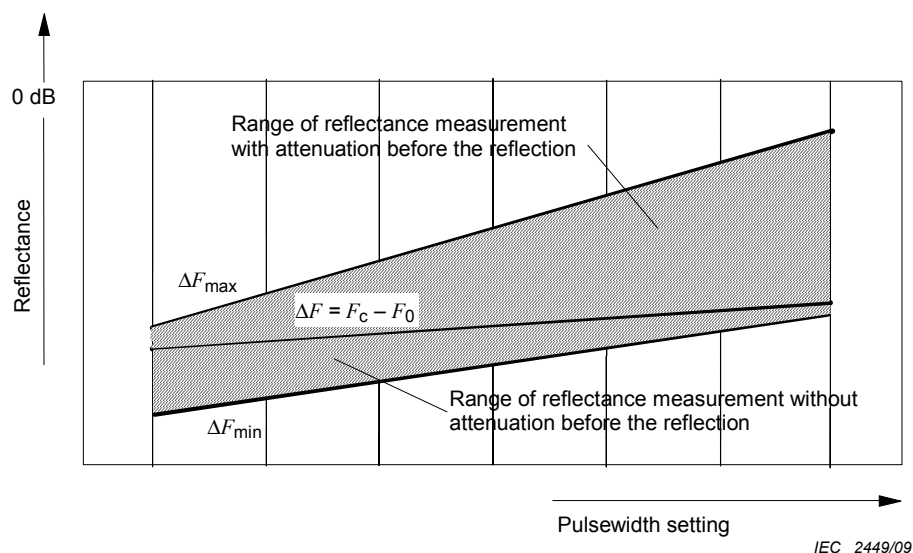


Figure 25 – Maximum and minimum values for the pulse amplitude, ΔF

NOTE Some OTDR are able to internally reduce the backscattered power. In this case, the maximum value of the pulse amplitude also depends on the internal attenuation.

The range of reflectance measurements can be illustrated by an area shown in Figure 26. The lower limit of the area is defined by the minimum value of the pulse amplitude. The maximum values of the pulse amplitude give the upper limits of the range of reflectance measurements.



NOTE Reflectance values are negative.

Figure 26 – Range of reflectance measurement

9.5 Development of a test plan

The principle of reflectance calibration is to apply a set of known (or reference) reflectances, $R_{ref,j}$, to the OTDR, and to measure the displayed reflectance, $R_{otdr,j}$, for a given OTDR setting and for different values of location and power level. $R_{ref,j}$ should be chosen as a set of regularly spaced reflectance values within the range of reflectance measurement.

This leads to a very large quantity of measurements.

To help in reducing the quantity of measurements, this standard defines default values for the location, the displayed power level and reflectance sample spacing.

For the purpose of this standard, the default values of location L_d and displayed power level F_d are defined regarding the dynamic range, as illustrated in Figure 27.

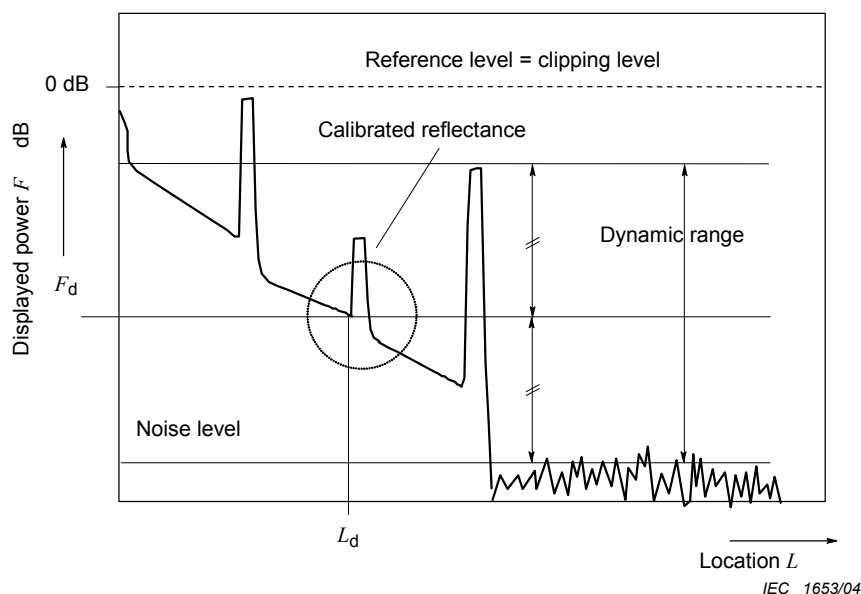


Figure 27 – Determining the default displayed power level and the default location

The default value for displayed power level F_d is defined as the half value of the dynamic range over the noise level. The location L_d , is defined in order to keep the reflection location as close as possible to middle of the A region defined in 7.3.

The default for reference reflectance sample spacing is 2 dB.

Build a test plan providing selection for appropriate reference reflectances, R_{ref} , using previously defined default values.

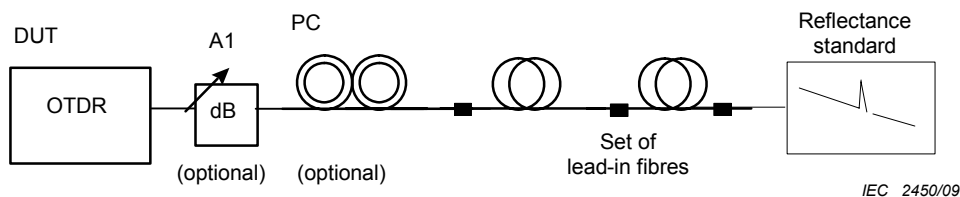
Calibration data for other values of location, displayed power level and reflectance sample spacing can be provided on a voluntary basis.

9.6 Equipment

In addition to the test OTDR, the measurement equipment includes

- a) a reflectance standard as described in Annex E or Annex F;
- b) a set of lead-in fibres. The lead-in fibres shall be type B IEC 60793-2-50 type B1.1 or B1.3;
- c) a variable attenuator, if necessary;
- d) optionally, a polarization controller.

The test set-up is shown in Figure 28.



Key

- A1 attenuator
PC polarization controller

Figure 28 – Set-up for reflectance calibration

The purpose of the lead in fibre (and the optional attenuator) is to place the reflectance standard at the displayed power level and location previously defined. The purpose of the optional polarization controller is to reduce the possible influence of the OTDR's polarization dependence.

9.7 Measurement procedure

9.7.1 Preparation

Select the appropriate combination of lead-in fibres and attenuator setting in order to place the reflectance standard within test plan conditions.

Set the backscatter parameter to the one stated in the documentation of the reflection standard. Check that the instrument setting is appropriate.

Record the displayed power level, F_d , (or F if the default value is not used) the location, L_d , (or L if the default value is not used), the used backscatter coefficient, C , and other OTDR's setting parameters.

9.7.2 Taking reflectance measurements

For each values of $R_{ref,j}$, follow the instructions of the OTDR manufacturer to take reflectance measurements of the reflectance standard. Alternatively define a measurement procedure to measure the reflectance. Annex G can be used to prepare this procedure.

If the reflectance measurement was not carried out using the manufacturer's instructions, then the used procedure has to be added in the documentation.

Record $R_{otdr,j}$ for each value of $R_{ref,j}$.

9.7.3 Calculation and results

Calculate the reflectance deviation $\Delta S_{R,j}$

$$\Delta S_{R,j} = R_{otdr,j} - R_{ref} \quad \text{dB} \quad (69)$$

9.7.4 Uncertainties

Note that the following list of uncertainties may not be complete. Additional contributions may have to be taken into account depending on the measurement set-up.

The standard deviation characterising the reflectance uncertainty, $u_{\Delta S_R}$, of an individual ΔS_R can be calculated from Equation (69), using the standard formula for propagation of uncertainties:

$$u_{\Delta S_R} \cong \frac{1}{R_{\text{ref}}} (u_{R_{\text{ref}}}^2 + u_{R_{\text{otdr}}}^2)^{1/2} \quad \text{dB/dB} \quad (70)$$

The uncertainty $u_{R_{\text{ref}}}$ can be extracted from the calibration certificate of the reflectance standard (see Annex E or F). The uncertainty $u_{R_{\text{otdr}}}$ should be accumulated, by root-sum-squaring, from the following contributions:

$u_{R, \text{type A}}$	is the reflectance uncertainty type A in dB, to be experimentally determined. It will depend on the displayed power level and OTDR averaging time;
$u_{R, l}$	is the reflectance uncertainty introduced by the uncertainty of the reflectance location, e.g. caused by the location of the cursor (s) used to measure ΔF ;
$u_{R, \lambda}$	is the uncertainty in dB caused by the difference between the centroidal wavelengths of the OTDR and the optical source used in the calibration of the reflectance standard;
$u_{R, \text{pdl}}$	is the uncertainty in dB introduced by changing the polarization state of the light reflected from the reflectance standard;
$u_{R, \text{attenuator}}$	is the loss uncertainty of the optical attenuator caused by the possibility that the backscatter signal is polarized and the attenuator may have polarization-dependent loss.

Annex A (normative)

Recirculating delay line for distance calibration

A.0 Introduction

In this annex, a fibre-type recirculating delay line to be used as calibration artefact for OTDR distance calibration is described.

A.1 Construction

As illustrated in Figure A.1, the device is constructed from the following:

- a four-port coupler, preferably with a wavelength-insensitive coupling ratio, with a long fibre (the length of which depends on the distance range to be calibrated) fusion spliced between a pair of input and output ports;
- an input fibre (typically around 1 km) which is spliced to the second input port with the input equipped with a connector;
- an output fibre which is spliced to the second output port of the coupler and which is terminated in a flat reflecting connector. The length of this fibre piece is kept short, for example <1 m. The reflection from the flat termination is sufficient to give an accurately define location.

The length of the lead-in fibre is defined by the input fibre, one branch of the coupler and the output fibre. The length of the loop is defined by the long fibre specified in item a) and the second half of the coupler. All fibres shall be single-mode according to IEC 60793-2-50.

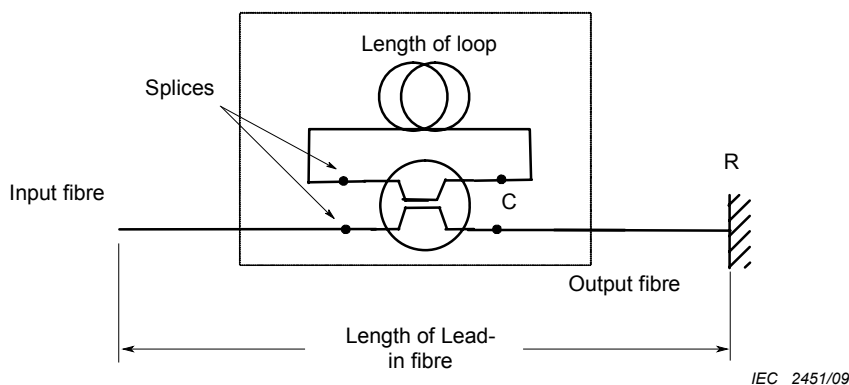


Figure A.1 – Recirculating delay line

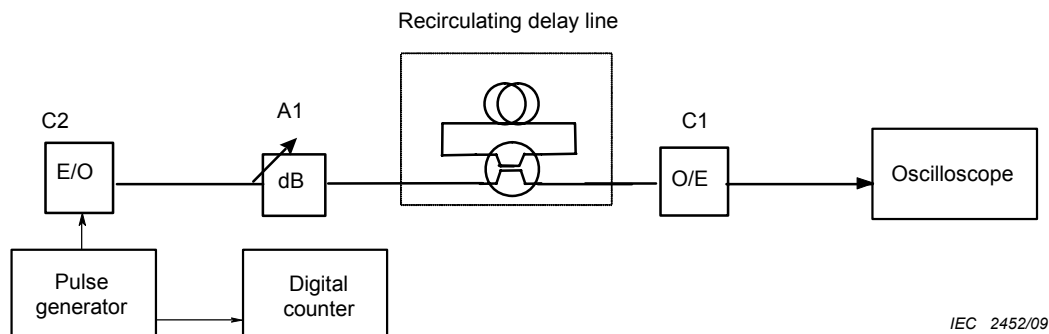
and O/E converters, an oscilloscope and an optical attenuator. The centroidal wavelength of the E/O converter shall be known.

A.1.1 Procedure

A.1.1.1 Loop transit time measurement

To determine the loop transit time, set up the system shown in Figure A.2. Set the pulse generator to give pulses of suitable width, and the attenuator to generate a suitable amplitude for the E/O converter. Set the pulse repetition rate, in kilohertz, to $200/L_b$, where L_b is the approximate length of the fibre in the loop in kilometres. View the output pulses from the O/E converter on the oscilloscope and then adjust the repetition rate until the following two pulses

are superimposed on the oscilloscope trace: first, the optical pulse which has been transmitted directly from the E/O converter to the O/E converter without going round the loop, and second, the fraction of the preceding optical pulse which has travelled once round the loop are detected together. Record the repetition period with the digital counter. This is the loop transit time T_b . Estimate the uncertainty σ_{T_b} by introducing small changes to the repetition rate.



Key

- A1 Attenuator
- C1 optical-to-electrical converter
- C2 electrical-to-optical converter

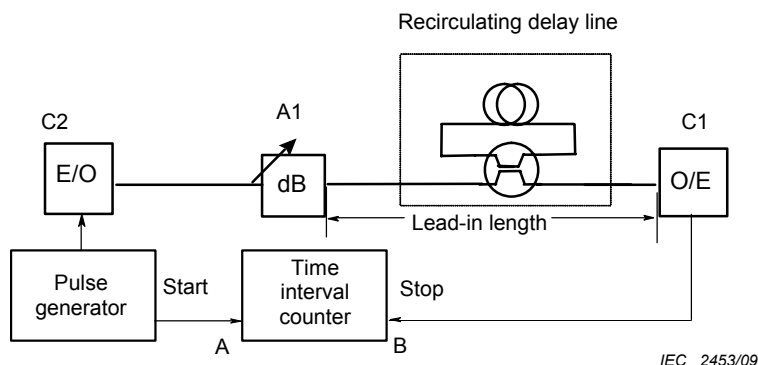
Figure A.2 – Measurement set-up for loop transit time T_b

A.1.1.2 Lead-in transit time measurement

To calibrate the transit time of the lead-in fibres, set up the system shown in Figure A.3. Adjust the pulse width, amplitude and repetition rate (approximately 1 kHz is suggested) of the pulse generator to suitable values and provide a trigger signal for the digital counter. It may be necessary to delay the output pulse to the E/O converter with respect to this trigger pulse as some counters have a "dead time" between start and stop trigger. Adjust the trigger level of channel B of the counter so that it is just reliably triggered by the output of the O/E converter; this is to prevent the counter being triggered by smaller pulses which have made one or more circuits of the loop. Record the transit time T_1 displayed by the counter. Temporarily disconnect the O/E converter from the counter, connect to an oscilloscope, view the directly transmitted pulses and note their amplitude.

Then remove the recirculating delay line and connect the O/E converter directly to the attenuator. Again view the pulses on the oscilloscope. Adjust the attenuator to give pulses of the same amplitude as before. Reconnect the O/E converter to the counter and measure the transit time T_2 . The transit time of the lead-in fibre T_a is then given by

$$T_a = T_1 - T_2 \quad (\text{A.1})$$



Key

- A1 Attenuator
- C1 optical-to-electrical converter
- C2 electrical-to-optical converter

Figure A.3 – Calibration set-up for lead-in transit time T_a

A.2 Uncertainties

The guidelines of the mathematical basis given in Annex D should be used to calculate and state the uncertainties.

The uncertainty of the loop transit time u_{T_b} should be accumulated, by root-sum-squaring, from the following contributions:

- $u_{T_b, \text{counter}}$ the time uncertainty, in seconds, due to the digital counter, as caused by clock frequency uncertainty and time interval resolution;
- $u_{T_b, \text{adjust}}$ the time uncertainty, in seconds, due to adjusting the repetition rate with the help of the oscilloscope;
- $u_{T_b, \lambda}$ the time uncertainty, in seconds, due to the uncertainty of the E/O converter's centroidal wavelength; it can be calculated by multiplying the wavelength uncertainty by the fibre length L_a and the chromatic dispersion;
- $u_{T_b, \Theta}$ the time uncertainty due to the fibre's temperature coefficient; typical value: 1 cm/(km °C) within the allowable temperature range.

The uncertainty of the lead-in transit time u_{T_a} should be accumulated, by root-sum-squaring, from the following contributions:

- $u_{T_a, \text{counter}}$ the time uncertainty, in seconds, due to the digital counter, as caused by clock frequency uncertainty, time interval resolution and setting the trigger amplitude;
- $u_{T_a, \text{type A}}$ the time uncertainty type A, in seconds, for example as caused by timing jitter; it can be obtained from a series of successive counter readings;
- $u_{T_a, \lambda}$ the time uncertainty, in seconds, due to the uncertainty of the E/O converter's centroidal wavelength; it can be calculated by multiplying the wavelength uncertainty by the fibre length L_a and the chromatic dispersion;
- $u_{T_a, \Theta}$ the time uncertainty due to the fibre's temperature coefficient; typical value: 1 cm/(km °C) within the allowable temperature range.

Additional contributions may have to be taken into account, depending on the measurement set-up and procedure.

A.3 Documentation

The following calibration results shall be supplied with the recirculating delay line:

- a) approximate lengths of lead-in fibre and loop;
- b) measured transit times of lead-in fibre and loop;
- c) centroidal wavelength of the E/O converter;
- d) time uncertainties $\pm 2 u_{Ta}$ and $\pm 2 u_{Tb}$ as calculated according to A.3.

Annex B (normative)

Optical fibre standard for loss calibration

B.0 Introduction

In this annex, a fibre-type calibration standard is defined for OTDR loss calibration with the fibre standard method.

B.1 Fibre requirements

The fibre shall be a single-mode fibre type B according to IEC 60793-2-50 and ITU-T Recommendations G.650.1 and G.650.2, with the following additional requirements:

- | | | |
|--|--------------------------|-------------------|
| – attenuation coefficient at 1 310 nm | α_{1310} | $\leq 0,40$ dB/km |
| – attenuation coefficient at OH-peak 1 385 nm | α_{1385} | $\leq 1,00$ dB/km |
| – attenuation coefficient at 1 550 nm | α_{1550} | $\leq 0,25$ dB/km |
| – bend loss at 1 550 nm using 60 mm diameter mandrel and 100 turns | ΔA_{bend} | $\leq 0,20$ dB |
| – total length of fibre | L | ≥ 3 km |

B.2 Suitability check of the fibre

B.2.1 General

The selection of a suitable fibre for the fibre standard is outlined in B.2.3 and B.2.4. The tests ensure that the fibre's backscatter trace is

- a) essentially independent of the direction of the signal flow – this ensures that the backscatter trace can be used as a measure for attenuation;
- b) essentially linear – this ensures that the backscatter loss and, in conjunction with a), the attenuation are linearly dependent on the length of fibre.

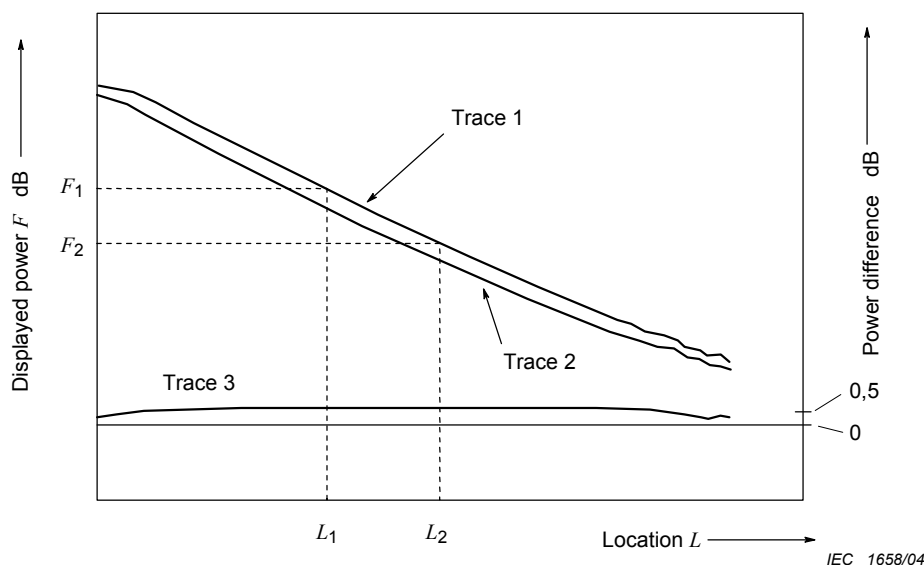
These tests are to be carried out with a selected OTDR which has a sufficiently long linear region of power levels. The linearity test is described in B.2.2.

B.2.2 Linearity test of the OTDR power scale

In order to check the longitudinal uniformity of the fibre standard, make sure that the range of the OTDR power scale used for characterizing the fibre standard is sufficiently linear. Determine this highly linear range by the following method:

- a) measure the OTDR trace of a single-mode optical fibre which is much longer than the fibre standard (trace 1 in Figure B.1). No specific requirements have to be met for this fibre;
- b) re-measure the same fibre for a reduced (for example, 0,5 dB) input power. For example, this can be achieved by adding an attenuation step using a variable optical attenuator (trace 2 in Figure B.1);
- c) subtract trace 2 from trace 1 to obtain trace 3 (Figure B.1);
- d) identify a region on trace 3, with the smallest possible slope and a length which is at least the length of the fibre standard;

- e) the highly linear power range is defined by the distance between L_1 and L_2 and by the corresponding displayed power levels F_1 and F_2 of trace 1 in Figure B.1.



NOTE It is expected that the OTDR polarization dependence has no influence on the determination of the highly linear power range; if adding a small attenuation step with a variable attenuator does not change the polarization state, then the OTDR polarization dependence will introduce the same perturbation to both traces 1 and 2, with no perturbation on trace 3.

Figure B.1 – Determination of a highly linear power range

B.2.3 Directional dependence of the backscatter loss

Connect the total length of fibre to the OTDR. Adjust the OTDR pulse width, for example to 1 μ s, so that a sufficiently large number of data points and an appropriate signal-to-noise ratio are obtained. Determine the backscatter traces from both fibre ends with alternating OTDR measurements, in order to average out any effects from the OTDR polarization dependence. It is preferable to insert a polarization controller between the OTDR and the fibre. This allows the OTDR polarization dependence to be cancelled out in a more systematic way, as described in 8.1.

For each direction A and B, calculate the average backscatter traces $F_A(L)$ et $F_B(L)$. Add the two backscatter traces so that the backscatter signals from the same location coincide:

$$\Delta A_{AB}(L) = F_A(L) + F_B(L_{\text{total}} - L) \quad (\text{B.1})$$

Analyse only the centre section of 2 km to 3 km, to avoid influences from the fibre ends. The difference between the maximum and the minimum of $\Delta A_{AB}(L)$ shall be stated. This ensures that the backscatter trace can be used as a measure for attenuation, as outlined above. Note that the accuracy of this test depends on the non-linearity measurement results from B.2.2.

B.2.4 Longitudinal attenuation uniformity of the fibre

This test should be carried out from one side only. Adjust the OTDR pulse width to the pulse width used in B.2.3, for example to 1 μ s. Make sure that the backscatter trace lies within the highly linear power range determined in B.2.2, for example with a lead-in fibre of appropriate length. For best accuracy, insert a polarization controller, measure the backscatter trace at several states of polarization, average the thereby obtained traces and determine the regression line from the averaged trace (see Figure B.2). Calculate the deviation ΔA_{reg} of the averaged trace from the regression line. Also, make sure that the uncertainty type A in this measurement does not contribute significantly to the value of ΔA_{reg} .

Note that testing longitudinal attenuation uniformity usually requires measuring the backscatter trace from both ends and calculating an averaged backscatter trace. This is not necessary here because the test in B.2.3 ensures nearly equal backscatter traces.

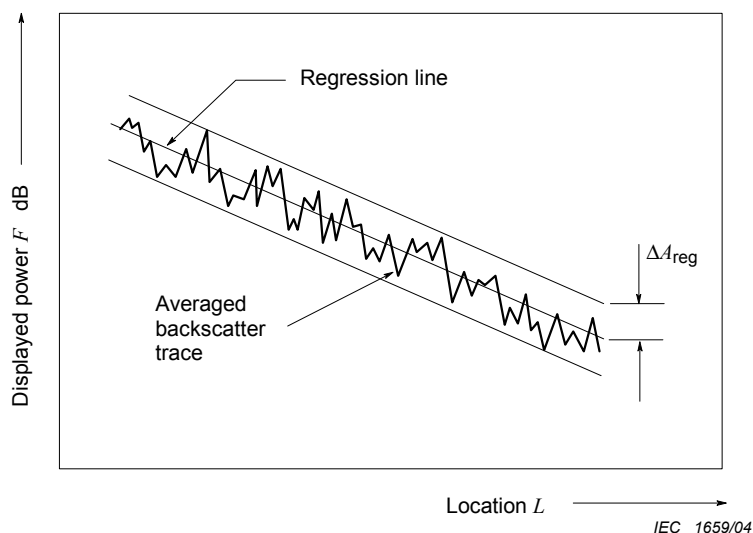


Figure B.2 – Testing the longitudinal backscatter uniformity of the fibre standard

Record the peak deviation ΔA_{reg} of the trace from the regression line as an estimate of the longitudinal attenuation uniformity. This ensures that the attenuation is linearly dependent on the length of fibre, as outlined above.

B.3 Preparation and calibration of the fibre standard

To prepare the fibre standard for use, transport and storage, proceed as follows:

- cut a centre section having a length of 2 km to 3 km from the total length of fibre;
- protect the fibre suitably for transportation and handling (for example by a loose tube, secondary coating or a cable);
- wind the fibre loosely so that no additional loss will occur from winding and transportation.

If the fibre standard is intended to be used as a mobile calibration tool and frequent shipment must be considered, its attenuation stability against temperature cycling should be tested referring to IEC 60794-1-2, method F1. The measurement results shall be identical with the initial values within the limits of the measurement accuracy. If this is not the case, then the temperature cycle should be repeated until no change in the attenuation at 23 °C is observed; it may be necessary to repeat the procedures of Clause B.2 to check that the fibre is still suitable.

Determine the fibre attenuation $A(\lambda)$ within the spectral ranges of 1 285 nm to 1 330 nm and 1 530 nm to 1 570 nm, for example with a filtered white-light source and in steps of 2 nm, referring to IEC 60793-1-40 (cut-back method). A fibre loop with $r = 30$ mm is integrated in the reference fibre length in order to avoid the propagation of higher modes. The temperature shall be stabilized within $23\text{ °C} \pm 3\text{ °C}$ during the measurement. Record the spectral width of the light source $\Delta\lambda_{\text{FWHM}}$.

Prepare the fibre with single-mode connectors on both ends. Alternatively, splicing of pigtails is permitted up to a maximum length of 5 m each.

B.4 Recalibration of the optical fibre standard

Remove the single-mode connectors or pigtails from the optical fibre standard. For the recalibration, use the attenuation procedure in Clause B.3. The attenuation recalibration result should not differ from the first calibration result by more than ± 4 times $u_{A, \text{std}}$ specified in Clause B.5 (this includes ageing).

B.5 Uncertainty of the fibre standard

The guidelines of the mathematical basis given in Annex D should be used to calculate and state the uncertainties.

The uncertainty of the fibre standard $u_{A, \text{std}}$ should be accumulated by root-sum-squaring from the following contributions:

$u_{A, \text{cutback}}$	the loss uncertainty due to the repeatability of the cut-back measurement;
$u_{A, \lambda}$	the loss uncertainty due to the uncertainty of the centroidal wavelength of the light source used in the attenuation measurement;
$u_{A, \text{meter}}$	the loss uncertainty due to the non-linearity of the power meter used in the cut-back measurement;
$u_{A, \theta}$	the loss uncertainty, in decibels, caused by the temperature dependence within the allowable temperature range.

Additional contributions may have to be taken into account, depending on the measurement set-up and procedure.

B.6 Documentation

The fibre standard shall be accompanied by the following documentation:

- approximate length of the fibre;
- fibre attenuation for the spectral ranges of 1 285 nm to 1 330 nm and 1 530 nm to 1 570 nm, in steps of 2 nm;
- spectral width of the light source used for the measurement;
- allowable temperature range ($23\text{ °C} \pm 3\text{ °C}$, as specified before);
- attenuation uncertainty in the expanded form $\pm 2 u_{A, \text{std}}$, as calculated in Clause B.5;
- longitudinal attenuation uniformity in the form of ΔA_{reg} , as described in B.2.4.
- bidirectional dependence of the backscatter loss in the form of ΔA_{AB} as described in B.2.3.

Annex C
(normative)

Standard splice simulator for loss calibration

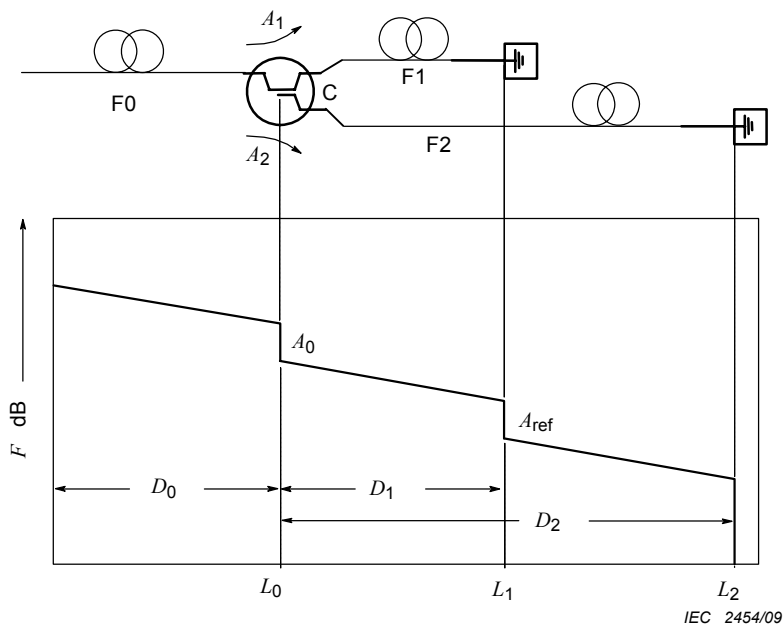
C.1 Structure

C.1.1 Description

The simulator is made with three fibres F0, F1 and F2, the length of which are D_0 , D_1 , D_2 . A wavelength-independent branching device C connects them.

The simulator and its theoretical backscatter signature are shown in Figure C.1. The small circles at the end of fibres F1 and F2 indicate devices which eliminate reflections.

From L_0 to L_1 the displayed backscatter signal is the contribution of the two fibres F1 and F2. After L_1 , only the backscatter signal from fibre F2 is displayed. The standard splice loss used for OTDR calibration is A_{ref} . This splice appears at the end of the fibre F1, at location L_1 .



Key

- | | |
|----------------------------|--|
| F0, F1 and F2 | fibres |
| D_0 , D_1 and D_2 | length of each fibres |
| L_0 , L_1 and L_2 | location of the end of each fibres |
| C | branching device |
| A_1 and A_2 | attenuation of each branch's of the branching device |
| A_0 and A_{ref} | excess loss of branching device and reference loss |

Figure C.1 – Splice simulator and idealized OTDR signature

C.1.2 Splice loss analysis

Assuming that the fibres have the same nature and that the attenuation of the branching device is the same in both directions, the theoretical description of the device is

$$A_1 = -10 \log X_1 \quad X_1 = 10^{-A_1/10} \quad (\text{C.1})$$

$$A_2 = -10 \log X_2 \quad X_2 = 10^{-A_2/10} \quad (\text{C.2})$$

A_1 is the branching device attenuation from F0 to F1 (which is the same as the branching device attenuation from F1 to F0). A_2 is the branching device attenuation from F0 to F2 (which is the same as the branching device attenuation from F2 to F0). Equivalently, X_1 and X_2 denote the coupling ratios.

The excess loss A_0 and the reference loss A_{ref} are given by the following equations; note that due to the OTDR round trip correction, the logarithmic conversion uses a factor of 5 instead of 10:

$$A_0 = -5 \log (X_1^2 + X_2^2) \quad (\text{C.3})$$

$$A_{\text{ref}} = -5 \log \left(\frac{X_2^2}{X_1^2 + X_2^2} \right) = 5 \log \left(1 + \frac{X_1^2}{X_2^2} \right) \quad (\text{C.4})$$

A_{ref} can also be expressed with A_1 and A_2 :

$$A_{\text{ref}} = 5 \log \left(1 + 10^{(A_2 - A_1)/5} \right) \quad (\text{C.5})$$

This equation defines the reference loss.

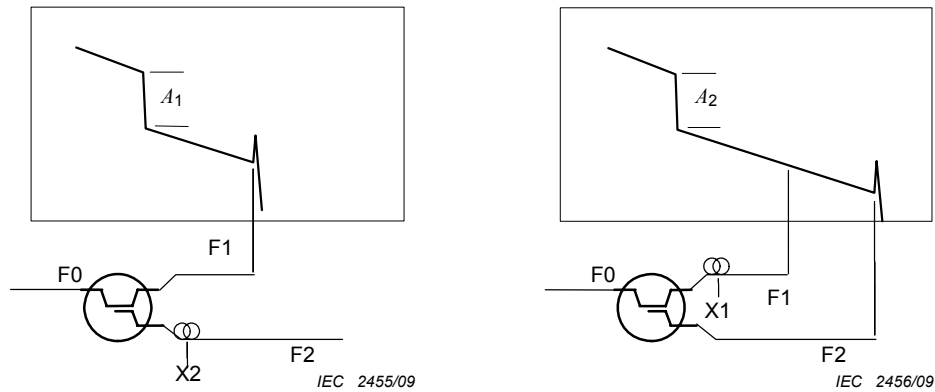
C.2 Preparation of the standard splice simulator

To prepare the standard splice simulator:

- a) select a spool of single-mode fibre according to IEC 60793-2-50;
- b) cut fibres F0, F1 and F2 from this same spool, with respective length D_0 , D_1 and D_2 , for example of about 1 000 m, 2 000 m and 4 000 m;
- c) wind the fibres loosely on different spools;
- d) select a 1×2 achromatic branching device with single-mode fibre pigtails and low wavelength and polarization dependence. Typical requirements are:
 - 1) coupling ratio = 50 % $\pm$ 1,5 % from 1 290 nm to 1 330 nm and 1 530 nm to 1 570 nm;
 - 2) polarization dependence of coupling ratio less than $\pm$ 0,5 %;
- e) splice the three fibres to the branching device;
- f) add a connector to the input of the splice simulator;
- g) place and fix the three fibre spools and the branching device in a rugged transit case to protect the standard from shock and vibration;
- h) if the splice simulator is intended to be used as a mobile calibration tool and frequent shipment must be considered, its splice loss stability against temperature cycling should be tested referring to IEC 60794-1-2, method F1. The measurement results shall be identical with the initial values within the limits of the measurement accuracy. If this is not the case, then the temperature cycle shall be repeated until no change in the attenuation at 23 °C is observed.

C.3 Calibration procedure

An OTDR is used for calibration: the calibrated value A_{ref} is only dependent on the coupling ratio of the branching device. When the coupling ratio is 50/50, then the backscatter signals due to each branch (F1 and F2) are equal. This phenomenon can be demonstrated by excluding one branch then the other, using an appropriate device to bend the fibre (see Figure C.2).



Key

F0, F1 and F2	fibres
X1 and X2	locations after the branching device
A_1 and A_2	attenuation of each branch of the branching device

Figure C.2 – Determination of the reference loss A_{ref}

To calibrate the splice simulator, proceed as follows:

- exclude branch F2 using an appropriate device to bend the fibre, at point X2; the necessary degree of bending can be determined by searching for the maximum achievable attenuation A_1 ;
- measure the attenuation A_1 with the OTDR;
- exclude branch F1 using an appropriate device to bend the fibre, at point X1;
- measure the attenuation A_2 with the OTDR. Be careful that the two attenuation measurements are made in the same condition (for example, rms slope measurement with the same points for start and stop, and at the same displayed power level).

The reference loss is obtained from Equation (C.5):

$$A_{\text{ref}} = 5 \log \left(1 + 10^{(A_2 - A_1)/5} \right)$$

Note that no specific requirements concerning the accuracy of the OTDR power scale have to be met, because the result of this computation only depends on the capability of the OTDR of measuring the small difference between A_1 and A_2 correctly. For an ideal coupler, $A_1 = A_2$, and $A_{\text{ref}} = 1,505$ dB.

C.4 Uncertainties

The guidelines of the mathematical basis given in Annex D should be used to calculate and state the uncertainties. Calculate the uncertainties using the following.

At the moment of the calibration, the losses A_1 and A_2 are nearly the same:

$$A_2 = A_1 + \mathcal{E} \quad (\text{C.6})$$

$$A_{\text{ref}} = -5 \log \left(1 + 10^{-\mathcal{E}/5} \right) \quad (\text{C.7})$$

When $\mathcal{E}$ approaches zero:

$$\partial A_{\text{ref}} \equiv \frac{1}{2} \partial \mathcal{E} \quad (\text{C.8})$$

$$\partial A_{\text{ref}} = \frac{1}{2} \left(\partial A_1 + \partial A_2 \right) \equiv \frac{2}{2} \partial A_1 \quad (\text{C.9})$$

Note that A_1 and A_2 are linearly added as they are correlated. The result is that the uncertainty of the reference loss is identical with the uncertainty of the branching loss:

$$u_{A_{\text{ref}}} = u_{A_1} \quad (\text{C.10})$$

The uncertainty u_{A_1} should be accumulated, by root-sum-squaring, from the following uncertainties of the branching loss:

$u_{A,\lambda}$	the loss uncertainty, in decibels, caused by the uncertainty of the OTDR's centroidal wavelength;
$u_{A,L}$	the linearity uncertainty, in decibels, caused by the non linearity of the OTDR used to measure A_1 and A_2 ;
$u_{A,\theta}$	the loss uncertainty, in decibels, caused by the temperature dependence within the allowable temperature range;
$u_{A,\text{coupler pdl}}$	the loss uncertainty, in decibels, introduced by the coupler's polarization dependence (PDL) and the fact that the incident wave is polarized. IEC 61300-3-2 discusses PDL measurements;
$u_{A,\text{otdr pdl}}$	the uncertainty of A_1 (or A_2), in decibels, caused by the OTDR's polarization dependence (PDL) and the fact that the backscatter signal is polarized. This uncertainty can be reduced by inserting a polarization controller between the OTDR and the splice loss simulator, measuring A_1 and A_2 at different states of polarization and averaging the results;
$u_{A,\text{uni}}$	the branching loss uncertainty caused by the fibre's non-uniformity.

C.5 Documentation

The splice simulator shall be accompanied by the following documentation:

- approximate fibre lengths D_0 , D_1 and D_2 ;
- approximate distance of the reference loss from the input connector of the splice simulator $D_0 + D_1$; this distance will be used as splice location;
- reference loss A_{ref} ;
- centroidal wavelength used for the calibration;
- allowable temperature range;
- uncertainty of the reference loss as calculated according to Clause C.4.

Annex D (normative)

Mathematical basis

D.0 Introduction

This annex summaries the form of evaluating, combining and reporting the uncertainty of measurement. It is based on the "Guide to the expression of uncertainty in measurement" (GUM) [12]. It does not relieve the need to consult this guide for more advice.

This standard distinguishes two types of evaluation of uncertainty of measurement. Type A is the method of evaluation of uncertainty by the statistical analysis of a series of measurements on the same measurand. Type B is the method of evaluation of uncertainty based on other knowledge.

D.1 Type A evaluation of uncertainty

The type A evaluation of standard uncertainty can be applied when several independent observations have been made for a quantity under the same conditions of measurement.

For a quantity X estimated from n independent repeated observations X_k , the arithmetic mean is:

$$\bar{X} = \frac{1}{n} \sum_{k=1}^n X_k \quad (\text{D.1})$$

This mean is used as the estimate of the quantity, that is $x = \bar{X}$. The experimental standard deviation of the observations is given by:

$$s(X) = \left[\frac{1}{n-1} \sum_{k=1}^n (X_k - \bar{X})^2 \right]^{1/2} \quad (\text{D.2})$$

where

$\bar{X}$ is the arithmetic mean of the observed values;

X_k are the measurement samples of a series of measurements;

n is the number of measurements; it is assumed to be large, for example, $n \geq 10$.

The type A standard uncertainty $u_{\text{typeA}}(x)$ associated with the estimate x is the experimental standard deviation of the mean:

$$u_{\text{typeA}}(x) = s(\bar{X}) = \frac{s(X)}{\sqrt{n}} \quad (\text{D.3})$$

D.2 Type B evaluation of uncertainty

The type B evaluation of standard uncertainty is the method of evaluating the uncertainty by means other than the statistical analysis of a series of observations. It is evaluated by scientific judgement based on all available information on the variability of the quantity.

If the estimate x of a quantity X is taken from a manufacturer's specification, calibration certificate, handbook, or other source and its quoted uncertainty $U(x)$ is stated to be a multiple k of a standard deviation, the standard uncertainty $u(x)$ is simply the quoted value divided by the multiplier.

$$u(x) = U(x) / k \quad (\text{D.4})$$

If only upper and lower limit $X_{\max}$ and $X_{\min}$ can be estimated for the value of the quantity X (for example a manufacturer's specifications or a temperature range), a rectangular probability distribution is assumed, the estimated value is

$$x = \frac{1}{2}(X_{\max} + X_{\min}) \quad (\text{D.5})$$

and the standard uncertainty is

$$u(x) = \frac{1}{2\sqrt{3}}(X_{\max} - X_{\min}) \quad (\text{D.6})$$

The contribution to the standard uncertainty associated with the output estimate y resulting from the standard uncertainty associated with the input estimate x is

$$u(y) = c \times u(x) \quad (\text{D.7})$$

where c is the sensitivity coefficient associated with the input estimate x , that is the partial derivative of the model function $y(x)$, evaluated at the input estimate x .

$$c = \frac{\partial y}{\partial x} \quad (\text{D.8})$$

The sensitivity coefficient c describes the extent to which the output estimate y is influenced by variations of the input estimate x . It can be evaluated by Equation (D.8) or by using numerical methods, that is by calculating the change in the output estimate y due to a change in the input estimate x from a model function. Sometimes it may be more appropriate to find the change in the output estimate y due to the change of x from an experiment.

D.3 Determining the combined standard measurement uncertainty

The combined standard measurement uncertainty is used to collect a number of individual uncertainties into a single number. The combined standard measurement uncertainty is based on statistical independence of the individual uncertainties; it is calculated by root-sum-squaring all standard uncertainties obtained from type A and type B evaluation:

$$u_c(y) = \sqrt{\sum_{i=1}^n u_i^2(y)} \quad (\text{D.9})$$

where

i is the current number of individual contribution;
 $u_i(y)$ are the standard uncertainty contributions;
 n is the number of uncertainties.

NOTE It is acceptable to neglect uncertainty contributions to this equation that are smaller than 1/10 of the largest contribution, because squaring them will reduce their significance to 1/100 of the largest contribution.

When the quantities above are to be used as the basis for further uncertainty computations, then the combined standard measurement uncertainty, u_c , can be re-inserted into Equation (D.9). Despite its partially type A origin, u_c should be considered as describing an uncertainty of type B.

D.4 Reporting

In calibration reports and technical data sheets, combined standard measurement uncertainties shall be reported in the form of expanded measurement uncertainties, together with the applicable level of confidence. Correction factors or deviations shall be reported. The expanded measurement uncertainty U is obtained by multiplying the standard uncertainty $u_c(y)$ by a coverage factor k :

$$U = k \times u_c(y) \quad (D.10)$$

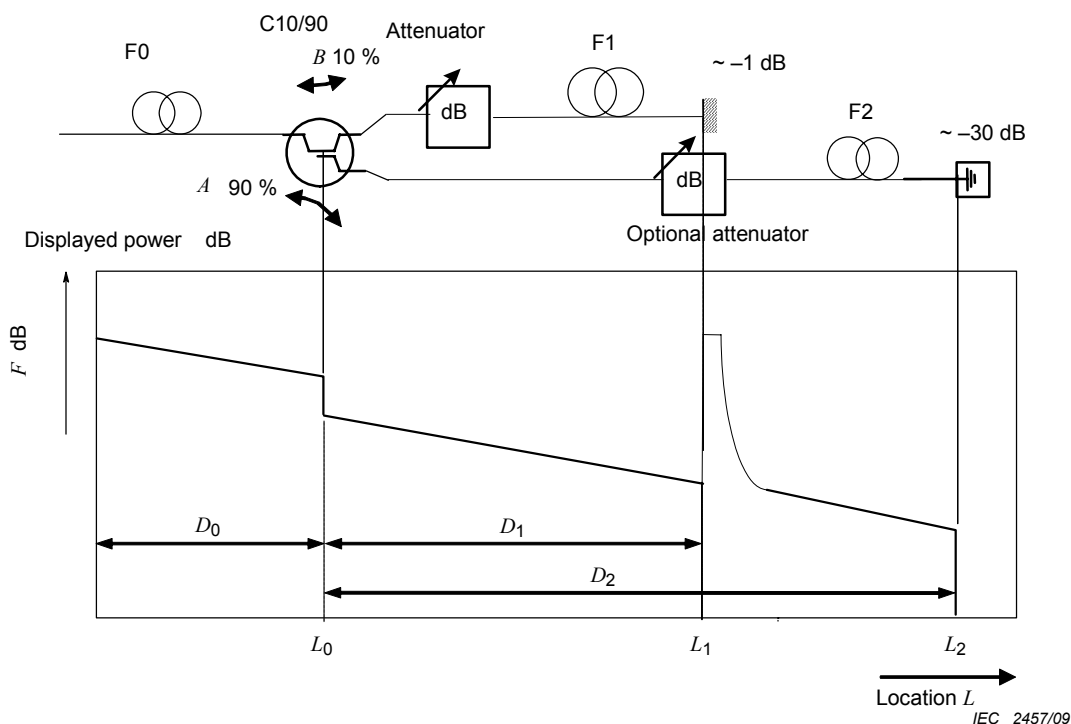
For a coverage probability of approximately 95 %, the default level, then $k = 2$. If a level of confidence of approximately 99 % is chosen, then $k = 3$. The above values for k are valid under some conditions, see GUM [12]; if these conditions are not met, larger coverage factors are to be used to reach these levels of confidence.

Annex E (normative)

Reflectance standard

E.1 Description

The reflectance standard is composed of a four port coupler, two fibres F1 and F2, having lengths D_1 and D_2 with identical or similar optical characteristics, and a variable attenuator (Figure E.1). The fibre F1 is terminated with a high reflector and the fibre F2 with a low reflection. Typical fibre lengths for D_1 , D_2 would be 1 km and 2 km. The coupler should have a split ratio $\sim 90:10$ with low wavelength and polarization dependence. The coupler shall be connected to give 90 % coupling ratio to fibre F2. The OTDR trace produced by such a device is also shown in Figure E.1.



Key

F0, F1 and F2	fibres
D_0 , D_1 and D_2	length of each fibre
L_0 , L_1 and L_2	location of the end of each fibres
C10/90	branching device
A and B	attenuation of each branch of the branching device

NOTE Optionally, a non reflective attenuator is added on fibre F2 at location L_1 , to associate an attenuation with the reference reflectance.

Figure E.1 – Reflectance standard description and trace

E.2 Reflectance analysis

NOTE 1 Following analysis includes both linear and logarithm expressions of different parameters. Appropriate care should be applied using following formulas. The analysis is based on that of reference [14] of the bibliography.

Let the CW input power to the branching device be P_0 . Then the backscattered power returned to the input coupler port, P_{bs} is:

$$P_{bs} = P_0 \left\{ \frac{AK}{\alpha v} (1 - e^{-2\alpha D_2}) + \frac{BK}{\alpha v} (1 - e^{-2\alpha D_1}) \right\} \text{ W} \quad (\text{E.1})$$

where

α is the fibre attenuation in m^{-1} ;

v is the velocity of light in the fibre ($\Rightarrow c / N$);

A is the two way transmission coefficient of the coupler for the 90 % arm;

B is the two way transmission coefficient of the coupler and attenuator in the 10 % arm;

K is the fibre backscatter parameter,

B is composed of two parts

$$B = B_0 10^{-AT/5} \text{ dB} \quad (\text{E.2})$$

Where B_0 is the two way coupler and attenuator losses when the attenuator is set to 0 dB and AT is the attenuator setting in dB. (AT is taken as being positive.)

The power reflected, P_ρ , from the highly reflecting termination of reflectivity ρ is

$$P_\rho = P_0 \rho B e^{-2\alpha D_1} \text{ W} \quad (\text{E.3})$$

In pulsed input condition, and assuming a rectangular pulse, the backscatter power just before the reflection peak, $P_{bs,D1}$, is

$$P_{bs,D1} = P_0 \frac{K}{\alpha v} e^{-2\alpha D_1} (e^{\alpha v \Delta T} - 1) (A + B) \text{ W} \quad (\text{E.4})$$

where ΔT is the OTDR pulse width. The power at the peak, P_P , is the sum of the reflected and backscatter powers

$$P_P = P_0 \rho B e^{-2\alpha D_1} + P_{bs,D1} \text{ W} \quad (\text{E.5})$$

Hence the height of the peak above the backscatter, ΔF is

$$\Delta F = 5 \log \left\{ 1 + \frac{\rho B \alpha v}{K(A+B)} (e^{\alpha v \Delta T} - 1)^{-1} \right\} = 5 \log \left\{ 1 + \frac{\rho B}{A} \frac{\alpha v}{K} \frac{(e^{\alpha v \Delta T} - 1)^{-1}}{(1+B/A)} \right\} \text{ dB} \quad (\text{E.6})$$

$$= 5 \log \left\{ 1 + \frac{\rho B}{A} \frac{\alpha v \Delta T}{K \Delta T (1+B/A) (e^{\alpha v \Delta T} - 1)} \right\} \text{ dB} \quad (\text{E.6a})$$

as $\lim_{x \rightarrow 0} \left(\frac{e^x - 1}{x} \right) = 1$, when $\alpha v \Delta T \ll 1$ then

$$\Delta F = 5 \log \left\{ 1 + \frac{\rho B}{A K \Delta T (1+B/A)} \right\} \text{ dB} \quad (\text{E.6b})$$

NOTE 2 The error is less than 0,04 dB at $\Delta T = 1 \mu\text{s}$ and $\alpha = 0,35 \text{ dB/km}$.

Thus the coupler/attenuator/reflector appears to the OTDR as a reflective event with a reflectivity of $\rho B/(A(1 + B/A))$ (see Equation (G.9)) and hence a reflectance, R of:

$$R = 10 \log \left\{ \rho B / (A(1 + B/A)) \right\}$$

$$R \approx 10 \log \left\{ \frac{\rho B_0}{A} \right\} - 2AT - 10 \log \left(1 + \frac{B}{A} \right) \text{ dB} \quad (\text{E.7})$$

Since $B/A < 0,01$ the last term in Equation (E.7) can usually be neglected (error is less than 0,04 dB) and

$$R \approx 10 \log \left\{ \frac{\rho B_0}{A} \right\} - 2AT \text{ dB}$$

$$R \approx 10 \log \{ \rho_{\text{ref}} \} - 2AT \text{ dB} \quad (\text{E.8})$$

From Equations (E.1) and (E.3), the ratio of the reflected to the CW backscattered power with attenuation set to 0 dB is

$$\frac{P_{\rho,0}}{P_{\text{bs},0}} = \frac{P_0 \rho B_0 e^{-2\alpha D_1}}{P_0 \frac{K}{\alpha \nu} \left\{ A(1 - e^{-2\alpha D_2}) + B_0(1 - e^{-2\alpha D_1}) \right\}}$$

$$= \frac{\rho B_0}{A} \frac{\alpha \nu}{K} \frac{e^{-2\alpha D_1}}{\left\{ (1 - e^{-2\alpha D_2}) + B_0(1 - e^{-2\alpha D_1}) / A \right\}} \quad (\text{E.9})$$

and the ratio of the backscattered power from the 10 % port to the 90 % port, $P_{B/A,0}$, is

$$P_{B/A,0} = \frac{B_0}{A} \frac{(1 - e^{-2\alpha D_1})}{(1 - e^{-2\alpha D_2})} \quad (\text{E.10})$$

hence

$$\frac{B_0}{A} = P_{B/A,0} \frac{(1 - e^{-2\alpha D_2})}{(1 - e^{-2\alpha D_1})} \quad (\text{E.11})$$

Substitute for B_0/A in (E.8) yields

$$\frac{\rho B_0}{A} = \frac{P_{\rho,0}}{P_{\text{bs},0}} \frac{K}{\alpha \nu} e^{2\alpha D_1} (1 - e^{-2\alpha D_2}) (1 + P_{B/A,0}) \quad (\text{E.12})$$

All of the quantities on the right hand side of Equation (E.12) are measurable; α , D_1 and D_2 can be measured with sufficient accuracy using an OTDR. In fact, it is not essential to determine K and ν as these are parameters which are input to the OTDR or exist as default values in the OTDR's software. However, if default values for K and ν are used to calculate $\rho B_0/A$, then the same values shall be input to the OTDR before carrying out the reflectance calibration.

E.3 Preparation and calibration of the standard reflectance

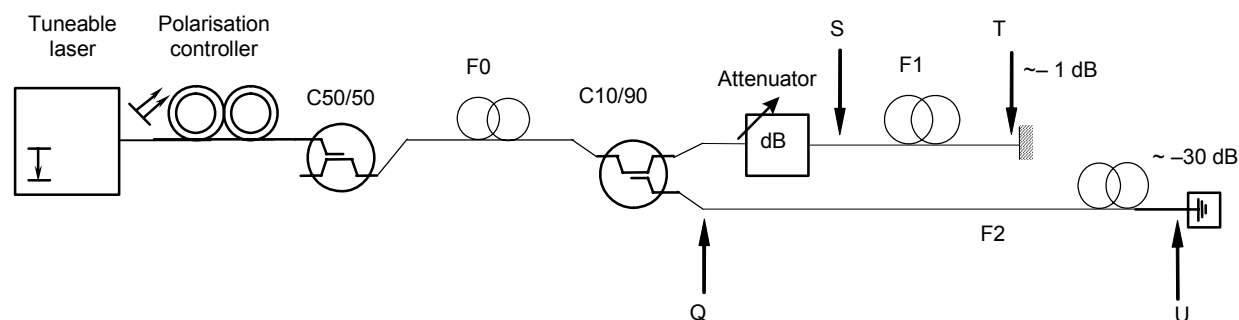
To prepare and calibrate the standard reflectance, proceed as follows:

- select a spool of single-mode fibre conforming to IEC 60793-2-50 type B1.1 or B1.3 with variation in attenuation coefficient of $< \pm 0,01$ dB/km along its length;
- measure the attenuation coefficient of this fibre;
- cut two fibres of length D_1 and D_2 from the spool and wind onto different spools, such that $D_2 = 2 D_1$. Typical lengths for D_1 and D_2 would be 1 000 m and 2 000 m;
- if required, measure the backscatter parameter, K ;
- measure the length D_1 and D_2 of the fibres. This can normally be done with sufficient accuracy using an OTDR, although time of flight may be used if necessary;

- f) select a 2 x 2 achromatic branching device with single-mode pigtails and low wavelength and polarization dependence. Coupling ratio should be approximately 90 %/10 %;
- g) select a variable attenuator with low polarization dependence and low reflectance, typically better than –50 dB;
- h) connect the attenuator to the 10 % output of the branching device using either a fusion splice or low reflectance (<–50 dB) connectors;
- i) connect the fibres of lengths D_1 and D_2 to the attenuator and the 90 % output port respectively of the coupler using either fusion splices or low reflectance connectors;
- j) fusion splice a high reflector onto the end of fibre F1, it should have a reflectivity >0,8;
- k) cleave the end of fibre F2 and apply matching gel to reduce the end reflection to better than –30 dB. Alternatively, cleave the end at an angle;
- l) fusion splice the lead-in fibre F0 to the input port of the coupler, the length of this fibre is not critical , ~ 100 m is suggested;
- m) fusion splice the output port of an achromatic 2 x 2 coupler with a 50/50 % coupling ratio to the end of the lead in fibre F0. Terminate the other output port with a low reflection;
- n) connect one of the input ports of the 50/50 coupler to a tuneable laser via a polarization state controller;

NOTE If it is not required to calibrate the two way loss of the attenuator, then steps o) to r) can be omitted.

- o) connect the other input port of the 90/10 coupler to a power meter;
- p) place a knot at the point Q in the 90 % port of this coupler to remove the backscatter and reflectance signal from this port. Here and in the following, the subscripts refer to the points at which knots are placed, e.g. P_{QS} . Figure E.2 shows reference point locations (Q, S, T, U);



IEC 2458/09

Key

F0, F1 and F2	fibres
C10/90 and C50/50	branching devices
S, T, Q and U	reference points

Figure E.2 – Calibration set up and reference points for calibration

- q) set the attenuator to 0 dB attenuation and put a knot at point S after the attenuator to remove the backscatter and reflected power after F1. Record this signal on the power meter $P_{QS,0}$. Repeat for other attenuator settings to obtain $P_{QS,AT}$. This represents the background return signal of the system as a function of the attenuator setting. **All power measurements should be in linear units;**
- r) remove the knot at S to obtain a signal from the reflector on the power meter, $P_{Q,AT}$. Calibrate the attenuator over the dynamic range and wavelength range required. The power

meter reading for each attenuator setting is corrected for any background signal by subtracting the background signal $P_{QS, AT}$ from the observed power $P_{Q, AT}$. Use this data to calculate a correction factor for the attenuator;

- s) remove the power meter from the input port and cleave the end of the coupler fibre pigtail to achieve a low reflectance (< -50 dB);
- t) connect the power meter to the unused input port of the 50/50 coupler;
- u) set the attenuator to 0 dB loss. Place knots at Q and S and measure the new background, P_{QS} ;
- v) remove knot at S and place knots in the fibre F1 at point T and in F2 at Q. Measure the power returned to the power meter, P_{QT} . Calculate the backscatter power from the 10 % arm as $P_{QT} - P_{QS}$;
- w) remove the knot at T and record the backscatter plus reflected power P_Q . Calculate the reflected power P_{ρ_0} , as $P_Q - P_{QT}$;
- x) place knots in the fibre F1 at S and F2 at U (remove knot at Q). Record the power returned to the power meter P_{SU} . Calculate the power returned from the 90 % arm as $P_{SU} - P_{QS}$;
- y) calculate $P_{B/A,0}$ as $(P_{QT} - P_{QS}) / (P_{SU} - P_{QS})$ and B_0/A from Equation (E.10);
- z) place knots at T and U and record the backscatter power P_{TU} . Calculate the backscatter power $P_{bs,0}$ as $P_{TU} - P_{QS}$;
- aa) use Equation (E.12) to calculate $\rho B_0/A$ using the values of P_{ρ_0} , $P_{B/A,0}$ and $P_{bs,0}$ from w) , y) and z);
- bb) use Equations (E.7) and (E.2) to calculate R_{ref} for different values of attenuation.

E.4 Uncertainties

Uncertainties in the reference reflectance R_{ref} are composed essentially of three parts (see Equation (E.7)), uncertainty in $\rho B_0/A$, u_p , uncertainty due to the attenuator u_{AT} and uncertainty in the quantity B/A , u_{BA} . The last term is negligible and can be neglected.

The uncertainty of the attenuator is made up of contributions:

u_{A1} arising from the repeatability of the attenuator value;

u_{A2} arising from the linearity of the power meter used to characterise the attenuator;

u_{A3} arising from any polarization dependence of the attenuator;

u_{A4} arising from the wavelength dependence of the attenuator combined with the uncertainty of the OTDR's actual wavelength. (May be negligible if the centroidal wavelength of OTDR is known).

The above contributions should be added in quadrature to obtain u_{AT} .

The uncertainty in $\rho B_0/A$ depends on the uncertainties in P_{ρ_0} , $P_{bs,0}$ and the fibre attenuation coefficient and lengths. Again, the error in P_{BA} can be neglected.

$$\rho_{ref} \approx \frac{\rho B_0}{A} = \frac{P_{\rho,0}}{P_{bs,0}} \frac{K}{\alpha v} e^{2\alpha D_1} (1 - e^{-2\alpha D_2}) \quad \text{dB} \quad (\text{E.13})$$

$$\approx \frac{P_{\rho,0} K}{P_{bs,0} v} (1 + 2\alpha D_1) (2D_2) \quad (\text{E.14})$$

From this, we obtain the contributions to the uncertainties in ρ_{ref} due to D_1 , D_2 , etc. as:

$$u_{\rho,D1} = \rho_{\text{ref}} \frac{2\alpha}{(1+2\alpha D_1)} u_{D1} \quad (\text{E.15})$$

$$u_{\rho,D2} = \rho_{\text{ref}} \frac{1}{D_2} u_{D2} \quad (\text{E.16})$$

$$u_{\rho,\alpha} = \rho_{\text{ref}} \frac{2D_1}{(1+2\alpha D_1)} u_{\alpha} \quad (\text{E.17})$$

$$u_{\rho,P\rho,0} = \rho_{\text{ref}} \frac{1}{P_{\rho,0}} u_{P\rho,0} \quad (\text{E.18})$$

$$u_{\rho,P_{\text{bs}},0} = -\rho_{\text{ref}} \frac{1}{P_{\text{bs},0}} u_{P_{\text{bs}},0} \quad (\text{E.19})$$

The errors in the quantities $P_{\rho,0}$, $P_{\text{bs},0}$ are composed of the errors in power measurements which comprise them. For each power measurement there will be contributions for detector and source noise and detector linearity. These should be added in quadrature to obtain the uncertainty in $P_{\rho,0}$ and $P_{\text{bs},0}$ which are $u_{P\rho,0}$ and $u_{P_{\text{bs}},0}$. Finally, these should be combined in quadrature with those for D_1 , D_2 etc., to obtain the total error in ρ_{ref} . This is then used to calculate the uncertainty in the reflectance R .

E.5 Documentation

The following calibration results shall be supplied with the reflectance standard:

- a) the approximate fibres lengths, D_0 , D_1 , and D_2 ;
- b) the approximate distance of the reflection from the input connector of the reflectance standard, $D_0 + D_1$; this distance will be used as reflection location;
- c) the backscatter parameter of the fibre (for both wavelengths);
- d) the reference reflectance value, R_{ref} for each value of the attenuator setting;
- e) the backscatter parameter (s) used for the calibration;
- f) the centroidal wavelength (s) used for the calibration;
- g) the uncertainty of the reflectances as calculated with E.4.

Annex F
(normative)

Simple version of reflectance standard

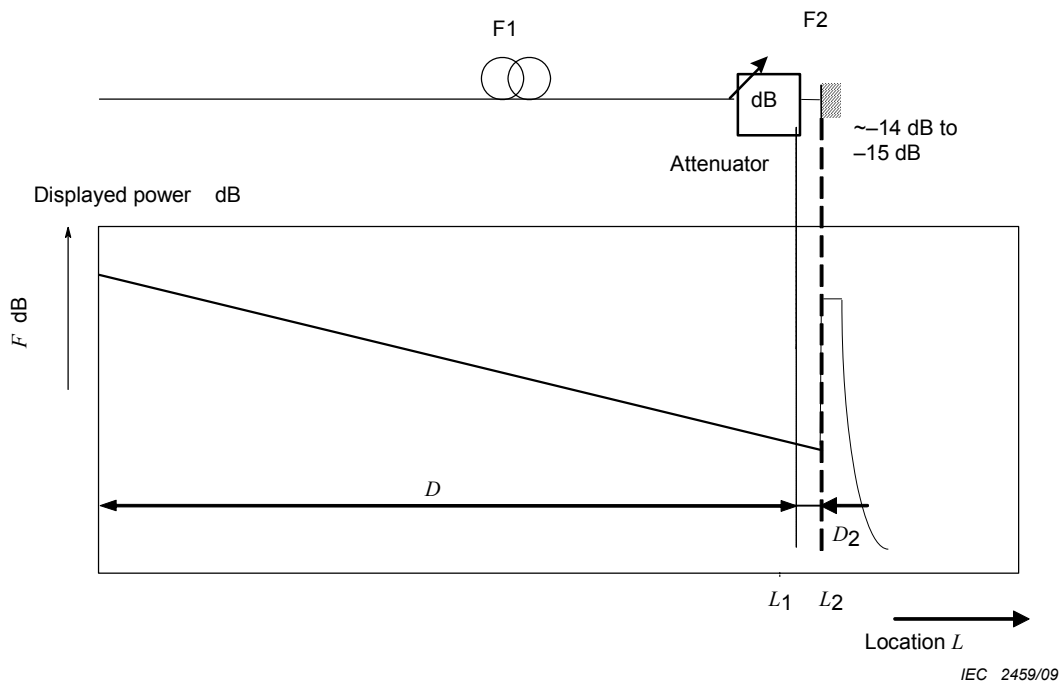
F.1 Description

The simple version of the reflectance standard can be used for high reflection (–20 dB to –35 dB) with an appropriate OTDR pulse width.

The simple reflections standard is composed of an optical fibre F1, a variable optical attenuator and an optical fibre F2 (Figure F.1).

Fibre F1 is 2 km or more and the variable optical attenuator has a low wavelength and polarization dependence. Fibre F2 should be less than 2 m and a length of 1 m is preferred.

The reflectance of both attenuator connectors should be –50 dB or less and the APC type connector is preferred. APC type connector is used at the far-end to provide high reflection.



Key

- F1 and F2 fibres
- D and D_2 length of each fibre
- L_1 and L_2 location of the end of each fibre

Figure F.1 – Reflectance standard description and trace

F.2 Reflectance analysis

Assuming $D \gg D_2$ ($L = L_2$) The power reflected, $P\rho$, from the highly reflecting termination of reflectivity ρ is

$$P_p = P_0 \rho E e^{-2\alpha D} \quad \text{W} \quad (\text{F.1})$$

Where E is the two ways attenuation coefficient of the attenuator. E is composed of two parts:

$$E = E_0 10^{-AT/5} \quad \text{dB} \quad (\text{F.2})$$

where E_0 is the two ways attenuation coefficient of attenuator losses when it is set to 0 dB and AT is the attenuator setting in dB. (AT is taken as being positive.)

In pulsed input condition, and assuming a rectangular pulse, the backscatter power just before the reflection peak, $P_{\text{bs,D}}$, is

$$P_{\text{bs,D}} = P_0 \frac{K}{\alpha \nu} e^{-2\alpha D} (e^{\alpha \nu \Delta T} - 1) \quad \text{W} \quad (\text{F.3})$$

where

- α is the fibre attenuation in m^{-1} ;
- ν is the velocity of light in the fibre ($\Rightarrow c / N$);
- K is the fibre backscatter parameter;
- ΔT is the OTDR pulse width.

The power at the peak, P_p is the sum of the reflected and backscatter powers

$$P_p = P_0 \rho E e^{-2\alpha D} + P_{\text{bs,D}} \quad \text{W} \quad (\text{F.4})$$

The height of the peak above the backscatter, ΔF is

$$\Delta F = 5 \log \left\{ 1 + \frac{\rho E \alpha \nu}{K} (e^{\alpha \nu \Delta T} - 1)^{-1} \right\} \quad \text{dB} \quad (\text{F.5})$$

Thus the attenuator/reflector appears to the OTDR as a reflective event with a reflectivity ρC and hence a reflectance, R of:

$$R = 10 \log \{ \rho E \} \\ R = 10 \log \{ \rho E_0 \} - 2AT = 10 \log \{ \rho_{\text{ref}} \} - 2AT \quad \text{dB} \quad (\text{F.6})$$

From Equation (F.1) the ratio of the reflected to the CW backscattered power with attenuation set to 0 dB is

$$\frac{P_{\text{p,0}}}{P_{\text{bs,0}}} = \frac{P_0 \rho E_0 e^{-2\alpha D}}{P_0 \frac{K}{\alpha \nu} (1 - e^{-2\alpha D})}$$

hence

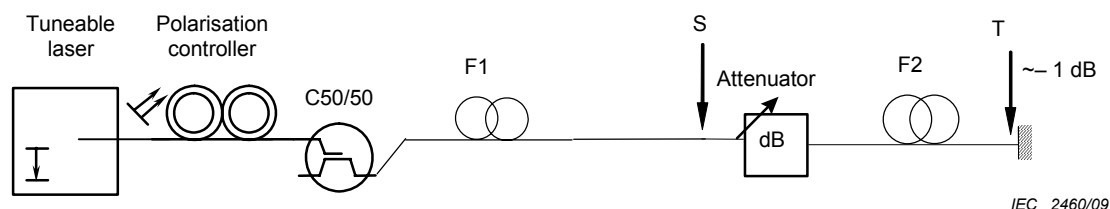
$$\rho_{\text{ref}} = \rho E_0 = \frac{P_{\text{p,0}}}{P_{\text{bs,0}}} \frac{K}{\alpha \nu} (e^{2\alpha D} - 1) \quad (\text{F.7})$$

All of the quantities on the right hand side of Equation (F.7) are measurable; α can be measured with sufficient accuracy using an OTDR. In fact it is not essential to determine K and ν as these are parameters which are input to the OTDR or exist as default values in the OTDR's software.

F.3 Preparation and calibration of standard reflectance simulator

To prepare and calibrate the standard reflectance simulator, proceed as follows:

- use a spool of single-mode fibre conforming to IEC 60793-2-50 with a variation in attenuation coefficient of $< \pm 0,01$ dB/km along its length;
- measure the attenuation coefficient of this fibre;
- select a variable attenuator with a low polarization dependence and high return loss, typically > 50 dB;
- connect a variable attenuator to fibre F1 and connect fibre F2 after the variable attenuator. Set the variable optical attenuator to 0 dB;
- fusion splice the output port of an achromatic 2 x 2 coupler with a 50/50 % coupling ratio to the end of the lead in fibre F1. Terminate the other output port with a low reflection;
- connect one of the input ports of the 50/50 coupler to a tuneable laser via a polarization state controller;
- connect the other input port of the 50/50 coupler to a power meter;
- put a knot at point S before the attenuator to remove the backscatter and reflected power after fibre F1. Record this signal on the power meter $P_{bs,0}$. This represents the background return signal of the system. **All power measurements should be in linear units.** Figure F.2 shows reference point location (S);
- remove the knot at point S to obtain a signal from the reflector on the power meter, $P_{Q,AT}$. Calibrate the attenuator over the dynamic range and wavelength range required. The power meter reading for each attenuator setting is corrected by subtracting the background signal $P_{bs,0}$ from the observed power $P_{Q,AT}$. Use this data to calculate a correction factor for the attenuator;
- set the attenuator to 0 dB. Record $P_{Q,0}$;
- calculate the reflected power $P_{\rho,0}$, as $P_{Q,0} - P_{bs,0}$;
- use Equation (F.7) to calculate ρ_{ref} using the values of $P_{\rho,0}$ $P_{bs,0}$ from k) and h);
- use Equations (F.6) and (F.2) to calculate R_{ref} for different values of attenuation.



Key

F1 and F2	fibres
C50/50	branching device
S	reference point
T	termination

Figure F.2 – Calibration set up and reference points for calibration

F.4 Uncertainties

The uncertainties in the reference reflectance R_{ref} have two main parts: uncertainty due to the attenuator, u_{AT} , and uncertainty due to the far-end reflection, u_R . The uncertainty of the attenuator is made up of:

u_{A1} from the repeatability of the attenuator;

- u_{A2} from the linearity of the power meter used to characterise the attenuator;
- u_{A3} from any polarization dependence of the attenuator;
- u_{A4} from the wavelength dependence of the attenuator combined with the uncertainty of the actual OTDR wavelength (negligible if the OTDR centroidal wavelength is known);
- u_{A5} from the internal reflection;
- u_K from the determination of K .

F.5 Documentation

The following calibration results shall be supplied with the reflections standard:

- a) the approximate lengths, D and D_2 ;
- b) the backscatter parameter of the fibre (for each wavelength);
- c) the reference reflectance value, R_{ref} for each attenuator setting;
- d) the backscatter level used for calibration;
- e) the centroidal wavelength used for calibration;
- f) the uncertainty of reflectance as calculated based on Equation (F.6).

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Annex G (informative)

OTDR basis: Backscatter theory – Reflectance measurements using an OTDR – Determination of fibre backscatter parameter

G.0 Introduction

In reflectance measurements with an OTDR, it is necessary to know the backscatter power at the point of the reflection. This annex discusses the measurement of the backscatter parameter with the help of a continuous wave (CW) source.

NOTE The analysis is based on that of reference [14] of the bibliography.

G.1 Backscatter theory

The backscattered power at the input port of a fibre is given by:

$$P_{bs}(z) = S\alpha_s \Delta z P_0 e^{-2\alpha z} \quad (G.1)$$

where

P_0 is the pulse input power;

S is the backscatter capture fraction;

α_s is the scattering coefficient, e.g. in m^{-1} ;

α is the attenuation in m^{-1} . Multiply α_{dB} by 0,000 23 to obtain α and α_{dB} is the attenuation in dB/km;

z is the distance at which the backscattered power is generated;

Δz is the fibre length contributing to the backscattered power at each time instant (depending on the pulse width setting: $\Delta z = v \Delta T/2$).

Alternately, the backscattered power can be defined as a time function: $P_{bs}(t)$.

$$2z = vt \Rightarrow P_{bs}(t) = S\alpha_s \frac{v}{2} \Delta T P_0 e^{-\alpha vt} \quad (G.2)$$

where

v is the group velocity, e.g.; in m/s;

$= c / N$ where c the speed of the light in vacuum, N the group index of the fibre;

ΔT is the OTDR pulse width, e.g. in seconds.

It is useful to define the backscatter coefficient C as the ratio of backscattered power at the input side of the fibre to the pulse input power. Equation (G.1) can be used to determine this ratio (it depends on the pulse width):

$$C(\Delta T) = \frac{P_{bs}(t=0)}{P_0} = S\alpha_s \frac{v}{2} \Delta T \quad (G.3)$$

Typical values for single-mode fibre are:

α_s (1 300 nm) $\approx \alpha = 0,000\ 08\ m^{-1}$, from multiplying 0,35 dB/km with 0,000 23

α_s (1 550 nm) $\approx \alpha = 0,000\ 046\ m^{-1}$, from multiplying 0,2 dB/km with 0,000 23

$S \approx (NA / n)^2 / 4 = 0,001\ 2$ from reference [13] of the bibliography.

$$S \approx \frac{3}{2n^2w^2(2\pi/\lambda)^2} \text{ based on the mode field diameter.}$$

For a pulse width of 1 ns, this leads to the following approximate values (however, exact values must be known):

$$\begin{aligned} C_{dB}(1\,300\text{ nm}, 1\text{ ns}) &= -78,4\text{ dB} \\ C_{dB}(1\,550\text{ nm}, 1\text{ ns}) &= -81,5\text{ dB} \end{aligned}$$

In order to have a value only depending on fibre parameters, the backscatter parameter K is defined from the previous Equation (G.3):

$$K = \frac{C(\Delta T)}{\Delta T} = S\alpha_s \frac{v}{2} \quad \text{s}^{-1} \quad (\text{G.4})$$

This leads to the following approximate values:

$$\begin{aligned} K(1\,300\text{ nm}) &= 14,4\text{ s}^{-1} \\ K(1\,550\text{ nm}) &= 7,1\text{ s}^{-1} \end{aligned}$$

G.2 Reflectance measurements using an OTDR

The reflectivity of an optical component is defined as the ratio of reflected to incident power:

$$\rho_{\text{OTDR}} = \frac{P_{\text{refl}}}{P_{\text{inc}}} \quad (\text{G.5})$$

In reflectance measurements with an OTDR, only the reflected power can be measured directly. The incident power must be measured indirectly through the backscatter signal. Figure G.1 shows the relevant OTDR signals.

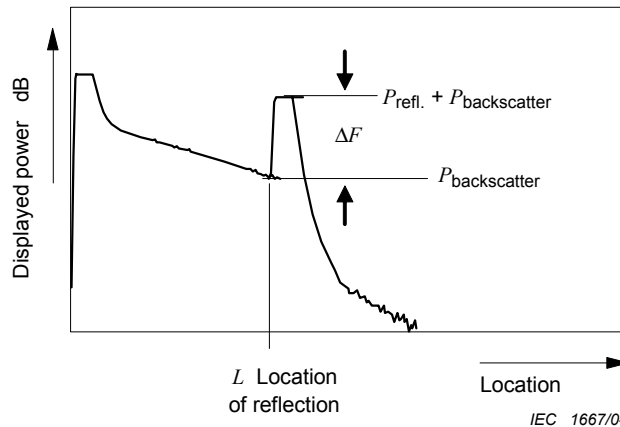


Figure G.1 – OTDR signals used for determining reflectance

At the distance L , the reflected power can be calculated as:

$$P_{\text{refl}}(L) = \rho_{\text{otdr}} P_{\text{inc}}(L) \quad (\text{G.6})$$

The backscattered power at the distance L can be calculated from Equations (G.1) and (G.4):

$$P_{\text{bs}}(L) = K \Delta T P_{\text{inc}}(L) \quad (\text{G.7})$$

The base of the pulse just prior to the reflection, (at location L) corresponds only to the backscatter power level, $P_{\text{bs}}(L)$. The peak just at the reflection corresponds to both the reflection and the backscattered power level, $P_{\text{refl}}(L) + P_{\text{bs}}(L)$.

An OTDR displays these signals in dB. Also, the vertical scale is compressed by a factor of 2. Therefore, the pulse amplitude can be expressed as follow:

$$\Delta F(L) = 5\log_{10}\{P_{\text{refl}}(L) + P_{\text{bs}}(L)\} - 5\log_{10}\{P_{\text{bs}}(L)\} \quad (\text{G.8})$$

Combining Equation (G.8) with (G.6) and (G.7) yields:

$$\begin{aligned} \Delta F(L) &= 5\log_{10}\left\{\frac{\rho P_{\text{inc}}(L) + K \Delta T P_{\text{inc}}(L)}{K \Delta T P_{\text{inc}}(L)}\right\} \\ \Delta F(L) &= 5\log_{10}\left\{\frac{\rho}{K \Delta T} + 1\right\} \end{aligned} \quad (\text{G.9})$$

Therefore:

$$R = 10\log_{10}(K) + 10\log_{10}(\Delta T) + 10\log_{10}\left(10^{\frac{\Delta F(L)}{5}} - 1\right) \quad (\text{G.10})$$

$$R = C(\Delta T) + 10\log_{10}\left(10^{\frac{\Delta F(L)}{5}} - 1\right) \quad (\text{G.11})$$

For high value of ΔF , e.g. more than 8 dB the following approximation can be used:

$$R \approx 10\log_{10}(K) + 10\log_{10}(\Delta T) + 2\Delta F(L) \quad (\text{G.12})$$

NOTE 1 From this Equation (G.12), the error is –0,11 dB at $\Delta F = 8$ dB and – 0,07 dB at $\Delta F = 9$ dB.

In summary, the reflectance of a test device can be calculated on the basis of a measured power difference, ΔF (see Figure G.1), and a known backscatter coefficient, K .

NOTE 2 Previous calculation should be done using the equivalent OTDR's pulsewidth, ΔT , which includes all parameters affecting the displayed pulsewidth. Other parameters, such as the associated attenuation, are not taken into account.

G.3 Determining the backscatter parameter

To determine the backscatter coefficient, C , it is useful to use a continuous wave source. To calculate the total backscattered power from a CW source, Equation (G.1) must be integrated over the total length of the fibre, L .

The result is:

$$P_{\text{bs,cw}} = \frac{S\alpha_s P_{\text{cw}}}{2\alpha} (1 - e^{-2\alpha L}) \quad (\text{G.13})$$

It is important to use a relatively long fibre (e.g. 1 000 m to 5 000 m) because the backscatter signal from a short fibre is likely to be disturbed by parasitic reflections. Assuming that the CW-backscatter signal, $P_{\text{bs,cw}}$, and the total transmission, αL , can be measured, then Equation (G.13) can be converted to:

$$S\alpha_s = \frac{P_{\text{bs,cw}}}{L P_{\text{CW}}} \frac{2\alpha L}{1 - e^{-2\alpha L}} = \frac{P_{\text{bs,cw}}}{P_{\text{CW}}} \frac{A}{L} \quad (\text{G.14})$$

where A is a dimensionless quantity characterising the round-trip transmission:

$$A = \frac{2\alpha L}{1 - e^{-2\alpha L}} \quad (\text{G.15})$$

Combining Equations (G.4) and (G.14), the backscatter coefficient can be calculated as:

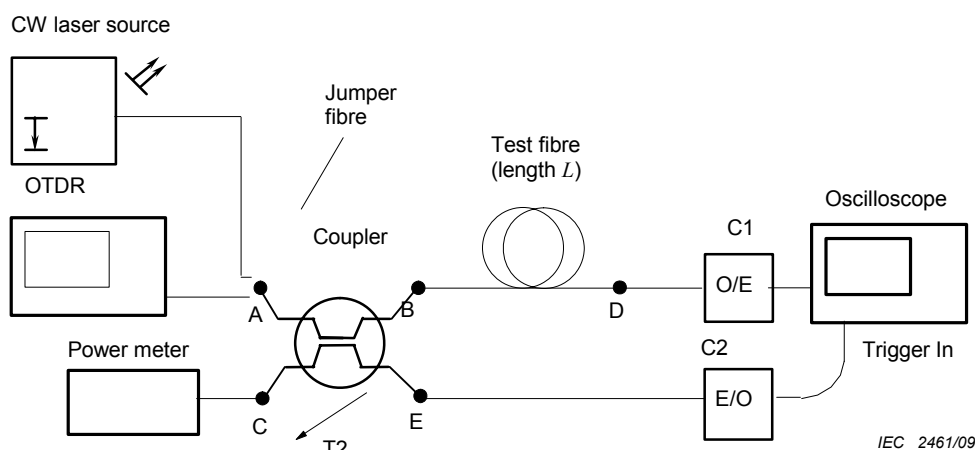
$$K = \frac{P_{\text{bs,cw}}}{P_{\text{cw}}} \frac{A}{L} \frac{v}{2} = \frac{P_{\text{bs,cw}}}{P_{\text{cw}}} \frac{A}{2T} \quad (\text{G.16})$$

This means that the backscatter coefficient can be calculated on the basis of the following measurements:

- CW input power, P_{CW} , to a test fibre;
- total transmission, αL , of the fibre (this is an attenuation measurement);
- backscatter power, $P_{\text{bs, CW}}$, from that fibre;
- transit time, T , through the test fibre.

Figure G.2 shows the set-up for determining the backscatter parameter, K . It consists of an OTDR, a CW laser source, an O/E converter, an oscilloscope, a power meter, a coupler (for example a 3 dB coupler) and a long test fibre of at least 1 000 m (5 000 m recommended) length.

Alternatively an OTDR which can be switched between pulse and CW mode can be used to replace the regular OTDR and the CW laser source.



Key

- A, B, C, D and E reference points
- C2 electrical-to-optical converter
- C1 optical-to-electrical converter
- T2 coupler transmission

Figure G.2 – Set-up for measurement of the backscatter coefficient

The measurement procedure is as follows:

- connect a jumper to the CW laser source. It is assumed that the jumper fibre remains connected to the source and that the CW power from the jumper fibre remains constant during the following measurements;
- to determine the transmission T_2 , splice the source jumper fibre to point B and measure the power P_C at point C;

- c) disconnect the jumper fibre from point B and measure the power at the output of the jumper fibre P_A at point A;
- d) calculate the coupler transmission $T2 = P_C / P_A$;
- e) splice the jumper fibre to point A and measure the incident power $P_{CW} = P_B$ at point B;
- f) make sure that points B and E are terminated in a non-reflective way;

Without the test fibre, measure the backscatter power, $P_{bs,C1}$, at port C;

- g) splice the test fibre to point B and make sure that point D is terminated in a non-reflective way.

Measure the backscatter power, $P_{bs,C2}$, at port C;

- h) calculate the backscatter power, $P_{bs, CW}$, at point B:

$$P_{bs, CW} = \frac{P_{bs,C2} - P_{bs,C1}}{T2} \quad (G.17)$$

- i) remove the terminations at points D and E;
- j) measure the CW output power at the end of the fibre, P_D , at point D;
- k) calculate the fibre transmission:

$$\alpha L = \ln \left(\frac{P_D}{P_B} \right) \quad (G.18)$$

- l) calculate quantity A in the second part of Equation (G.14) ;
- m) connect the source jumper to the OTDR. Set the OTDR with pulse width = 1 μ s.

Measure the transit time, T , through the test fibre, by measuring the pulse arrival time with a combination of O/E converter and oscilloscope, without and with the test fibre inserted;

- n) calculate the backscatter parameter with the help of Equation (G.16).

Alternatively, the ratio $P_{bs, CW} / P_{CW}$ can be measured with a well-calibrated continuous-wave return loss meter, in which case only the total transmission (attenuation) and the transit time remain to be measured.

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INTERNATIONAL
ELECTROTECHNICAL
COMMISSION

3, rue de Varembé
PO Box 131
CH-1211 Geneva 20
Switzerland

Tel: + 41 22 919 02 11
Fax: + 41 22 919 03 00
info@iec.ch
www.iec.ch